

Introduction to Natural Language Processing

Prof. Michael Yeomans

Who am I?



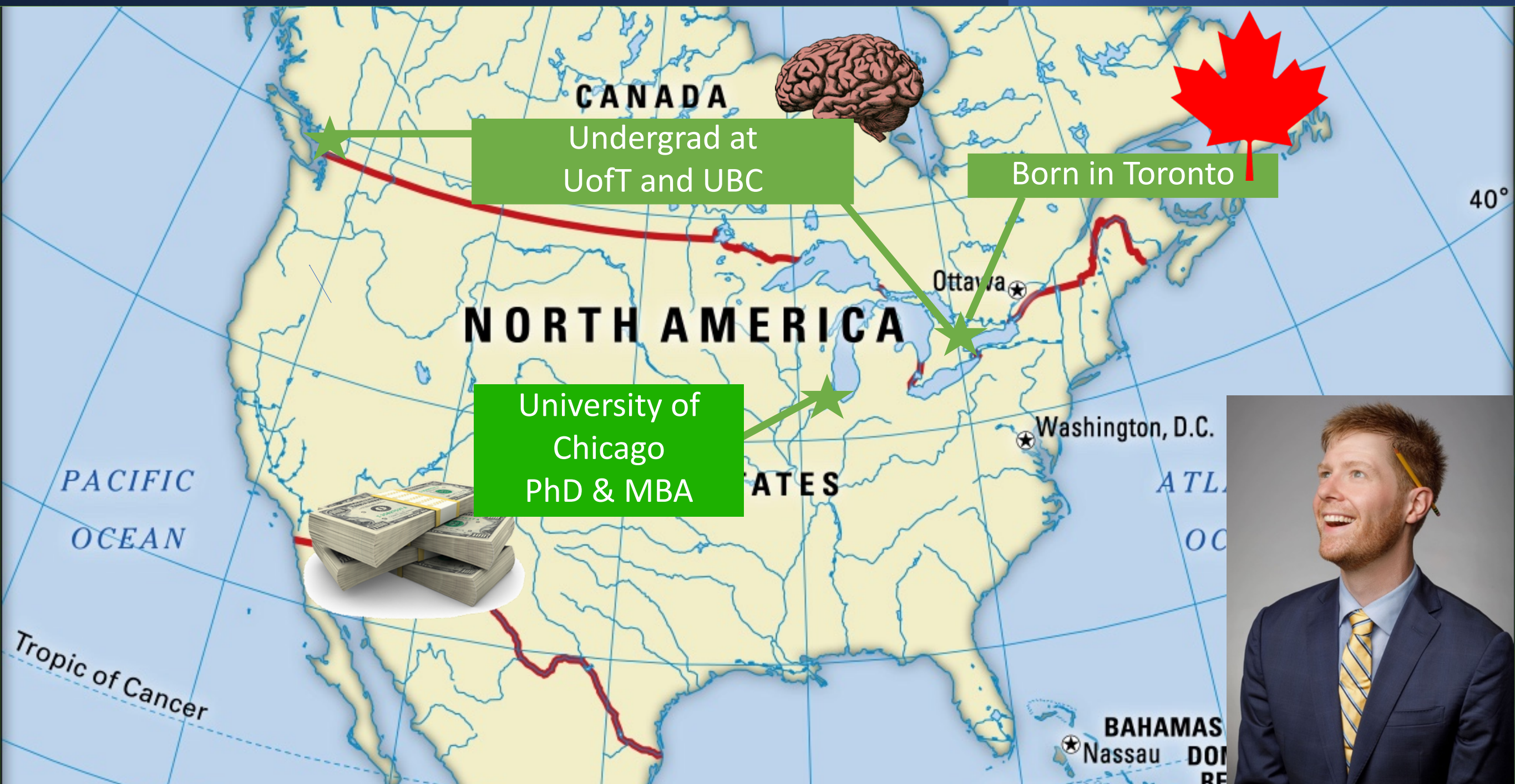
Who am I?



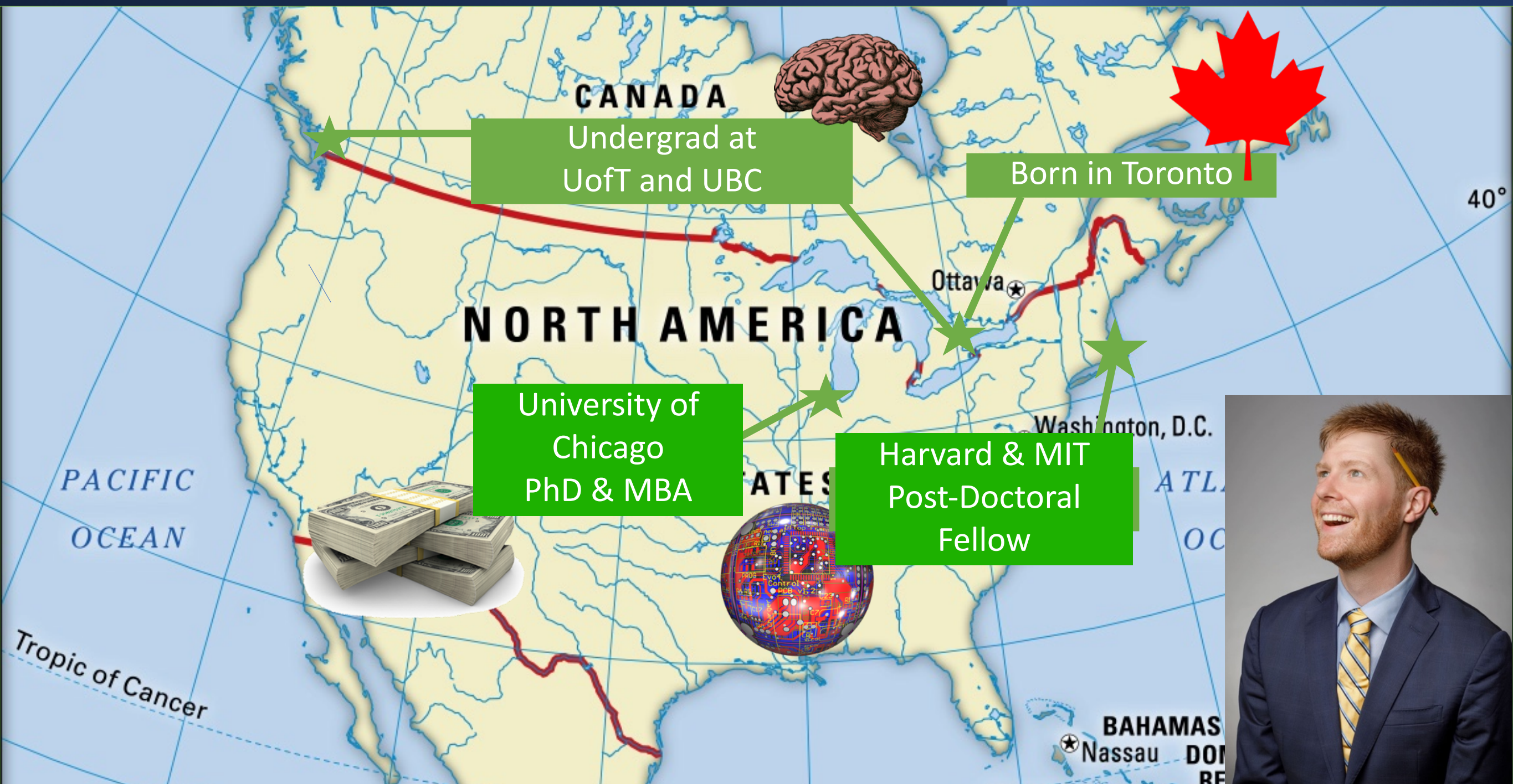
Who am I?



Who am I?



Who am I?



One Question for this Class

How do we turn words into numbers?



One Question for this Class

How do we turn words into numbers?

Measurement Validity

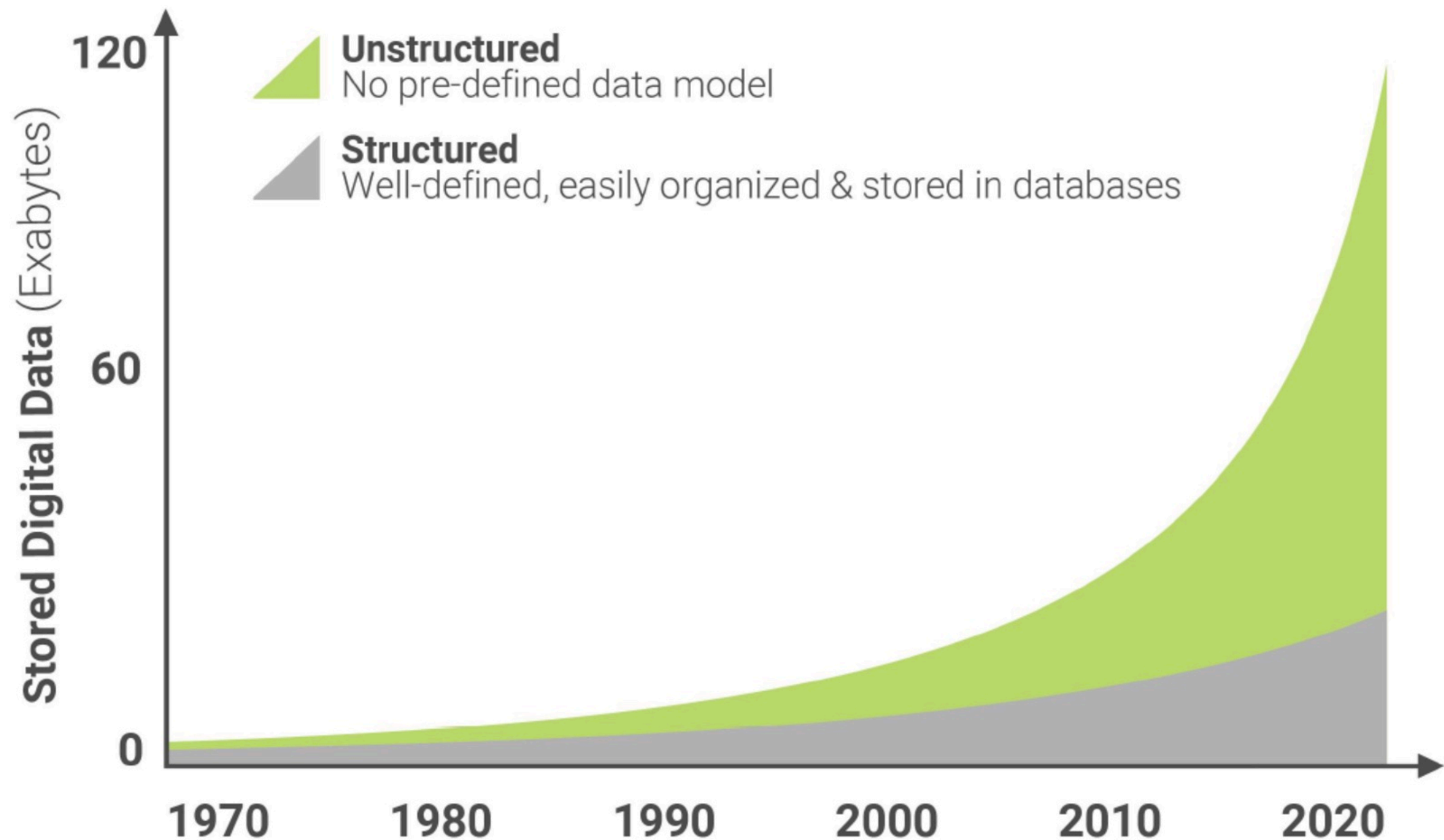
How well does a number represent the behaviour itself?

(Cronbach & Meehl, 1955; John & Benet- Martinez, 2000;
Flake, Pek & Hehman, 2017; Fried & Flake, 2018;
Mullainathan & Speiss, 2017)



Measurement Validity

One solution: structured questions



Measurement in Language

Measurement in Language



I think he needs to do a better job of caring about the work environment he is in. If he loses his job which he is on the verge of, he would be in trouble

I plan to tell him that his failure to politely interact with the customers has resulted in lower sales and is currently putting his job at risk.

I am going to give them critical feedback on the work they have done so far. This is a big project and each of us has important work to contribute. I don't feel as if they are doing their best. I want to steer them in the right direction.

I plan on telling her that she needs to work on her hygiene because she hasn't been smelling very good at the office.

from Blunden, Green & Gino, 2018

Measurement in Language



I think he needs to do a better job of caring about the work environment he is in. If he loses his job which he is on the verge of, he would be in trouble

-1.9

-0.7

I plan to tell him that his failure to politely interact with the customers has resulted in lower sales and is currently putting his job at risk.

1.37

I am going to give them critical feedback on the work they have done so far. This is a big project and each of us has important work to contribute. I don't feel as if they are doing their best. I want to steer them in the right direction.

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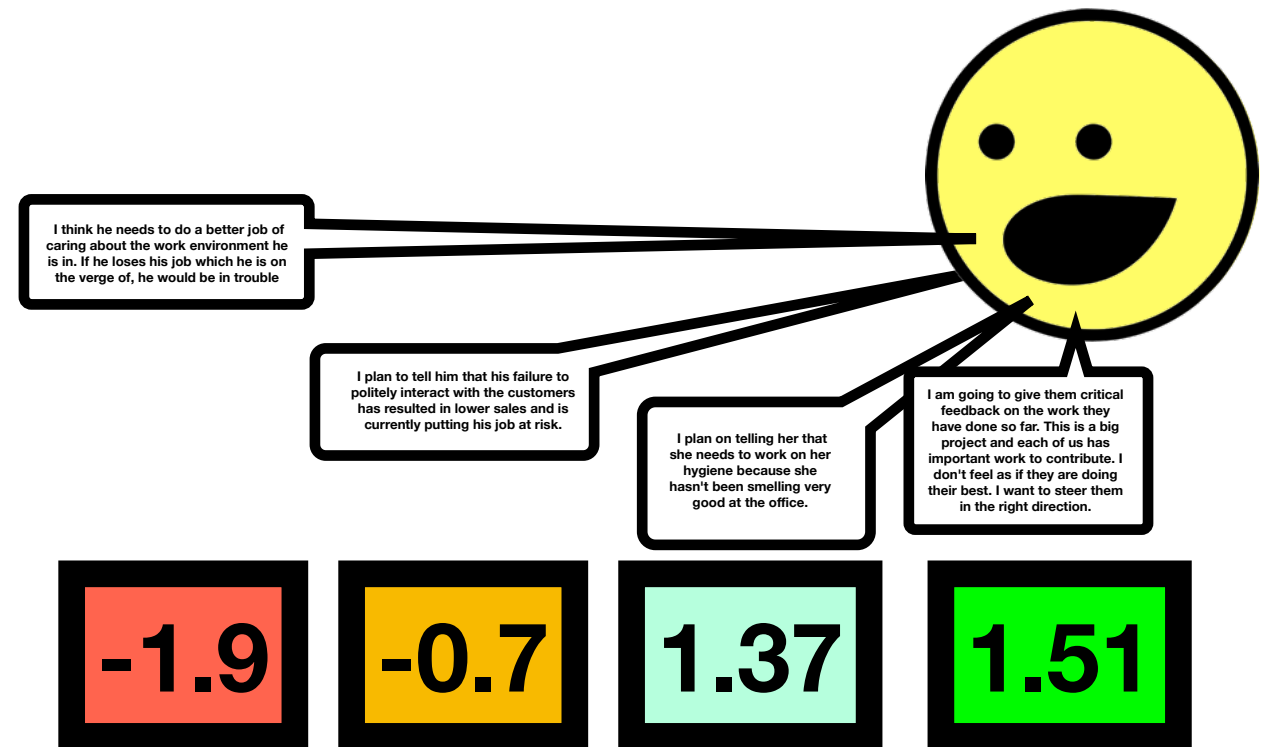
1.51

from Blunden, Green & Gino, 2018

Measurement in Language

Humans: Plusses

What we've always been doing



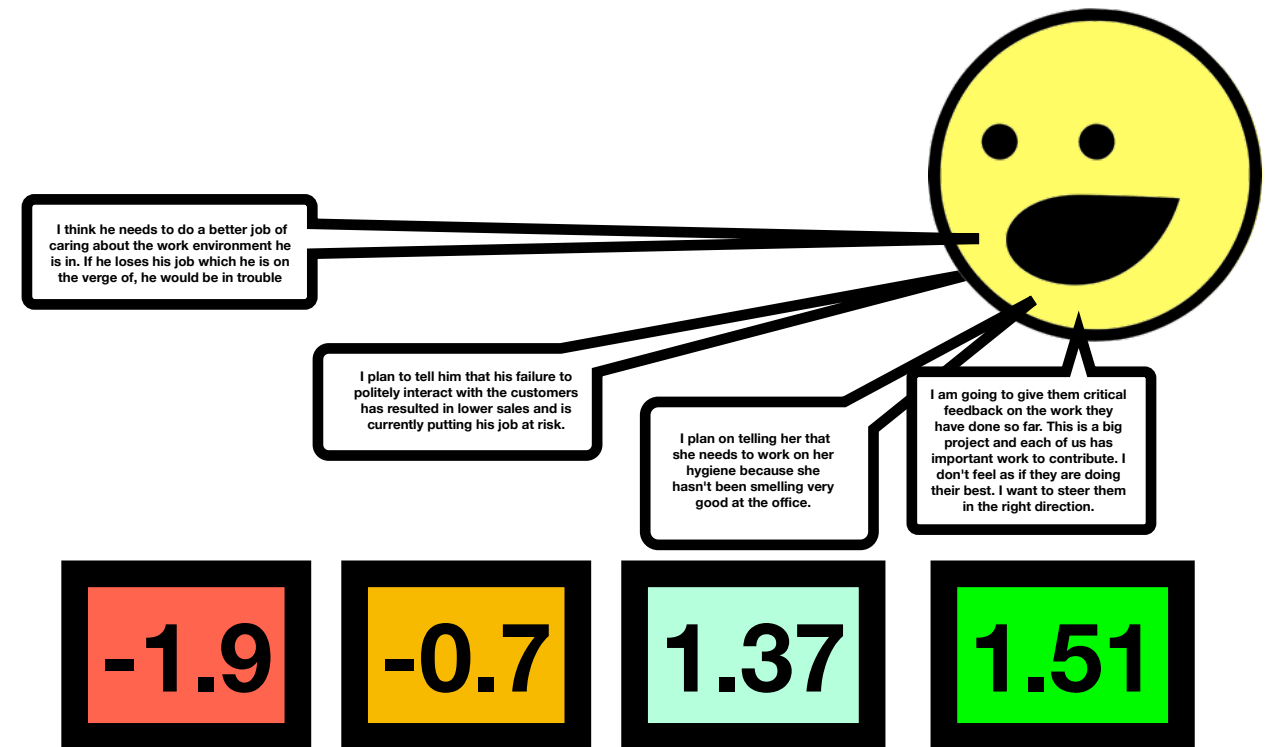
Measurement in Language

Humans: Plusses

What we've always been doing

More accurate than algorithms

for complex tasks



Measurement in Language

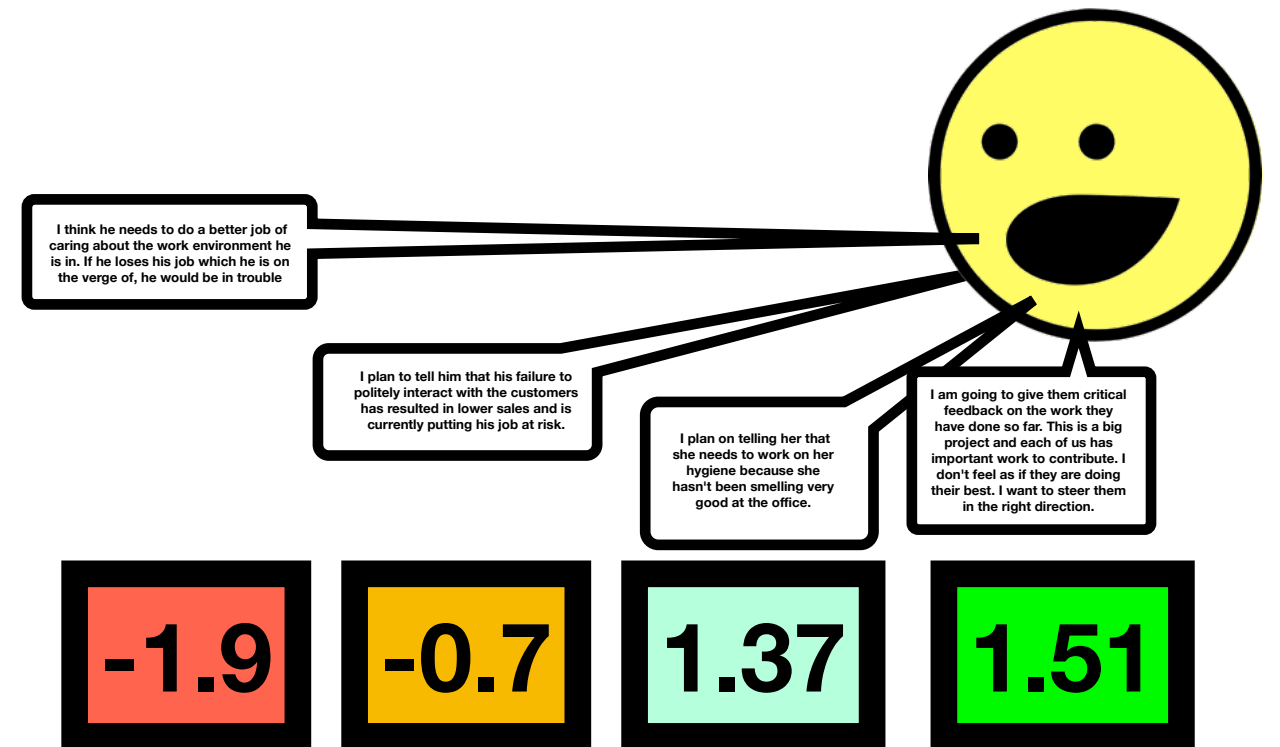
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Measurement in Language

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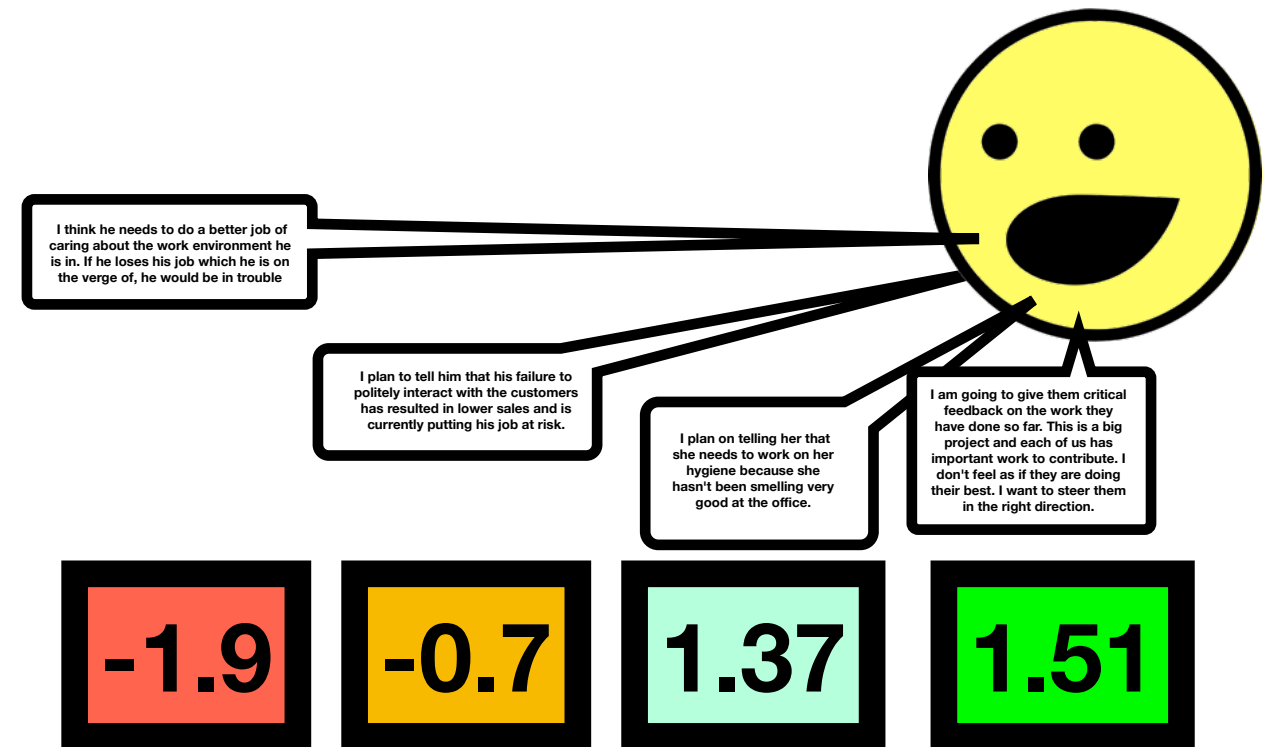
More accurate than algorithms

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Can understand context, nuance

Humans: Minuses

High marginal cost of labor



Measurement in Language

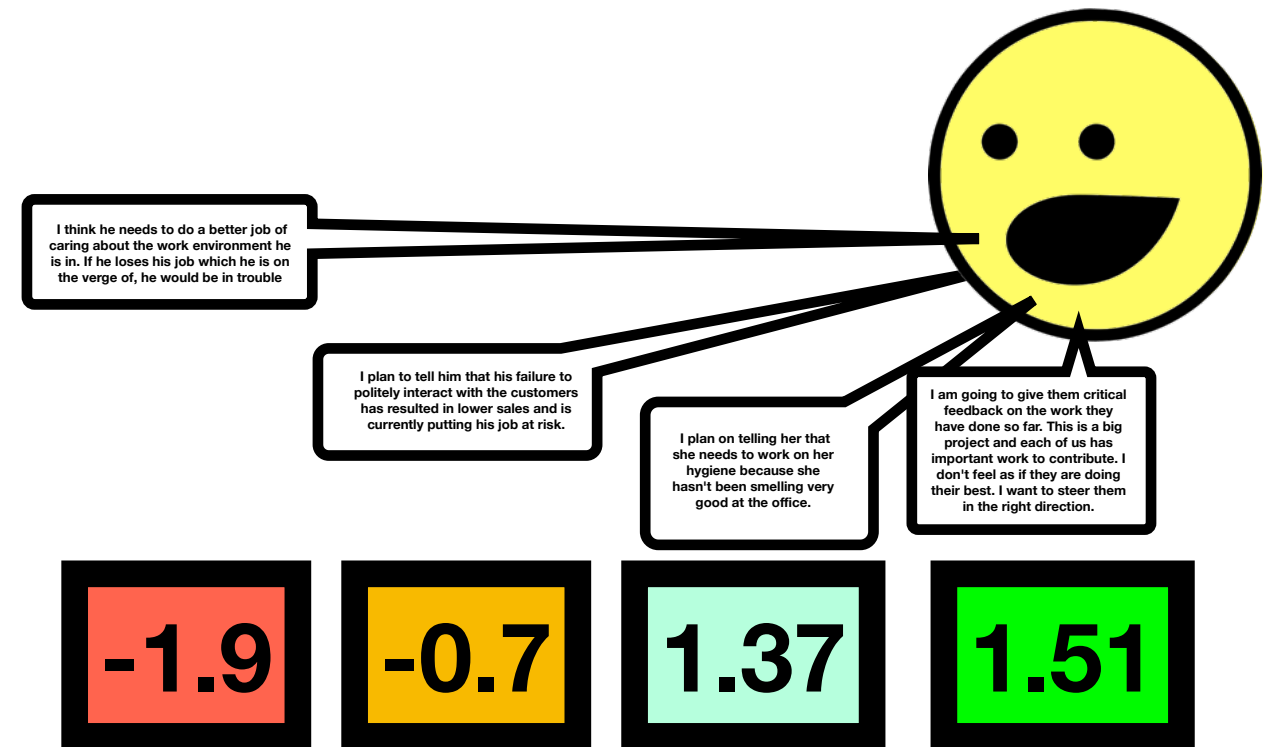
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Humans: Minuses

High marginal cost of labor

Not reliable



Measurement in Language

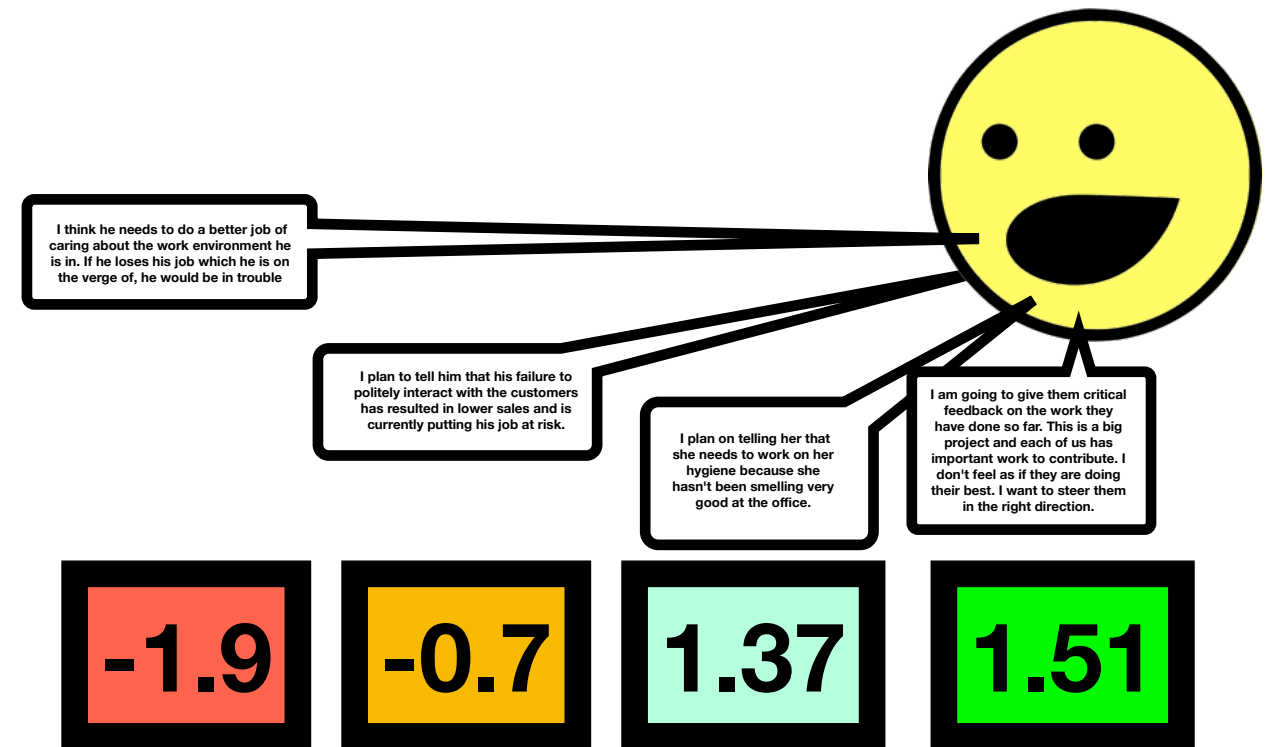
Humans: Plusses

What we've always been doing

More accurate than algorithms

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Humans: Minuses

High marginal cost of labor

Not reliable

Not interpretable



Goals for NLP in Social Science

Goals for NLP in Social Science

Approximate good things about humans

Validate measures in existing theoretical/empirical framework

Measure with roughly the same accuracy as trained RAs

Account for context-specificity

Goals for NLP in Social Science

Approximate good things about humans

Validate measures in existing theoretical/empirical framework

Measure with roughly the same accuracy as trained RAs

Account for context-specificity

Improve on bad things about humans

Decrease marginal cost of measurement in new texts

Increase replicability/reliability of measures

Increase interpretability of resulting models

How is an Idea Communicated?

How is an Idea Communicated?



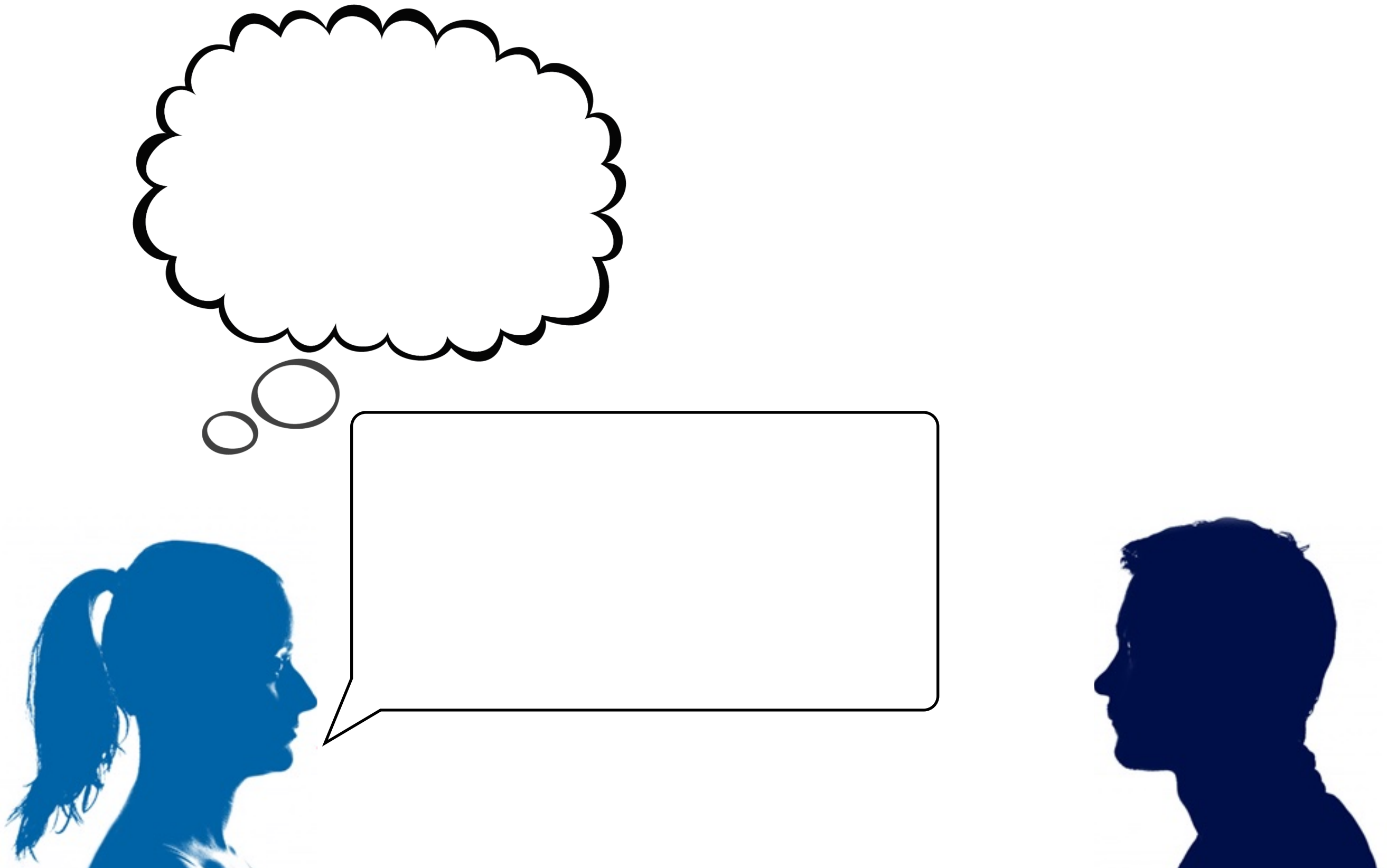
How is an Idea Communicated?



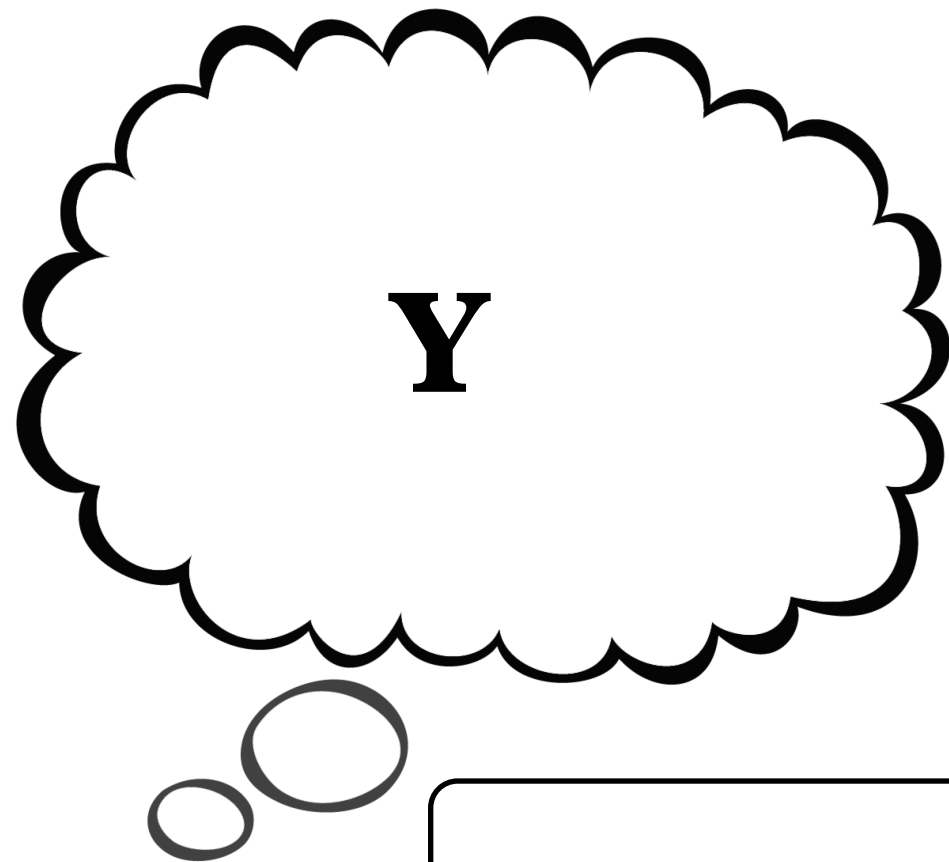
How is an Idea Communicated?



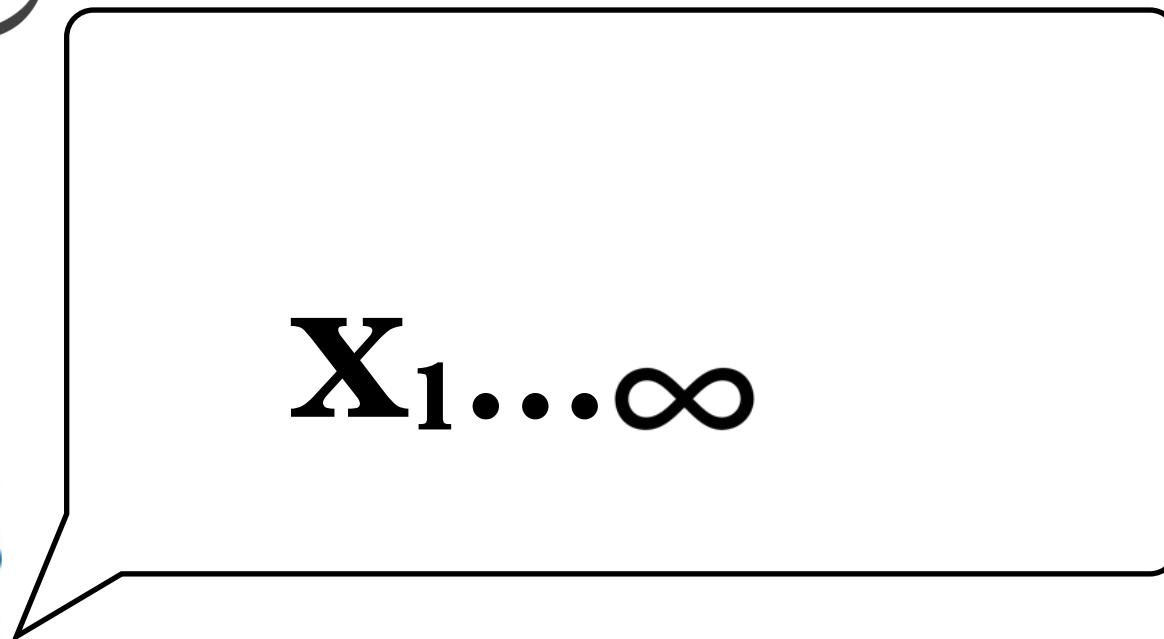
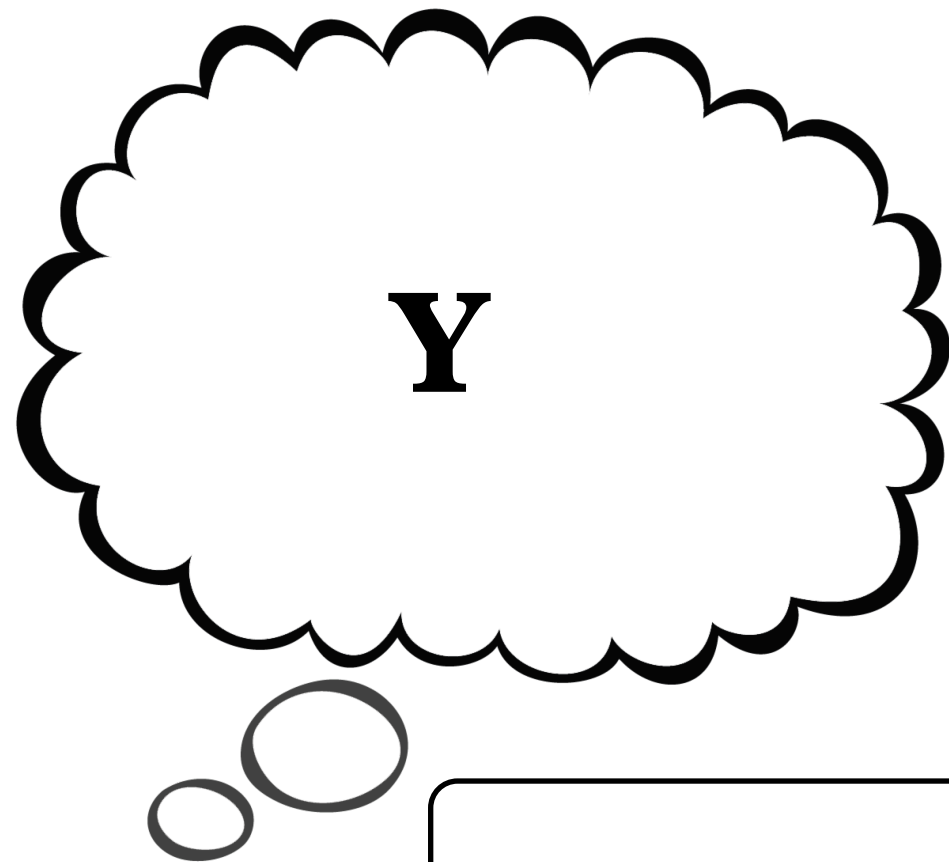
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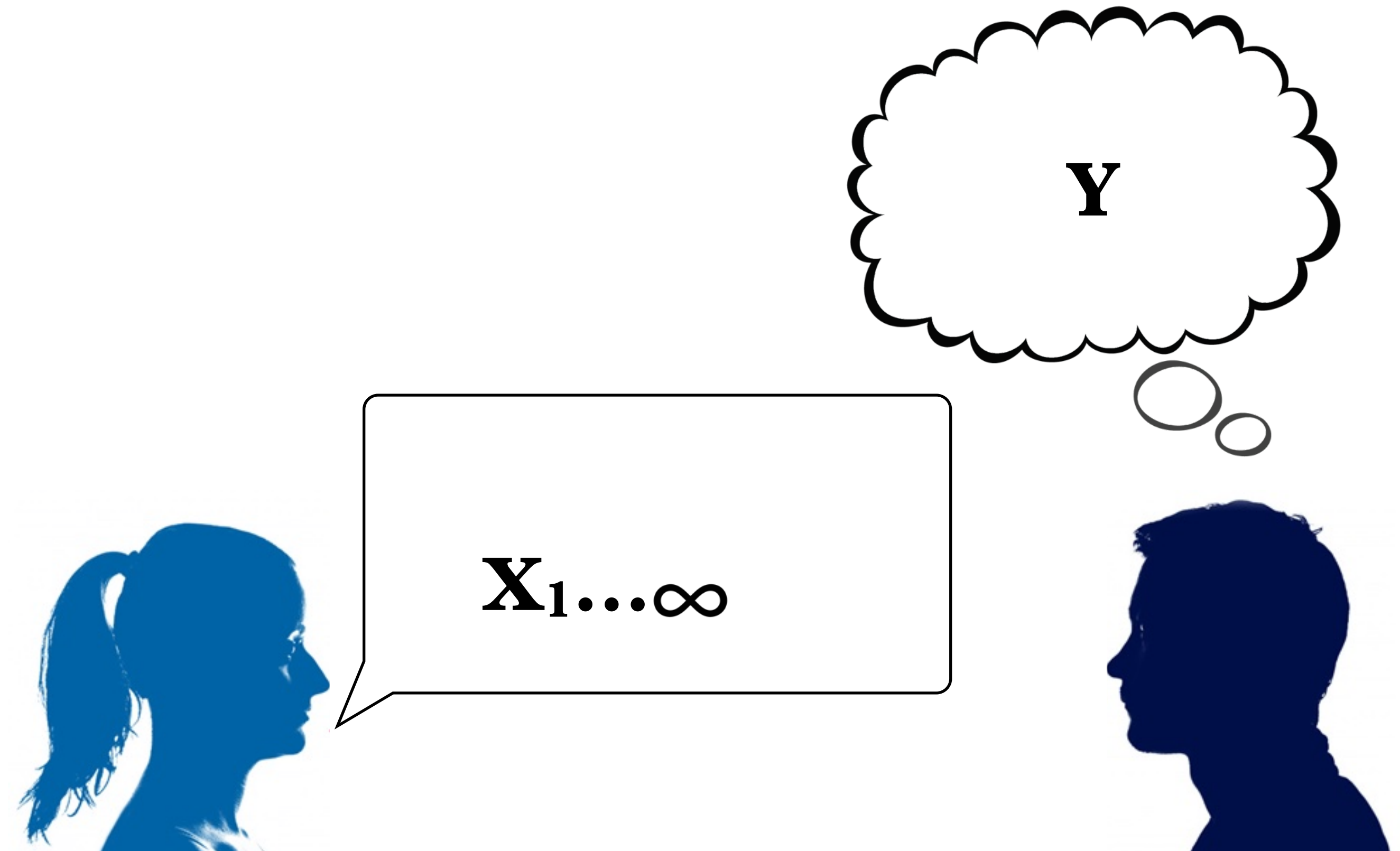
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How is an Idea Communicated?



How is an Idea Communicated?



A Linguistic Model

$$\hat{y} = a_0 + e$$

A Linguistic Model

$$\hat{y} = a_0 + X_1 + X_2 + X_3 + \dots + e$$

Feature Extraction

Select set of observables x

Usually from theory or pre-trained model

Sometimes estimated (e.g. topic model)

A Linguistic Model

$$\hat{y} = a_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + e$$

Feature Extraction

Select set of observables x

Usually from theory or pre-trained model

Sometimes estimated (e.g. topic model)

Feature Estimation

Determine β weights

A Linguistic Model

$$\hat{y} = a_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + e$$

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Select set of observables $\mathbf{x}$

Usually from theory or pre-trained model

Sometimes estimated (e.g. topic model)

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Determine β weights

Usually estimated empirically

A Linguistic Model

$$\hat{y} = a_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + e$$

Feature Extraction

Select set of observables $\mathbf{x}$

Usually from theory or pre-trained model

Sometimes estimated (e.g. topic model)

Feature Estimation

Determine β weights

Usually estimated empirically

Sometimes guesstimated (e.g. equal weights)

Some Basic Principles

Some Basic Principles

There is no "correct model"

Some Basic Principles

There is no "correct model"

Depends on context, domain, time frame,
research question, speakers, goals,
sample size, measurement error,
total compute, time pressure, audience

Some Basic Principles

There is no "correct model"

Depends on context, domain, time frame,
research question, speakers, goals,
sample size, measurement error,
total compute, time pressure, audience

Think economically - prioritize easy wins

"Is the juice worth the squeeze?"

Some Basic Principles

NLP is not a substitute for reading
it is a complement for reading

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If a human reader cannot learn the model,
a machine is unlikely to do better

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a machine is unlikely to do better

Better training data > better algorithm
valid sample
cleaned/structured
human checked/annotated

Feature Extraction Methods

Bag-of-words approaches

Structural approaches

Feature Extraction Methods

Bag-of-words approaches

Ngrams

Dictionaries

Topic Models

Structural approaches

Embedding Models

Parsing Grammar

Dialogue Acts

Feature Extraction Methods

Bag-of-words approaches

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Parsing Grammar

Dialogue Acts

The Bag of Words

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1

Document-Feature Matrix

Document-Feature Matrix

The fundamental workhorse of text mining

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The fundamental workhorse of text mining

Coercing unstructured text into structured data

Document-Feature Matrix

The fundamental workhorse of text mining

Coercing unstructured text into structured data

Imagine a matrix...

Document-Feature Matrix

The fundamental workhorse of text mining

Coercing unstructured text into structured data

Imagine a matrix...

One row for every document

Document-Feature Matrix

The fundamental workhorse of text mining

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Imagine a matrix...

One row for every document

One column for every feature of text

Document-Feature Matrix

The fundamental workhorse of text mining

Coercing unstructured text into structured data

Imagine a matrix...

- One row for every document

- One column for every feature of text

- Each cell counts the feature i in document j

Document-Feature Matrix

Consider this corpus:

"This is a sentence"

"This is also one sentence"

"This is a document"

"This is not a document"

Document-Feature Matrix

Consider this corpus:

"This is a sentence"

"This is also one sentence"

"This is a document"

"This is not a document"

This	is	a	sentence	also	one	document	not
------	----	---	----------	------	-----	----------	-----

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This	is	a	sentence	also	one	document	not

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1	1	1	1	0	0	0	0

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Consider this corpus:

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This	is	a	sentence	also	one	document	not
1	1	1	1	0	0	0	0
1	1	0	1	1	1	0	0

Document-Feature Matrix

Consider this corpus:

"This is a sentence"

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This	is	a	sentence	also	one	document	not
1	1	1	1	0	0	0	0
1	1	0	1	1	1	0	0
1	1	1	0	0	0	1	0
1	1	1	0	0	0	1	1

Pre-Processing the Bag

Pre-Processing the Bag

Punctuation

"This is a sentence!"

"This is a sentence?"

"This is a sentence."

"This is?"

Pre-Processing the Bag

Punctuation

"This is a sentence!"

"This is a sentence?"

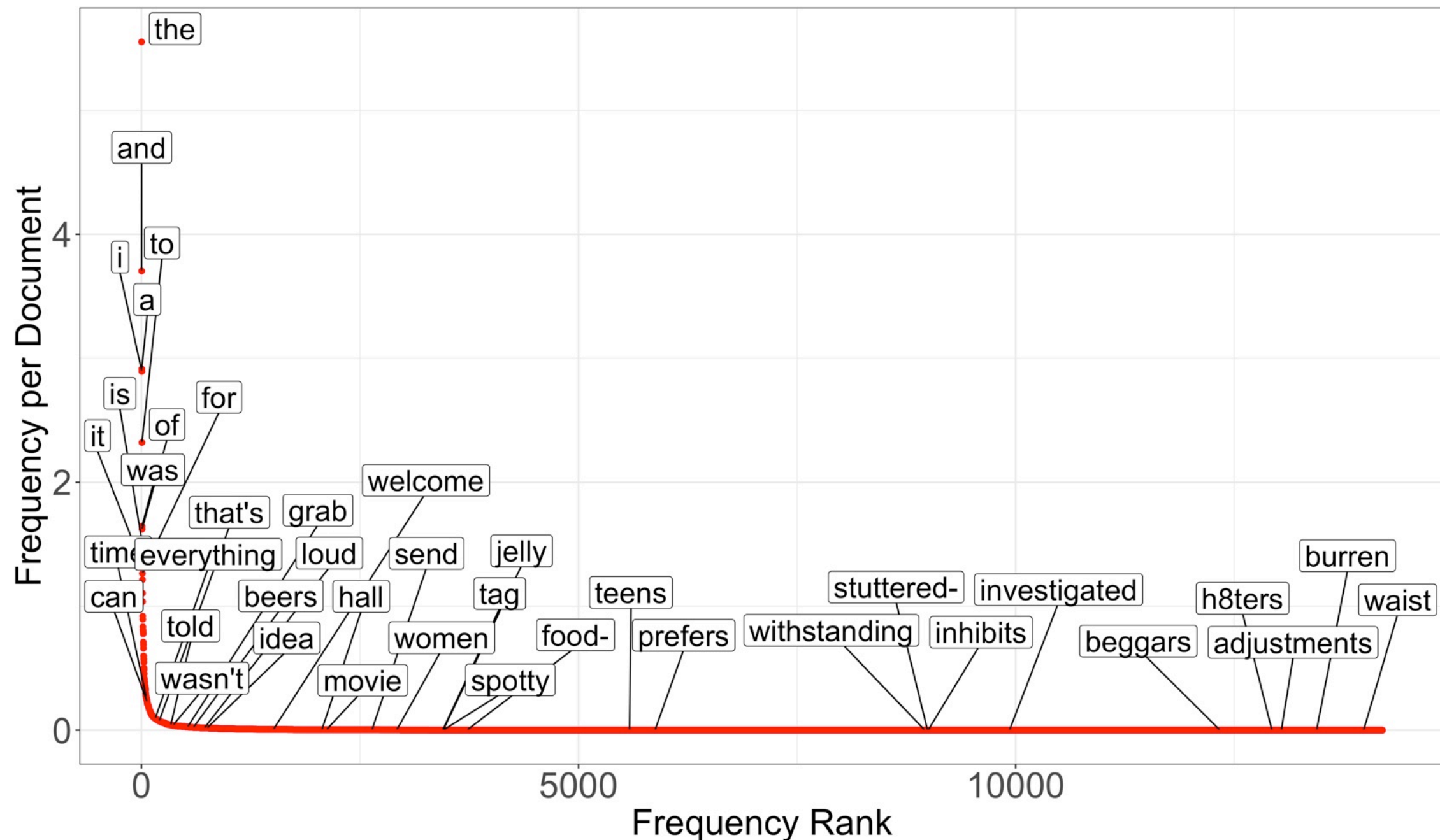
"This is a sentence."

"This is?"

This	is	a	sentence	xmark	qmark
1	1	1	1	1	0
1	1	1	1	0	1
1	1	1	1	0	0
1	1	0	0	0	1

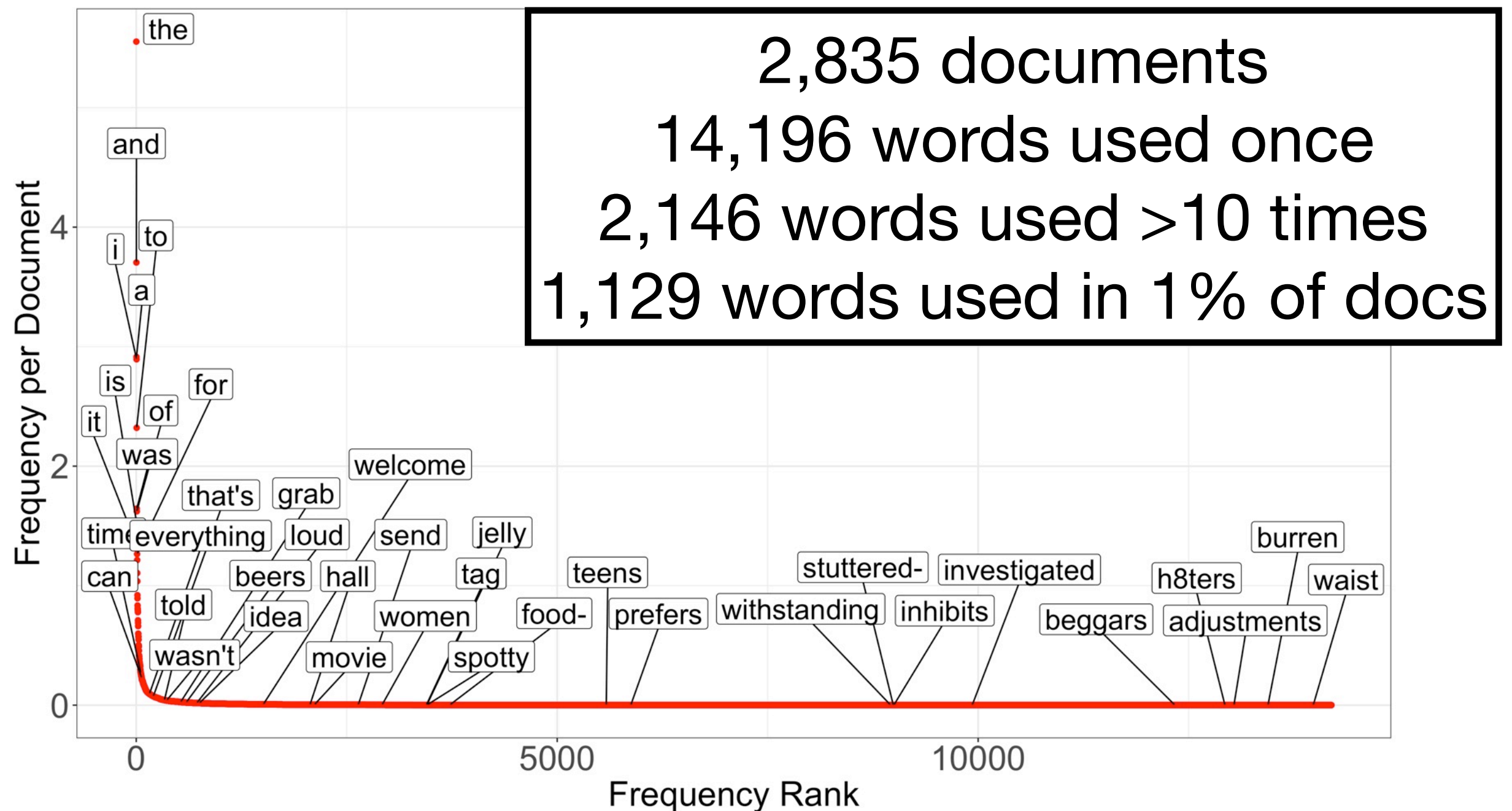
Pre-Processing the Bag

Filtering rare words



Pre-Processing the Bag

Filtering rare words



Pre-Processing the Bag

Capitalisation

"This is the sentence."

"The sentence is this."

"This is the mark."

"This is Mark."

Pre-Processing the Bag

Capitalisation

"This is the sentence."

"The sentence is this."

"This is the mark."

"This is Mark."

This	is	the	sentence	The	this	mark	Mark
1	1	1	1	0	0	0	0
0	1	0	1	1	1	0	0
1	1	1	0	0	0	1	0
1	1	0	0	0	0	0	1

Pre-Processing the Bag

Word Stemming

Pre-Processing the Bag

Word Stemming

changing
changed
change

stemming →

chang
chang
chang

studying
studies
study

stemming →

studi
studi
studi

Pre-Processing the Bag

Word Stemming

chang^{ing}
chang^{ed}
chang^e *stemming* → chang
chang
chang

"This is a sentence."
"This is sentenced."
"These are sentences."

stud^y^{ing}
stud^{ies}
stud^y *stemming* → studi
studi
studi

this	is	a	sentence	sentenced	these	are	sentences
1	1	1	1	0	0	0	0
1	1	0	0	1	0	0	0
0	0	0	0	0	1	1	1

Pre-Processing the Bag

Word Stemming

chang^{ing}
chang^{ed}
chang^e *stemming* → chang
chang
chang

"This is a sentence."
"This is sentenced."
"These are sentences."

stud^y^{ing}
stud^{ies}
stud^y *stemming* → studi
studi
studi

this	is	a	sentenc	these	are
1	1	1	1	0	0
1	1	0	1	0	0
0	0	0	1	1	1

Pre-Processing the Bag

Contractions

"This is a sentence."

"This isn't a sentence."

"This is not a sentence."

Pre-Processing the Bag

Contractions

"This is a sentence."

"This isn't a sentence."

"This is not a sentence."

this	is	a	sentence	isn't	not
1	1	1	1	0	0
1	0	1	1	1	0
1	1	1	1	0	1

Pre-Processing the Bag

Contractions

"This is a sentence."

"This isn't a sentence." -> "This is not a sentence"

"This is not a sentence."

Pre-Processing the Bag

Contractions

"This is a sentence."

"This isn't a sentence." -> "This is not a sentence"

"This is not a sentence."

this	is	a	sentence	not
1	1	1	1	0
1	1	1	1	1
1	1	1	1	1

Pre-Processing the Bag

Dropping common “stop” words

"This is a sentence."

"this is also a sentence."

"here is a rare word."

"here is another word."

"and another sentence."

Pre-Processing the Bag

Dropping common “stop” words

"This is a sentence."

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"and another sentence."

this	is	a	sentence	also	here	rare	word	another	and
1	1	1	1	0	0	0	0	0	0
1	1	1	1	1	0	0	0	0	0
0	1	1	0	0	1	1	1	0	0
0	1	0	0	0	1	0	1	1	0
0	0	0	1	0	0	0	0	1	1

Pre-Processing the Bag

Dropping common “stop” words

"This is a sentence."

"this is also a sentence."

"here is a rare word."

"here is another word."

"and another sentence."

sentence	also	rare	word	another
1	0	0	0	0
1	1	0	0	0
0	0	1	1	0
0	0	0	1	1
1	0	0	0	1

Pre-Processing the Bag

Dropping common “stop” words

i	down	is	should've	she's	more	because	haven	yourself	why	a	didn't
me	in	are	now	her	most	as	haven't	yourselves	how	an	doesn
my	out	was	d	hers	other	until	isn	he	all	the	doesn't
myself	on	were	ll	herself	some	while	isn't	him	any	and	hadn
we	off	be	m	it	such	of	ma	his	both	but	hadn't
our	over	been	o	it's	no	at	mightn	himself	each	if	hasn
ours	under	being	re	its	nor	by	mightn't	she	few	or	hasn't
ourselves	again	have	ve	itself	not	for	mustn	who	s	before	wasn
you	further	has	y	they	only	with	mustn't	whom	t	after	wasn't
you're	then	had	ain	them	own	about	needn	this	can	above	weren
you've	once	having	aren	their	same	against	needn't	that	will	below	weren't
you'll	here	do	aren't	theirs	so	between	shan	that'll	just	to	won
you'd	there	does	couldn	themselves	than	into	shan't	these	don	from	won't
your	when	did	couldn't	what	too	through	shouldn	those	don't	up	wouldn
yours	where	doing	didn	which	very	during	shouldn't	am	should		wouldn't

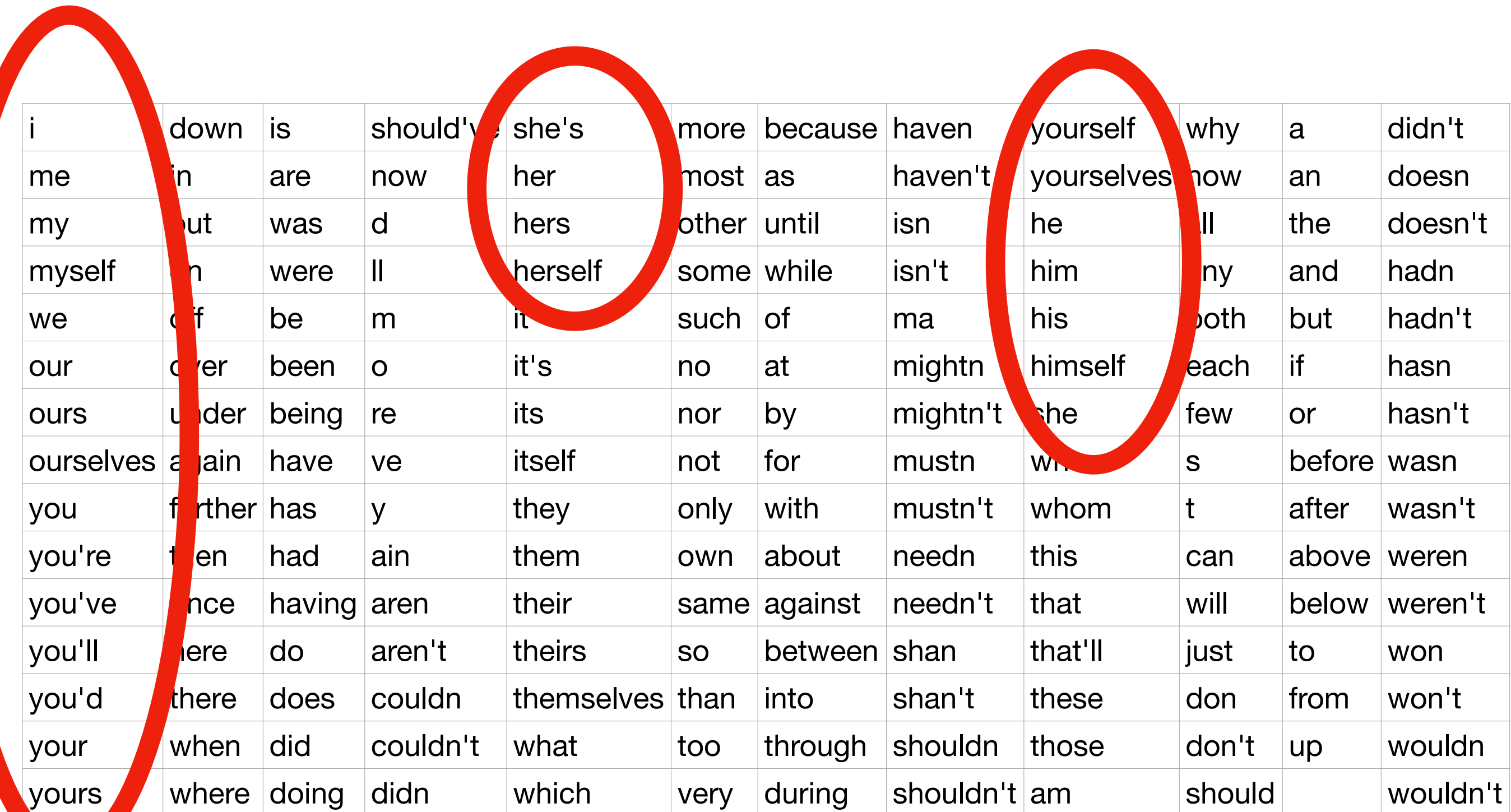
Pre-Processing the Bag

Dropping common “stop” words

i	down	is	should've	she's	more	because	haven	yourself	why	a	didn't
me	in	are	now	her	most	as	haven't	yourself	how	an	doesn
my	out	was	d	hers	other	until	isn	he	at	the	doesn't
myself	on	were	ll	herself	some	while	isn't	him	any	and	hadn
we	off	be	m	it	such	of	ma	his	both	but	hadn't
our	over	been	o	it's	no	at	mightn	himself	each	if	hasn
ours	under	being	re	its	nor	by	mightn't	one	few	or	hasn't
ourselves	again	have	ve	itself	not	for	mustn	who	s	before	wasn
you	further	has	y	they	only	with	mustn't	whom	t	after	wasn't
you're	then	had	ain	them	own	about	needn	it	can	above	weren
you've	once	having	aren	their	same	against	needn't	that	will	below	weren't
you'll	here	do	aren't	theirs	so	between	shan	that'll	just	to	won
you'd	when	does	couldn	themselves	than	into	shan't	these	don	from	won't
your	when	did	couldn't	what	too	through	shouldn	those	don't	up	wouldn
yours	where	doing	didn	which	very	during	shouldn't	am	should		wouldn't

Pre-Processing the Bag

Dropping common “stop” words



i	down	is	should've	she's	more	because	haven	yourself	why	a	didn't
me	in	are	now	her	most	as	haven't	yourselves	now	an	doesn
my	out	was	d	hers	other	until	isn	he	all	the	doesn't
myself	on	were	ll	herself	some	while	isn't	him	ny	and	hadn
we	off	be	m	it	such	of	ma	his	both	but	hadn't
our	over	been	o	it's	no	at	mightn	himself	each	if	hasn
ours	under	being	re	its	nor	by	mightn't	she	few	or	hasn't
ourselves	again	have	ve	itself	not	for	mustn	wh	s	before	wasn
you	further	has	y	they	only	with	mustn't	whom	t	after	wasn't
you're	then	had	ain	them	own	about	needn	this	can	above	weren
you've	nce	having	aren	their	same	against	needn't	that	will	below	weren't
you'll	ere	do	aren't	theirs	so	between	shan	that'll	just	to	won
you'd	there	does	couldn	themselves	than	into	shan't	these	don	from	won't
your	when	did	couldn't	what	too	through	shouldn	those	don't	up	wouldn
yours	where	doing	didn	which	very	during	shouldn't	am	should		wouldn't

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Dropping common “stop” words

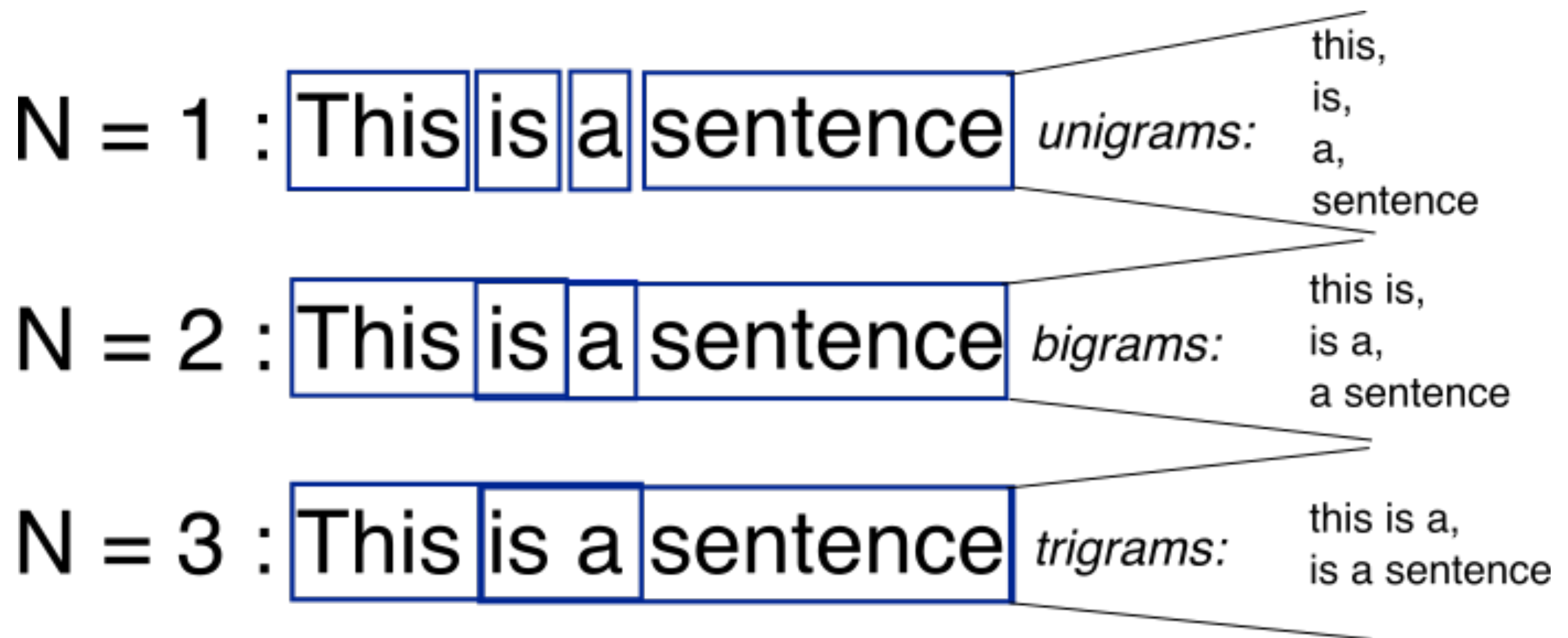
i	down	is	should've	she's	more	because	haven	yourself	why	a	isn't
me	in	are	now	her	most	as	haven't	yourselves	how	an	doesn
my	out	was	d	hers	other	until	isn	he	all	the	doesn't
myself	on	were	ll	herself	some	while	isn't	him	any	and	hadn
we	off	be	m	it	h	of	ma	his	both	but	hadn't
our	over	been	o	it's	no	t	mightn	himself	each	if	hasn
ours	under	being	re	its	nor	by	mightn't	he	few	or	hasn't
ourselves	again	have	ve	itself	not	or	mustn	who	s	before	wasn
you	further	has	y	they	only	with	mustn't	whom	t	after	wasn't
you're	then	had	ain	them	own	about	needn	his	can	above	weren
you've	once	having	aren	their	same	against	needn't	that	will	below	weren't
you'll	here	do	aren't	theirs	so	between	shan	that'll	just	to	won
you'd	there	does	couldn	themselves	than	into	shan't	these	don	om	won't
your	when	did	couldn't	what	too	through	shouldn	those	don't	p	wouldn
yours	where	doing	didn	which	very	during	shouldn'	am	should		wouldn't

Pre-Processing the Bag

Constructing phrases - “n-grams”

Pre-Processing the Bag

Constructing phrases - “n-grams”



Pre-Processing the Bag

Constructing phrases - “n-grams”

"This is a test."

"This is a guess."

Pre-Processing the Bag

Constructing phrases - “n-grams”

"This is a test."

"This is a guess."

this	is	a	test	this_is	is_a	a_test	this_is_a	is_a_test	guess	a_guess	is_a_guess
1	1	1	1	1	1	1	1	1	0	0	0
1	1	1	0	1	1	0	1	0	1	1	1

Pre-Processing the Bag

Punctuations (& emojis/emoticons)

Capitalisations

Contractions

Stemming

Common word weighting

Constructing phrases (ngrams)

Rare word filtering

Dictionaries

Dictionaries

Every word has its own score (from raters)

Dictionaries

Every word has its own score (from raters)

	Word	Score	Word	Score
Valence	love	1.000	toxic	0.008
	happy	1.000	nightmare	0.005
Arousal	elated	0.960	mellow	0.069
	frenzy	0.965	napping	0.046
Dominance	powerful	0.991	weak	0.045
	leadership	0.983	empty	0.081

(Osgood et al., 1957; Mohammad, 2018)

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Every word in a category has the same score

LIWC

(Tausczik & Pennebaker, 2010)

"a+"	"abound"	"abounds"	"achievement"
"abundance"	"abundant"	"accessible"	"acumen"
"accessible"	"acclaim"	"acclaimed"	"adequate"
"acclamation"	"accolade"	"accolades"	"achievements"
"accommodative"	"accommodative"	"accomplish"	"adaptable"
"accomplished"	"accomplishment"	"accomplishment"	"adjustable"

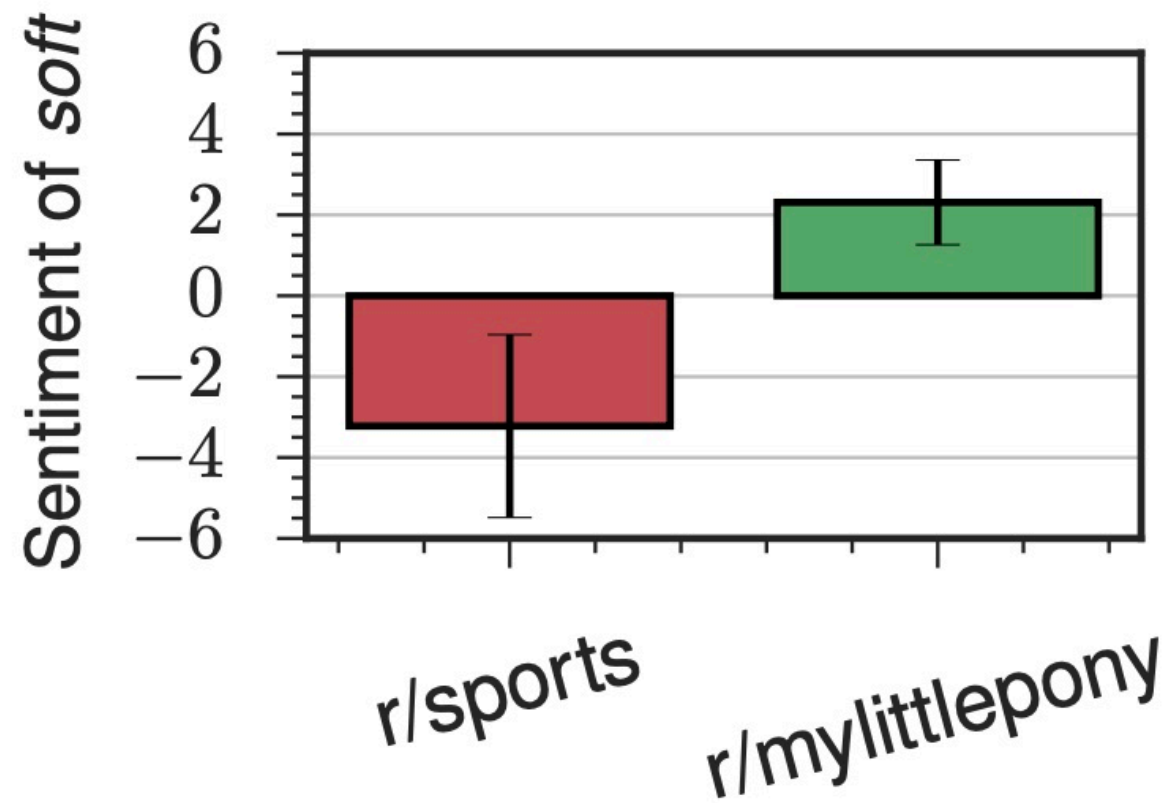
Sentiment

Context Specificity

“Inducing Domain-Specific Sentiment Lexicons from Unlabeled Corpora”
(Hamilton et al., 2016)

Sentiment

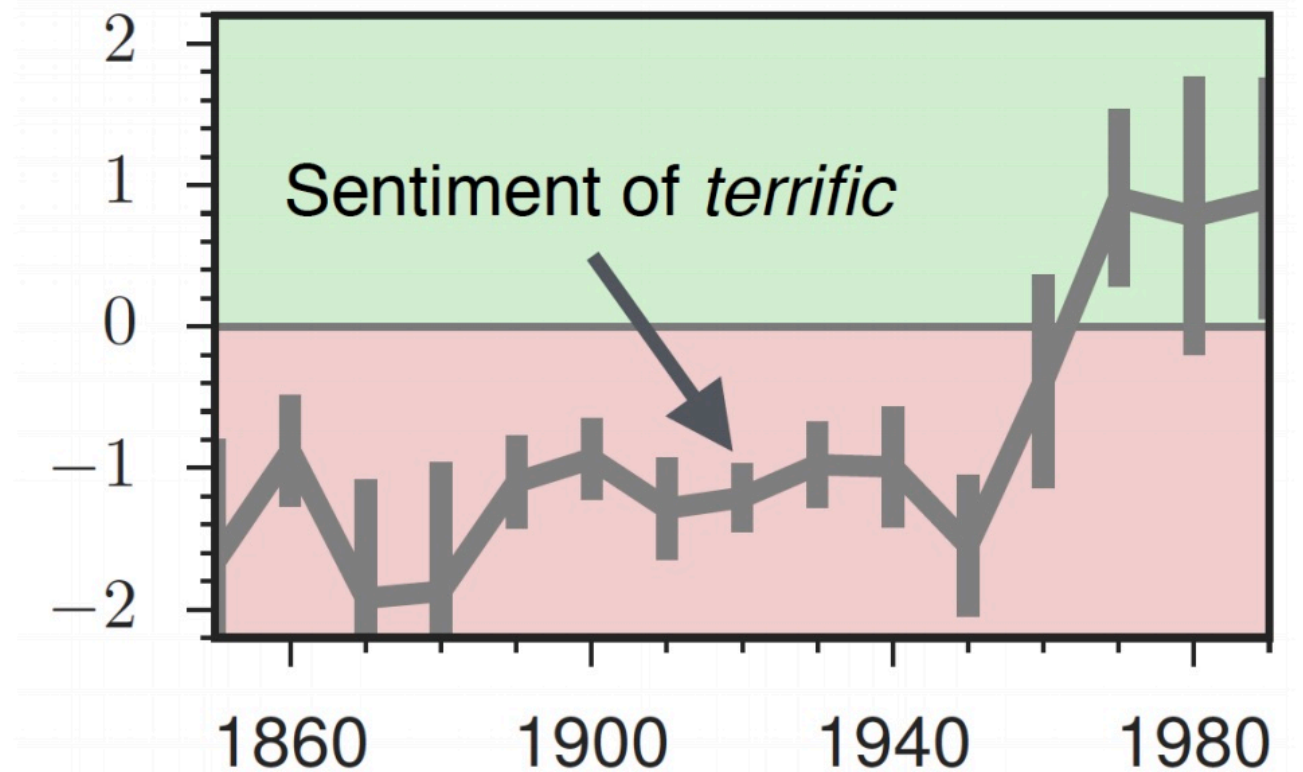
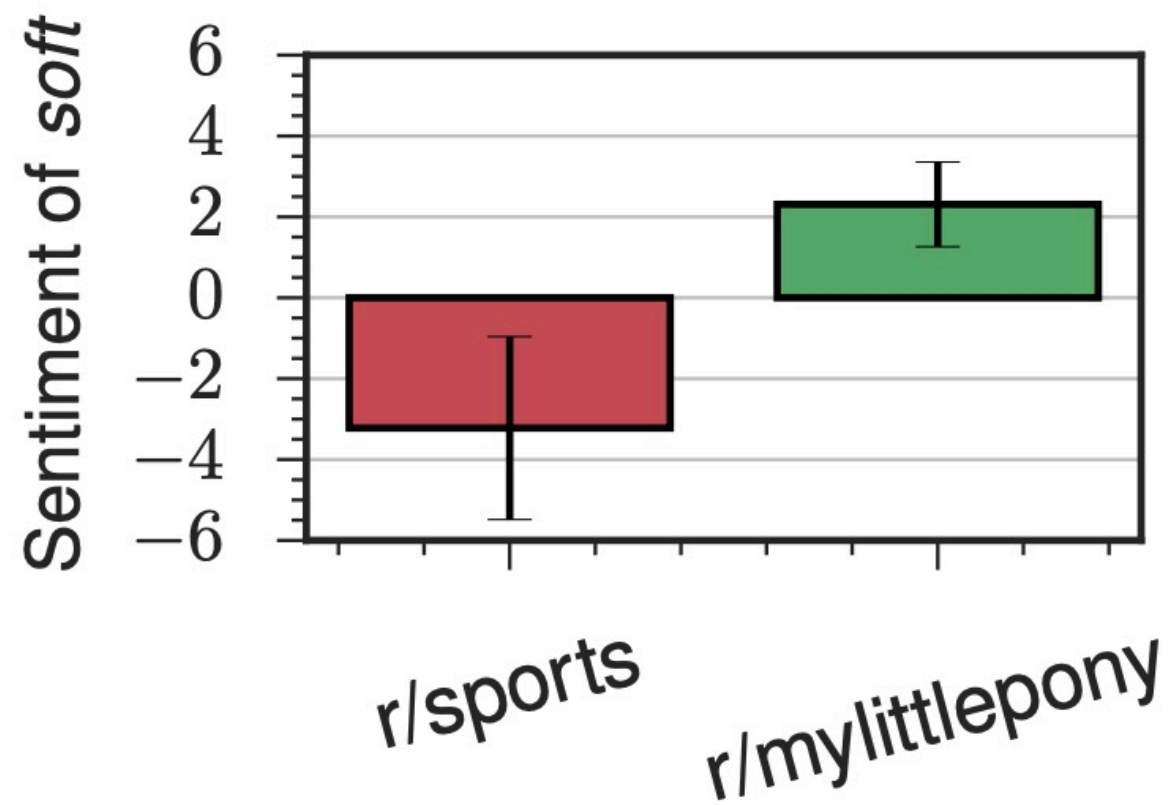
Context Specificity



“Inducing Domain-Specific Sentiment Lexicons from Unlabeled Corpora”
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Sentiment

Context Specificity



"Inducing Domain-Specific Sentiment Lexicons from Unlabeled Corpora"
(Hamilton et al., 2016)

Sentiment

Context Specificity

Random Forest (supervised ngram model)

Loughran Macdonald (finance-specific)

Harvard PsychoSocial dictionary (domain-general)

“Disclosure Sentiment: Machine Learning vs. Dictionary Methods”
(Frankel, Jennings & Lee, 2022)

Sentiment

Context Specificity

Random Forest (supervised ngram model)

Loughran Macdonald (finance-specific)

Harvard PsychoSocial dictionary (domain-general)

Figure 1. 10-K Filing Returns and Sentiment

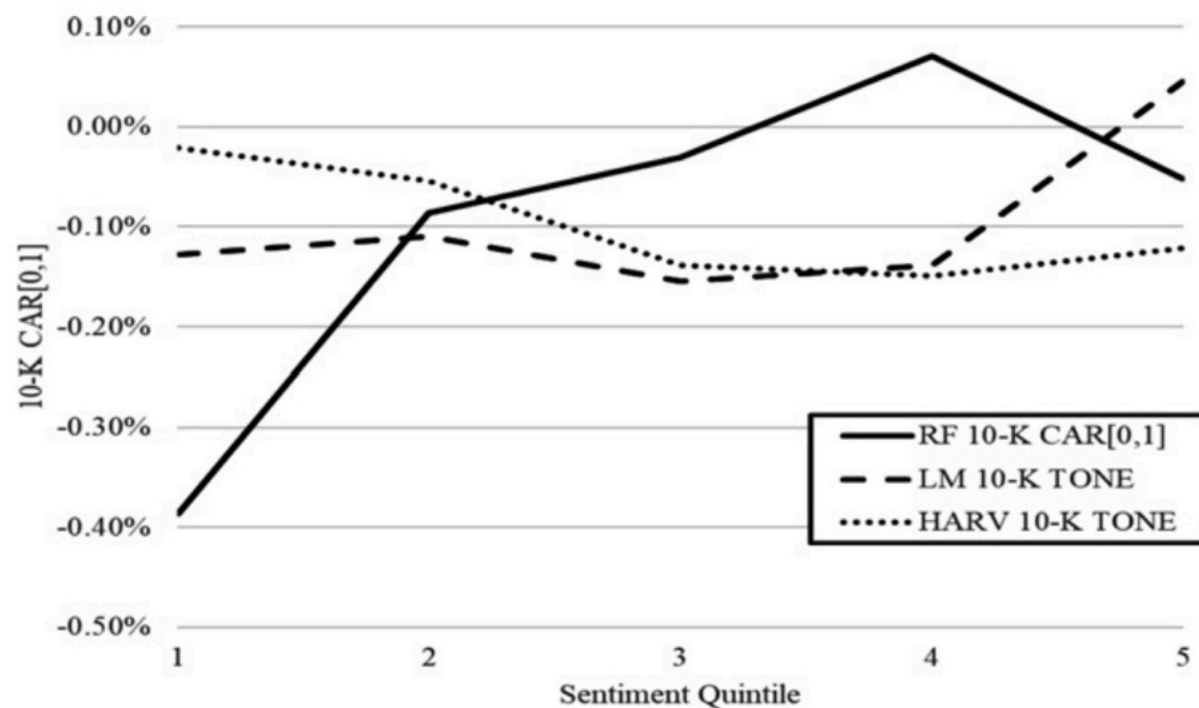
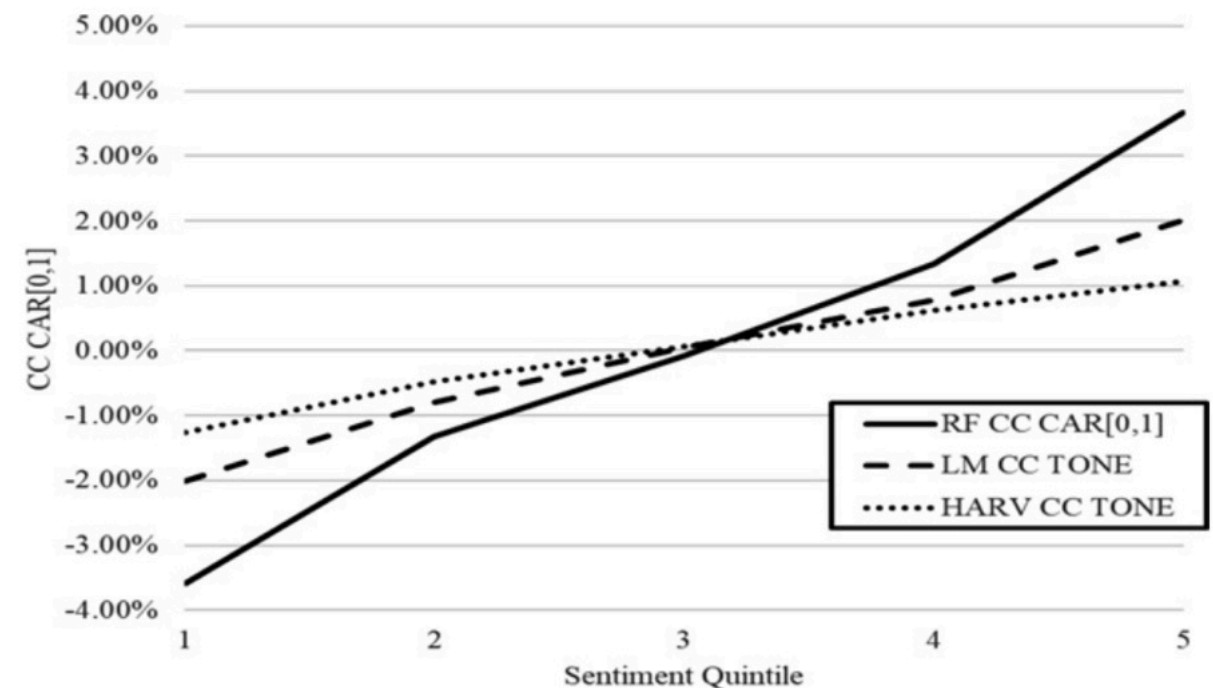


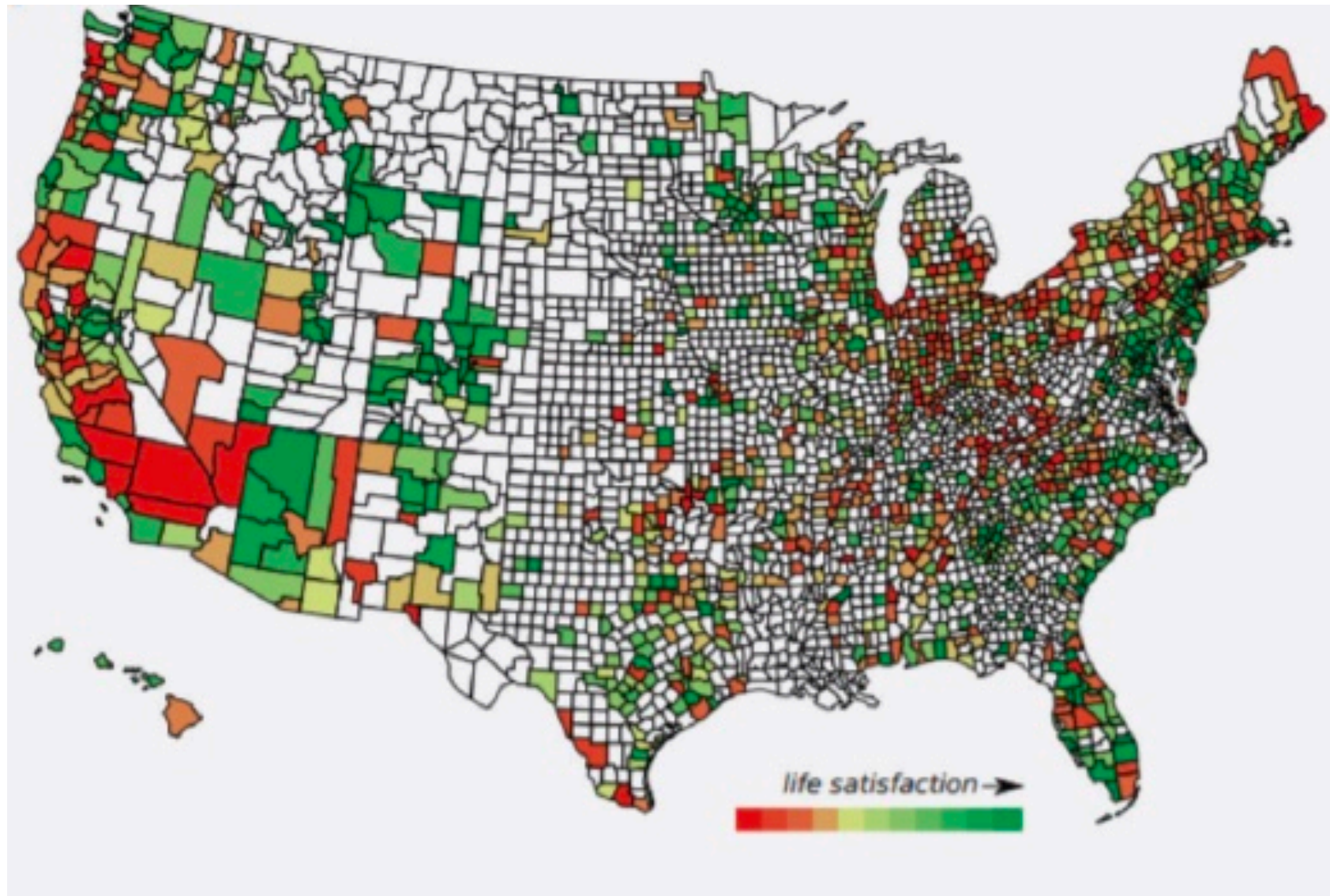
Figure 2. Conference Call Returns and Sentiment



“Disclosure Sentiment: Machine Learning vs. Dictionary Methods”
(Frankel, Jennings & Lee, 2022)

Sentiment

Topic vs Style

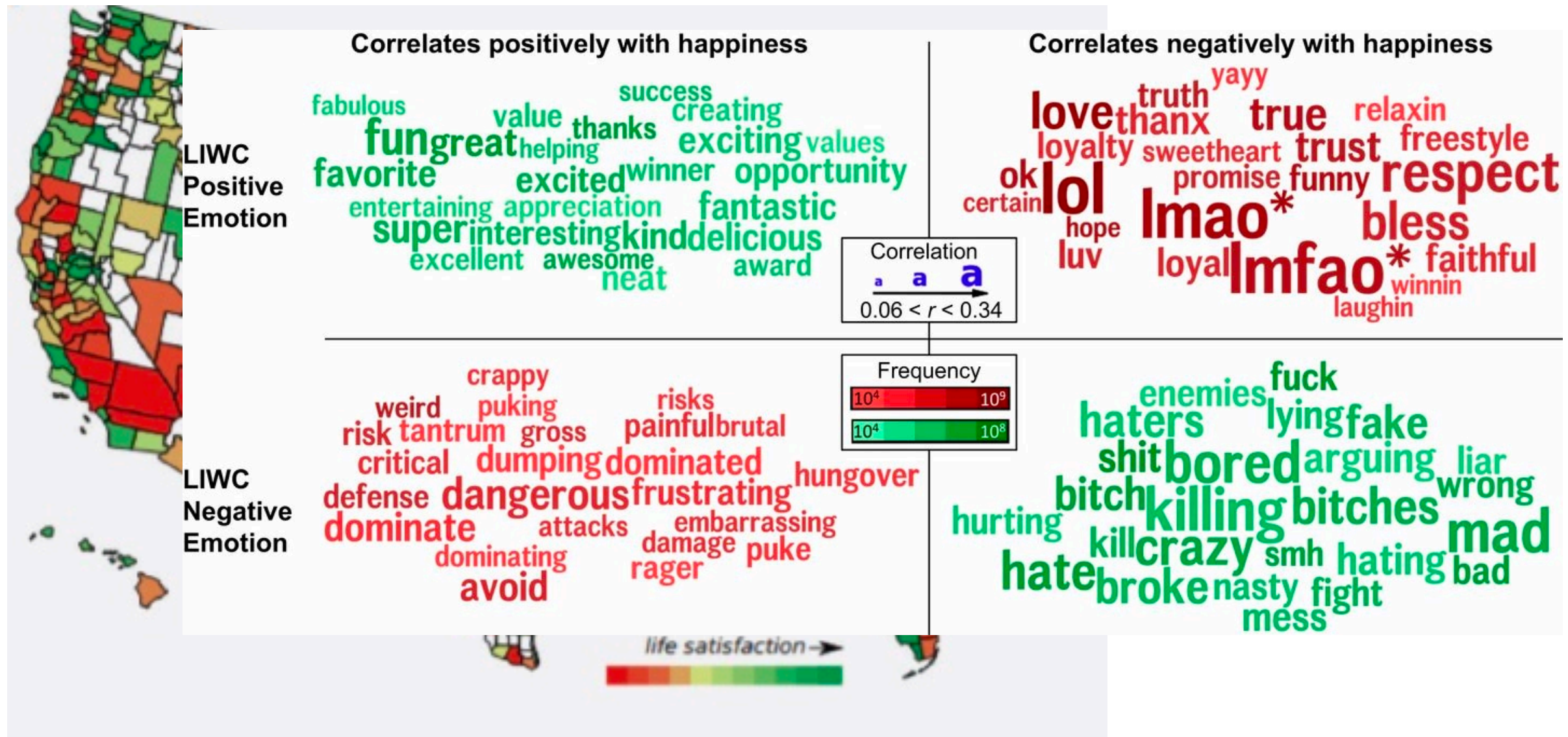


“Estimating geographic subjective well-being from Twitter”

(Jaidka et al., 2020)

Sentiment

Topic vs Style



“Estimating geographic subjective well-being from Twitter”

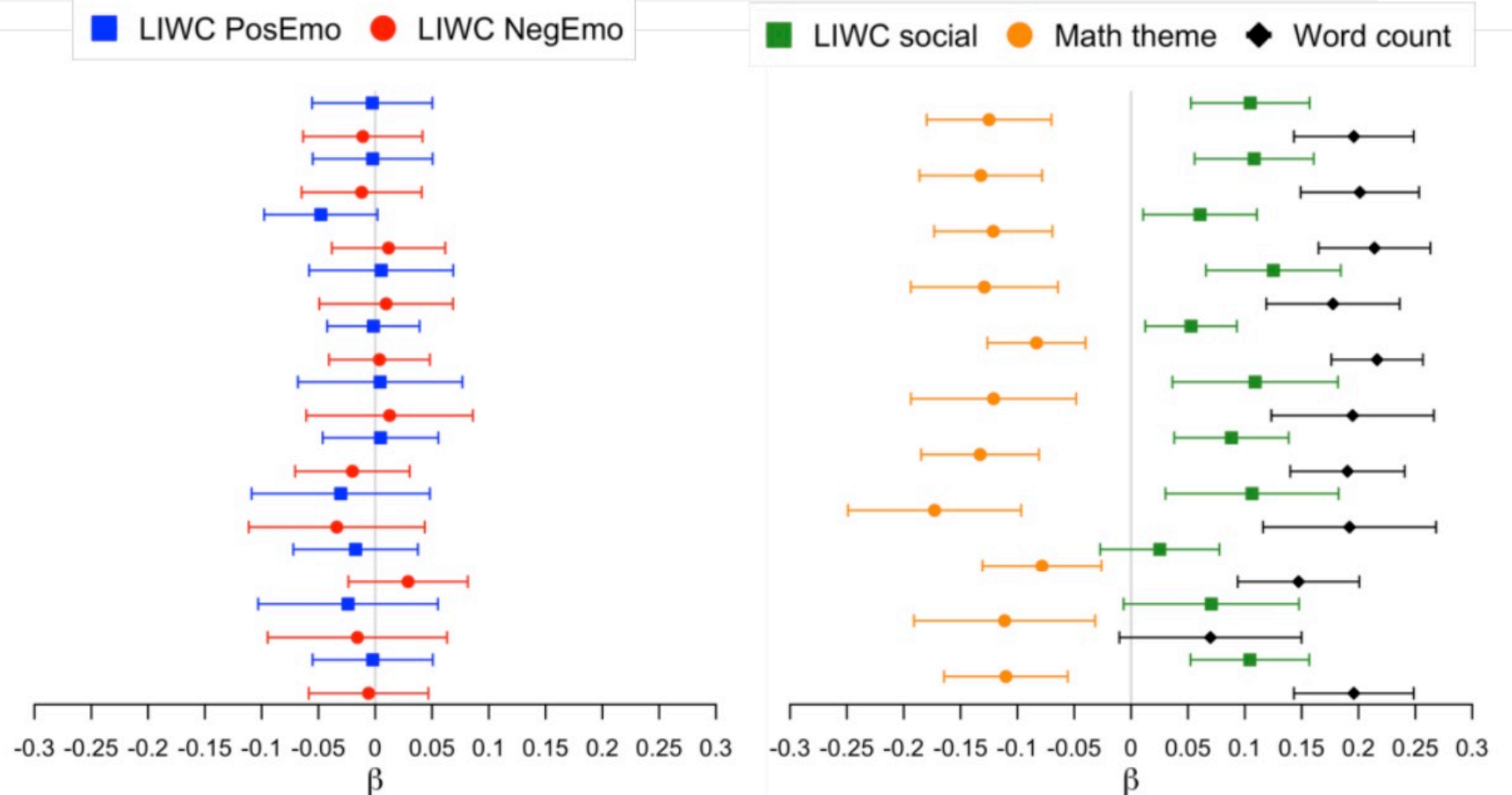
(Jaidka et al., 2020)

Sentiment

Topic vs Style

Specification	N_{people}	N_{obs}
Final analysis	185	1,579
No time covariate	185	1,579
Min. 3 words per file	194	1,734
Min. 10 words per file	159	1,255
Min. 1 file per 3-hour block	230	2,586
Min. 5 files per 3-hour block	116	844
Min. 3 observations	231	1,740
Min. 10 observations	58	723
Target hour (Min. 1 file)	187	1,601
Target hour (Min. 2 files)	106	757
Weighted by word count	185	1,579

Within-Person Associations with ESM Positive Emotion



“Emotion Fluctuations and Everyday Speech”
(Sun et al., 2019)

Sentiment

Best case scenario: descriptions of things

Sentiment

Best case scenario: descriptions of things



Travis M.

Sherman Heights, San
Diego, CA

👤👤 0 friends

★ 1 review

★☆☆☆☆ 12/31/2018

Got seated immediately wich is rare here. But after we sat down it took 20 minutes to be waited on. Anither 20 for our drink order. And nearly an hour after that we were still waiting on our appetizer. We cancelled our whole order and are now at a jack in the box just to get some decent service

Sentiment

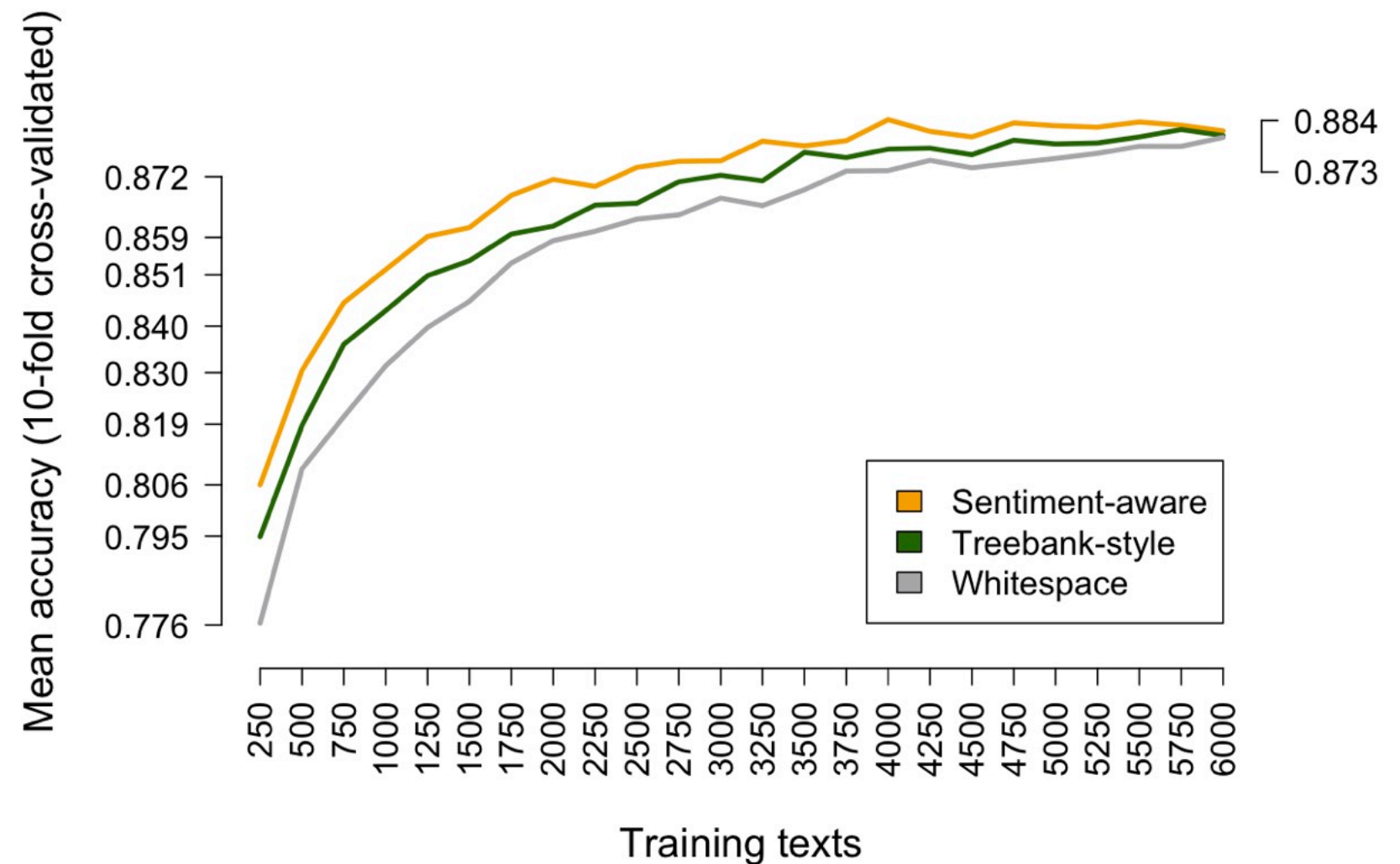
Other improvements

“Sentiment Analysis Symposium” Potts, 2011

Sentiment

Other improvements

Separate tokens for all caps [WOW], punctuation [!], emoticons [:]



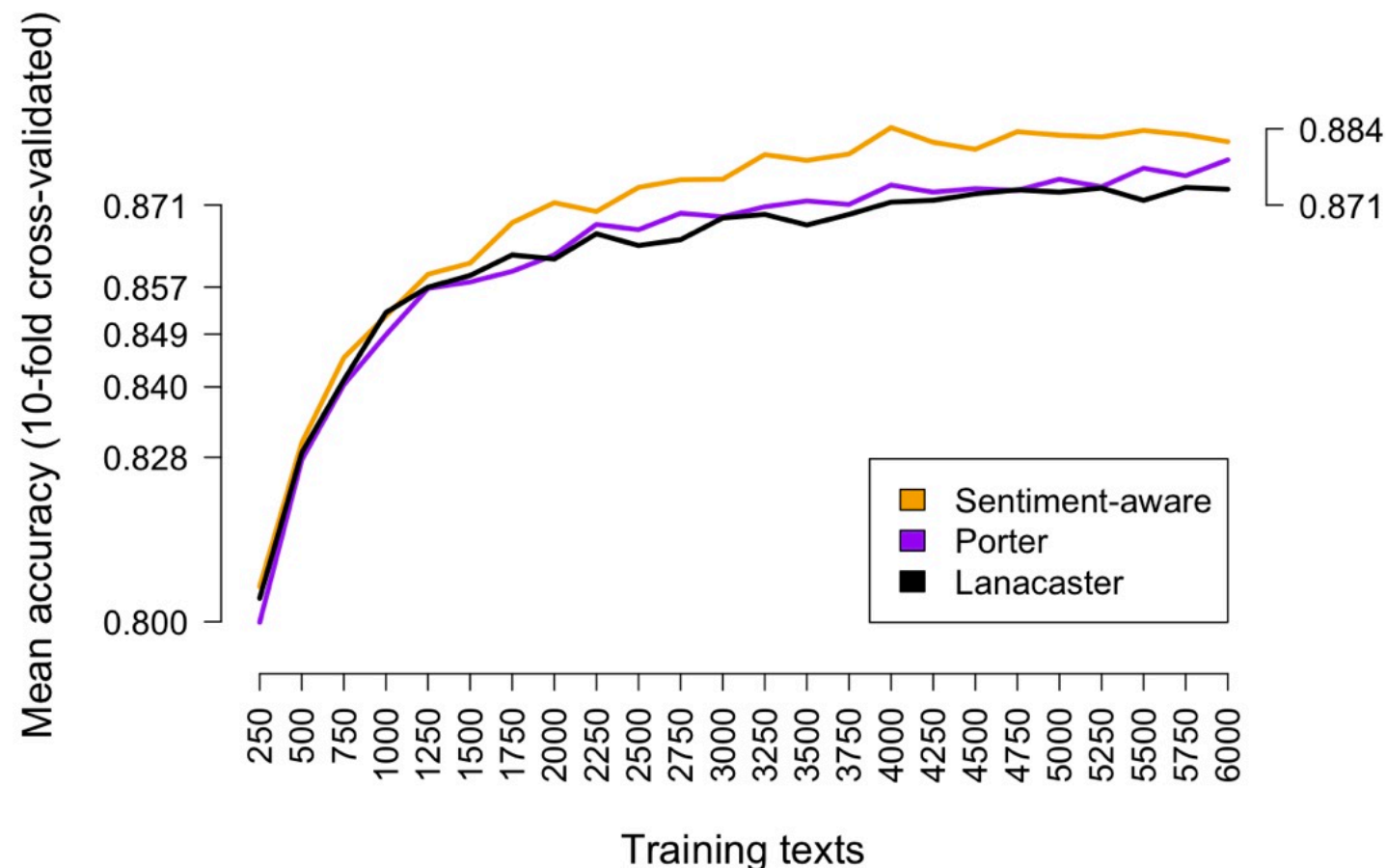
“Sentiment Analysis Symposium” Potts, 2011

Sentiment

Other improvements

Separate tokens for all caps [WOW], punctuation [!], emoticons [:]

Stemming only tends to help at low sample sizes



“Sentiment Analysis Symposium” Potts, 2011

Sentiment

Other improvements

Negation scoping

“It was that bad.” becomes [“it”, “was”, “that”, “bad”, “.”]

“It was not that bad, honestly.” Becomes....

[“it”, “was”, “not”, “that_NEG”, “bad_NEG”, “,”, “honestly”, “.”]

Sentiment

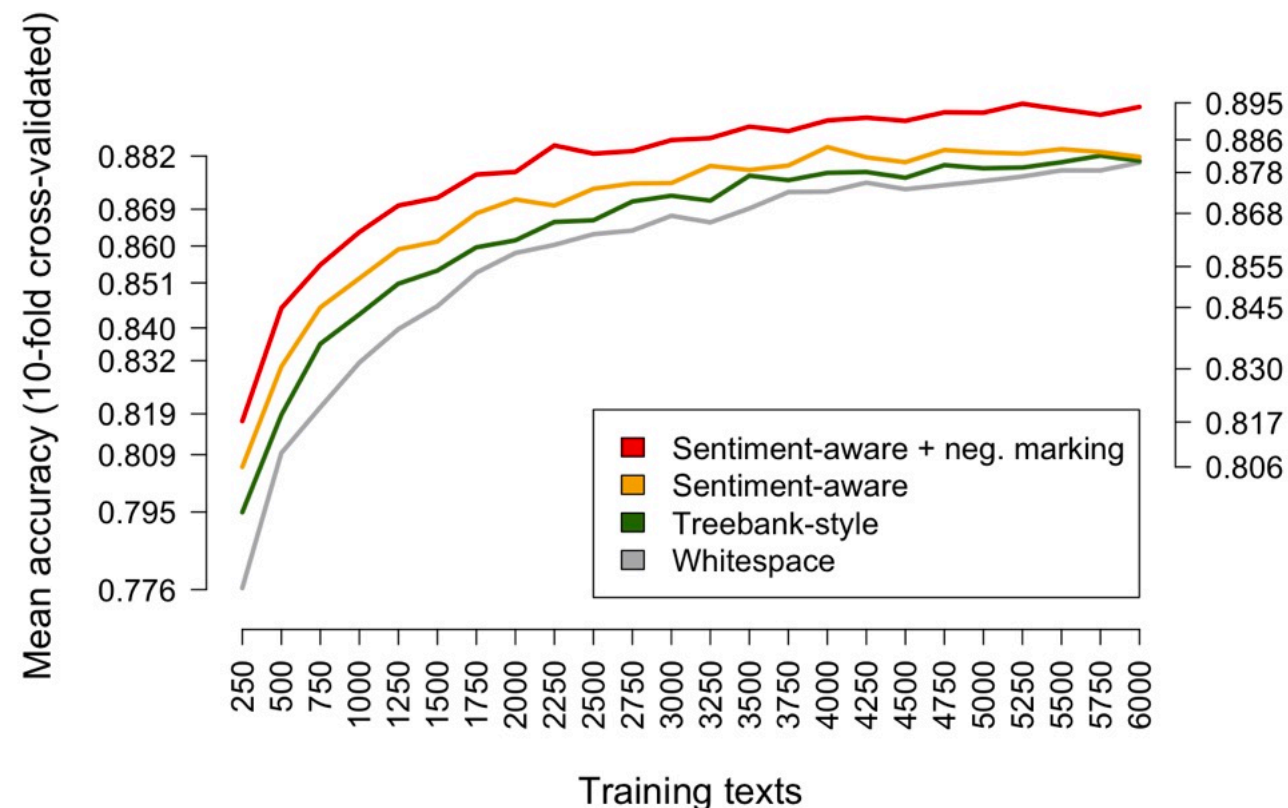
Other improvements

Negation scoping

“It was that bad.” becomes [“it”, “was”, “that”, “bad”, “.”]

“It was not that bad, honestly.” Becomes....

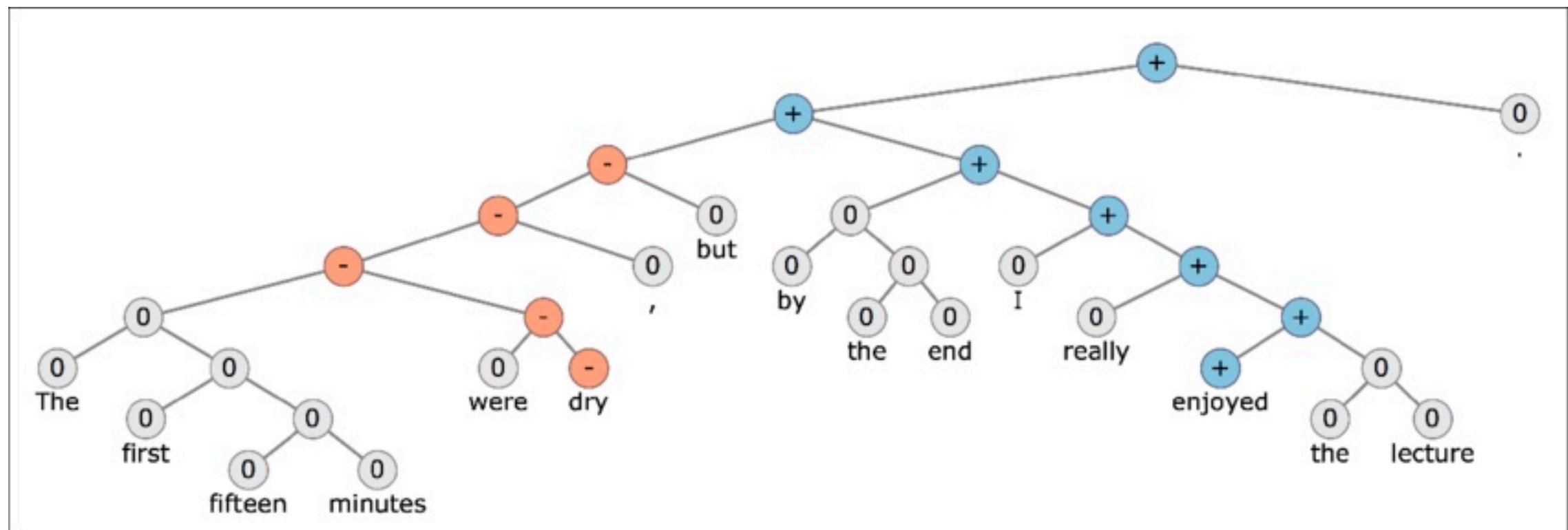
[“it”, “was”, “not”, “that_NEG”, “bad_NEG”, “,”, “honestly”, “.”]



“Sentiment Analysis Symposium” Potts, 2011

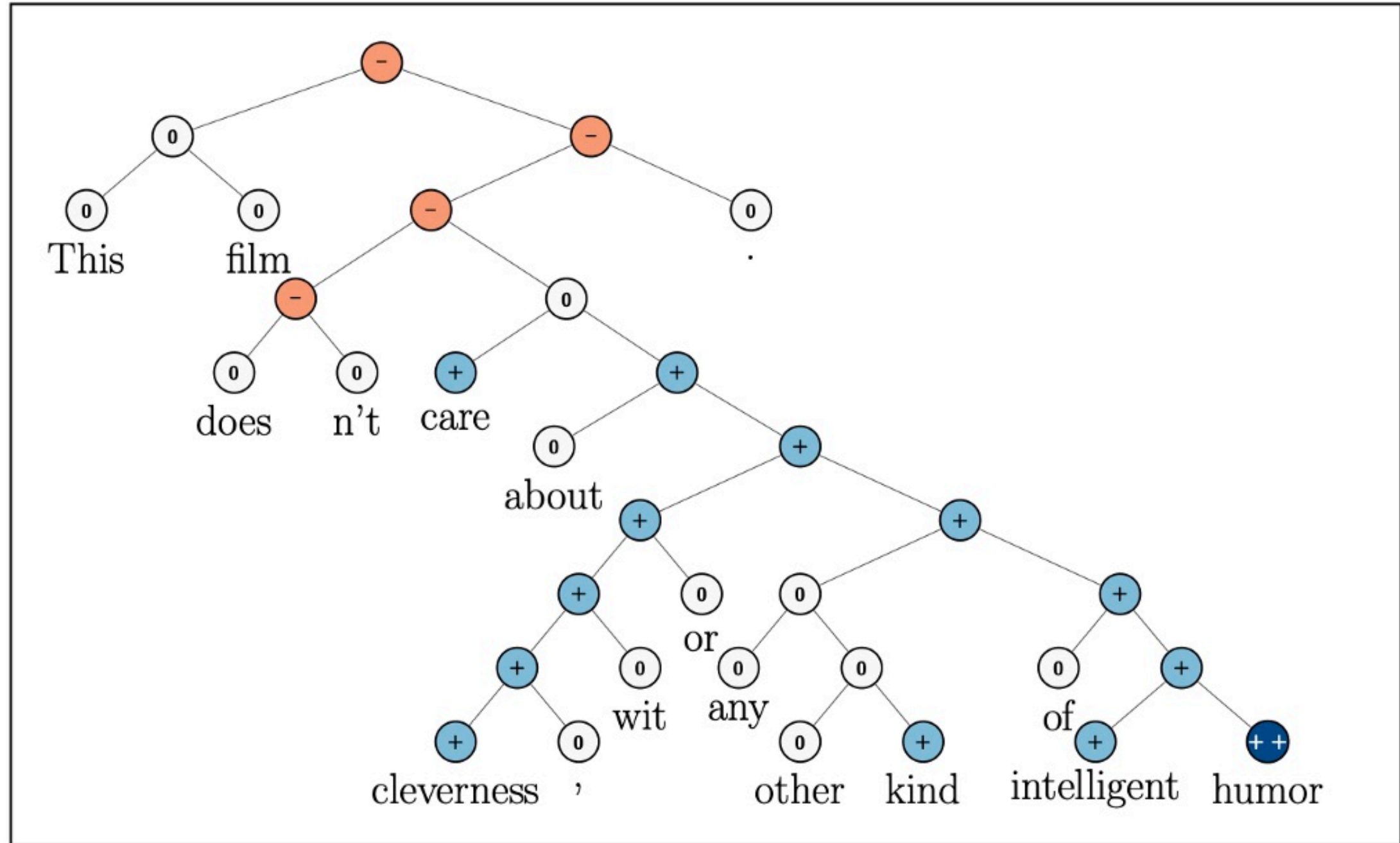
Sentiment

State of the art - Stanford Sentiment treebank



Sentiment

State of the art - Stanford Sentiment treebank



Socher et al., 2013

Sentiment

Good enough for most: VADER, sentimentr...

Word
This
is
not
good
news
,
it
is
very
bad
.

Sentiment

Good enough for most: VADER, sentimentr...

Word	Sentiment
This	0
is	0
not	0
good	+1
news	0
,	0
it	0
is	0
very	0
bad	-1
.	0

Sentiment

Good enough for most: VADER, sentimentr...

Word	Sentiment	Modifier
This	0	
is	0	
not	0	
good	+1	
news	0	
,	0	
it	0	
is	0	
very	0	
bad	-1	
.	0	

Sentiment

Good enough for most: VADER, sentimentr...

Word	Sentiment	Modifier
This	0	
is	0	
not	0	Negation
good	+1	
news	0	
,	0	
it	0	
is	0	
very	0	Intensifier
bad	-1	
.	0	

Sentiment

Good enough for most: VADER, sentimentr...

Word	Sentiment	Modifier
This	0	
is	0	
not	0	Negation
good	+1	*(-1)
news	0	*(-1)
,	0	
it	0	
is	0	
very	0	Intensifier
bad	-1	*(2)
.	0	

Sentiment

Good enough for most: VADER, sentimentr...

Word	Sentiment	Modifier	Score
This	0		0
is	0		0
not	0	Negation	0
good	+1	*(-1)	-1
news	0	*(-1)	0
,	0		0
it	0		0
is	0		0
very	0	Intensifier	0
bad	-1	*(2)	-2
.	0		0

Sentiment

Good enough for most: VADER, sentimentr...

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This	0		0
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news	0	*(-1)	0
,	0		0
it	0		0
is	0		0
very	0	Intensifier	0
bad	-1	*(2)	-2
.	0		0

-4

Your optional assignment

Create new project from version control

New Project Wizard

[Back](#) **Clone Git Repository**



Repository URL:

Project directory name:

Create project as subdirectory of:
 [Browse...](#)

☐ Open in new session

[Create Project](#) [Cancel](#)

Your optional assignment

Open TableOfContents.R

```
# these packages are required, install if you don't have them
library(quanteda)
library(textclean)
library(tidyverse)
library(ggrepel)
library(glmnet)

# a function we use to build a model later
source("BA_dfm.R")

# learn about handling text data in R
source("text_basics.R")

# load data - restaurant reviews
review_dat<-readRDS("review_dat.RDS")

# First thing - check variables
names(review_dat)
dim(review_dat)

# build an NLP model
source("BA_ngrams.R")
```

Your optional assignment

Open TableOfContents.R

Load packages

Load function script

Learn basics

Load data

Train and test model

```
# these packages are required, install if you don't have them
library(quanteda)
library(textclean)
library(tidyverse)
library(ggrepel)
library(glmnet)
```

```
# a function we use to build a model later
source("BA_dfm.R")
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```
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```

```
# load data - restaurant reviews
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```

```
# First thing - check variables
names(review_dat)
dim(review_dat)
```

```
# build an NLP model
source("BA_ngrams.R")
```

Your optional assignment

Your goals

The script build two models:

- Predicting price of restaurant
- Predicting star rating of review

Your assignment:

- build a model to predict the gender of the writer (using the “male” variable)
- evaluate accuracy on a separate test set, and build a plot showing the selected features

Your optional assignment

Some tips

Make sure every line of code works, in order, before making your own edits

If you have a problem, clear the environment and start over using the original raw files I sent you

Explore the data! Inspect distributions
(use `hist()` and `table()` to look at one variable at a time)

Work as a team!!!!

Your optional assignment

Due date

None

But if you send me a write-up,
I am happy to give feedback!

m.yeomans@imperial.ac.uk

Application in Your Code

Meta-data



Application in Your Code

Meta-data

Review characteristics:

Application in Your Code

Meta-data

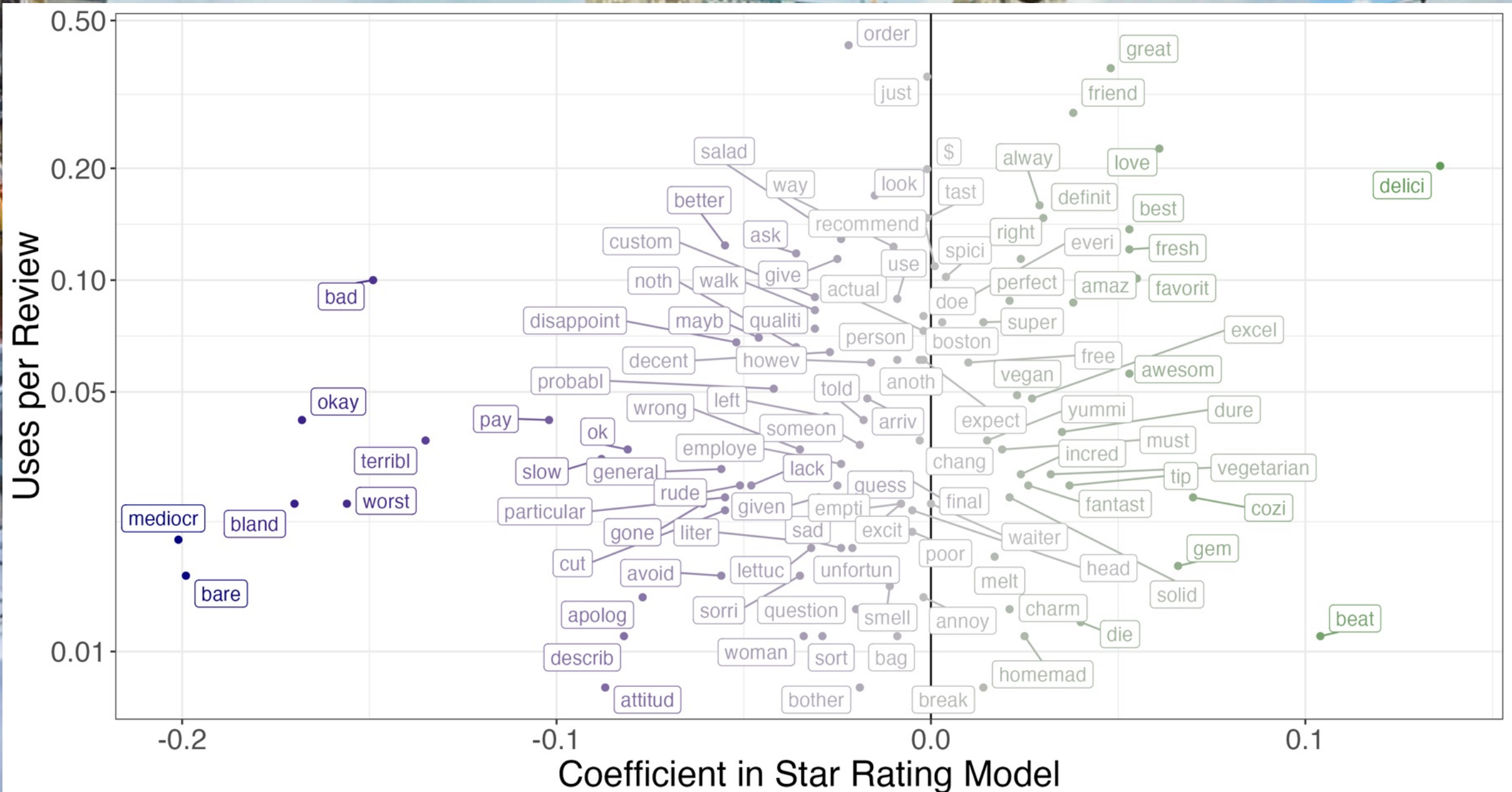
Review characteristics:

Star rating, time, likes from others, replies

Application in Your Code

Meta-data

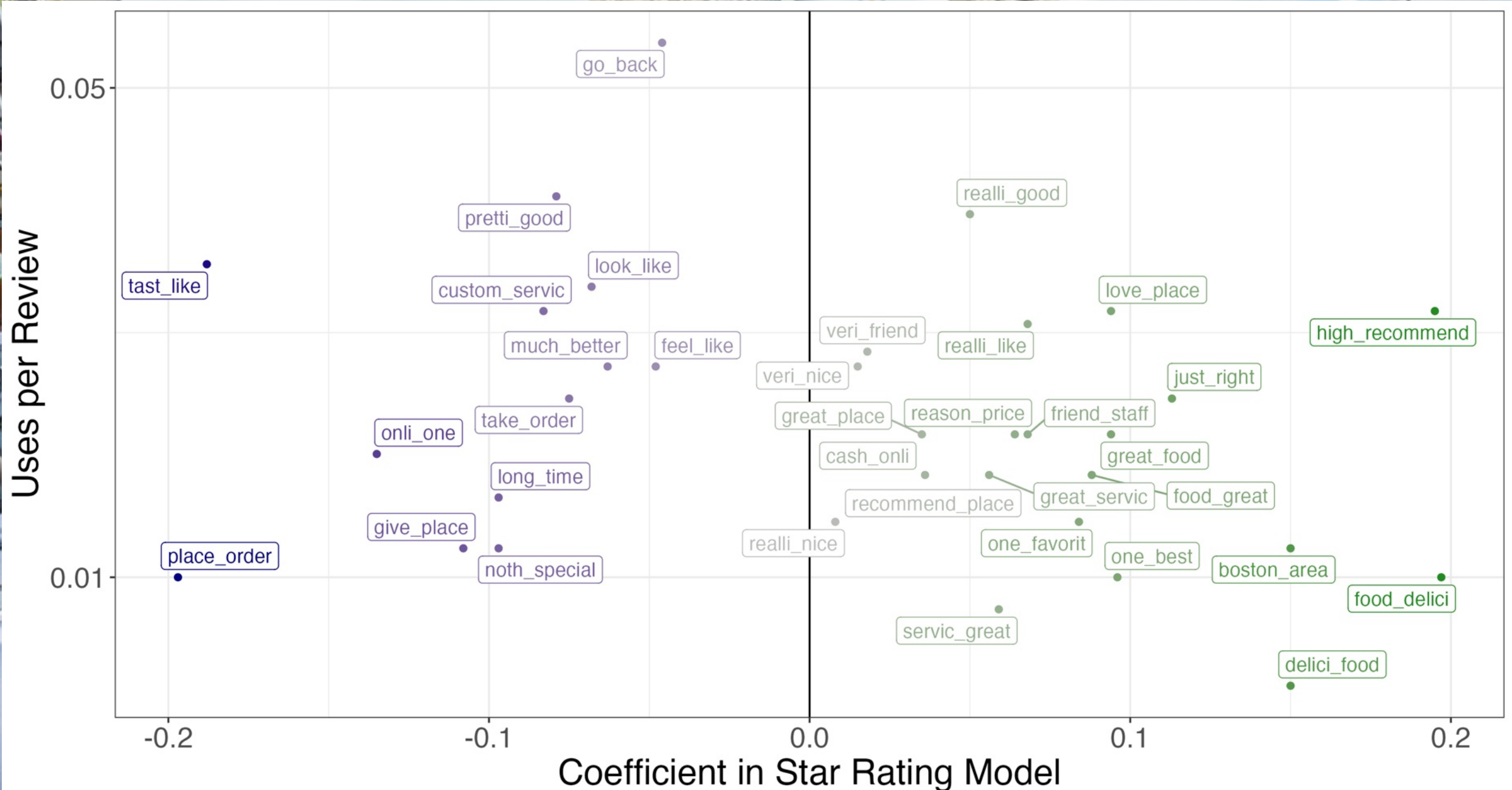
$$\hat{y} = a_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + e$$



Application in Your Code

Meta-data

$$\hat{y} = a_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + e$$



Application in Your Code

Meta-data

Review characteristics:

Star rating, time, likes from others, replies

Reviewer characteristics:

Application in Your Code

Meta-data

Review characteristics:

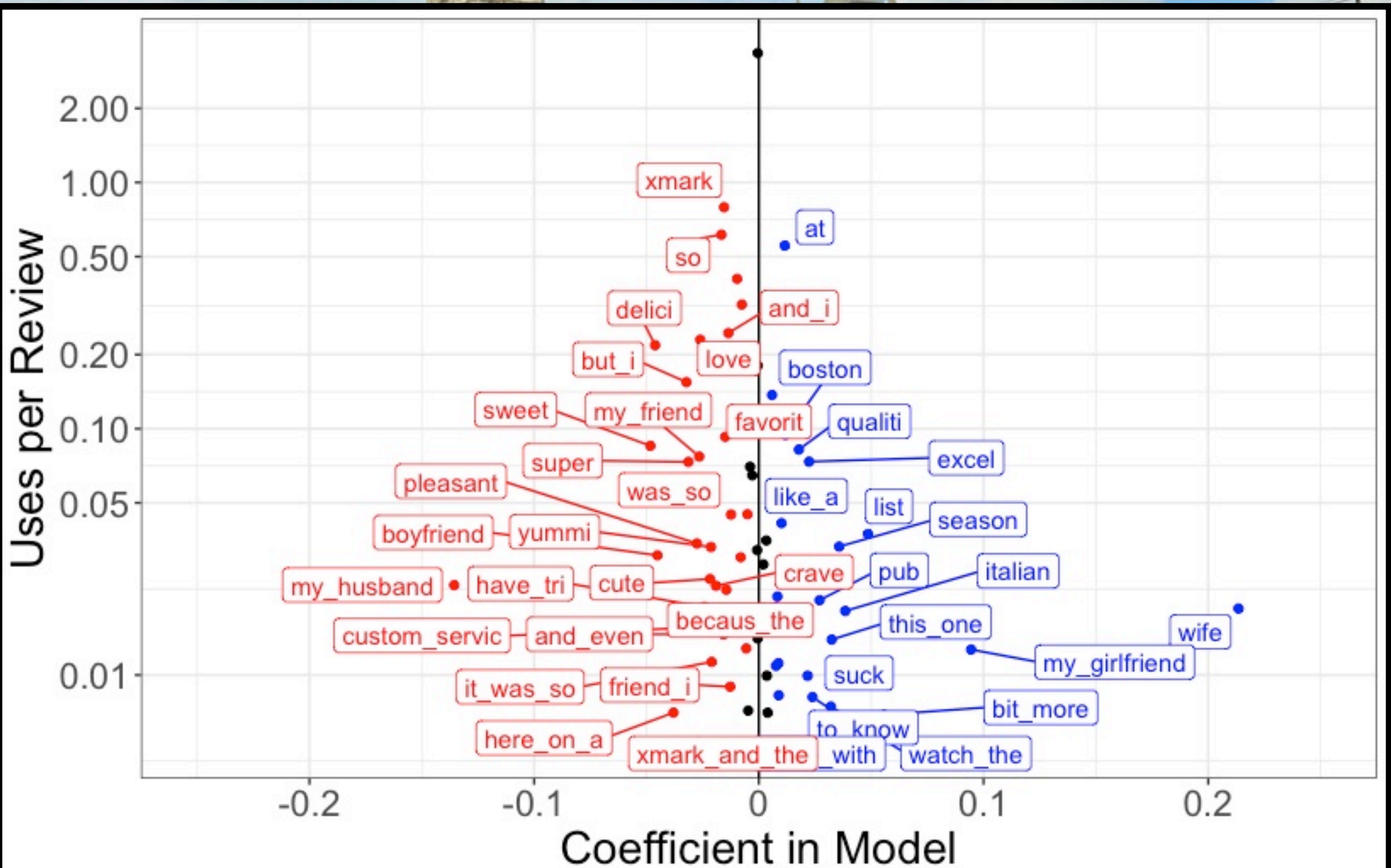
Star rating, time, likes from others, replies

Reviewer characteristics:

Demographics - gender, ethnicity, age, SES, education

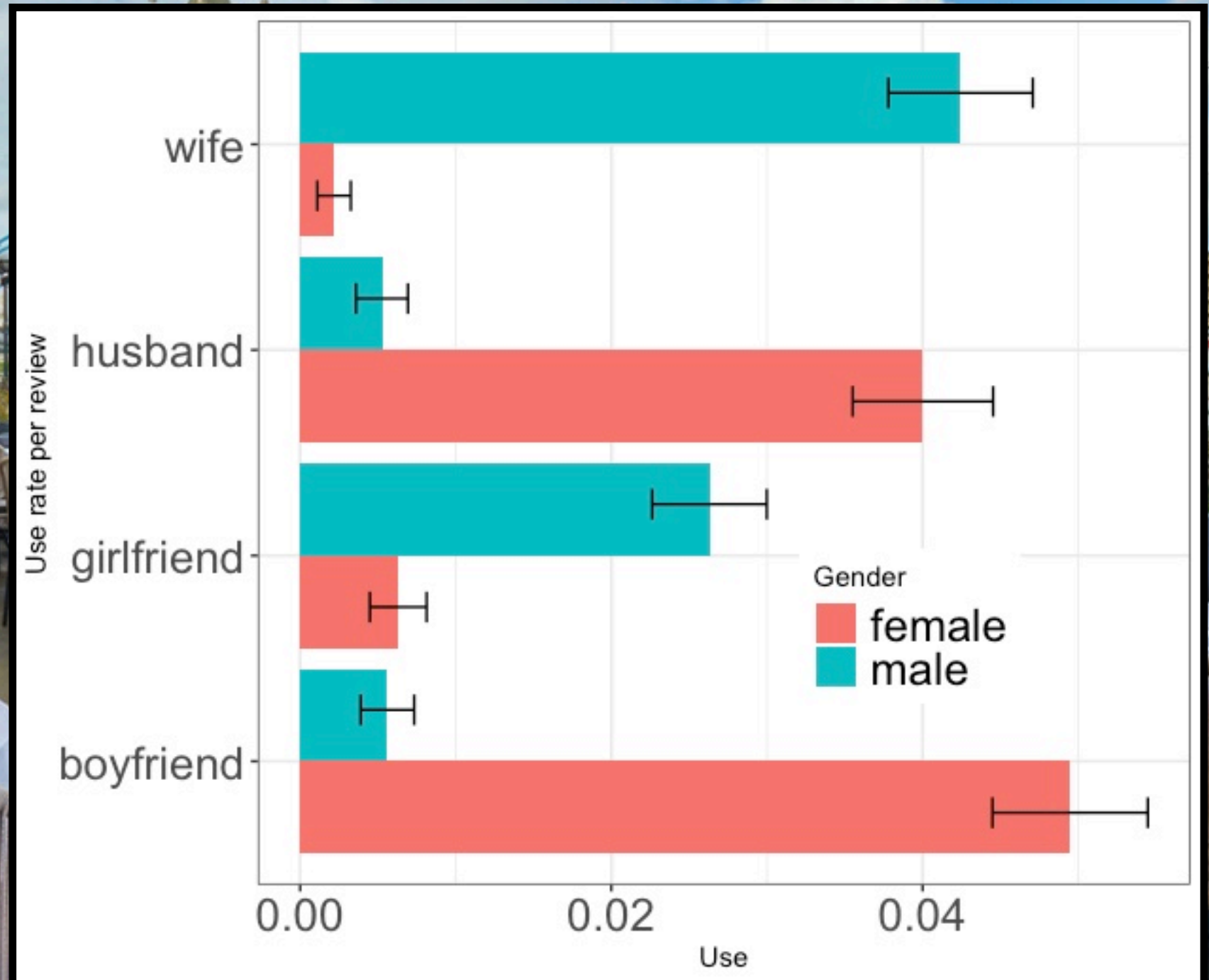
Application in Your Code

Meta-data



Application in Your Code

Meta-data



Application in Your Code

Meta-data

Review characteristics:

Star rating, time, likes from others, replies

Reviewer characteristics:

Demographics - gender, ethnicity, age, SES, education

Yelp data - Experience, other ratings

Application in Your Code

Meta-data

Review characteristics:

Star rating, time, likes from others, replies

Reviewer characteristics:

Demographics - gender, ethnicity, age, SES, education

Yelp data - Experience, other ratings

Restaurant characteristics:

Yelp metadata - price range, food type, good for kids, etc.

Application in Your Code

Meta-data

Review characteristics:

Star rating, time, likes from others, replies

Reviewer characteristics:

Demographics - gender, ethnicity, age, SES, education

Yelp data - Experience, other ratings

Restaurant characteristics:

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Location - city, high/low SES, partisanship, etc.

Application in Your Code

Meta-data

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Star rating, time, likes from others, replies

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Business metadata - size, rent, revenues, rent, ownership

Application in Your Code

Meta-data

Review characteristics:

Star rating, time, likes from others, replies

Reviewer characteristics:

Demographics - gender, ethnicity, age, SES, education

Yelp data - Experience, other ratings

Restaurant characteristics:

Yelp metadata - price range, food type, good for kids, etc.

Location - city, high/low SES, partisanship, etc.

Business metadata - size, rent, revenues, rent, ownership

Government metadata - taxes, health inspections

Health Inspections

(Kang et al., 2013)

Meta-data



Health Inspections

(Kang et al., 2013)

Meta-data

Hygienic:
Cooking Method & Garnish: brew, frosting, grill, crush, crust, taco, burrito, toast
Healthy or Fancier Ingredients: celery, calamity, wine, broccoli, salad, flatbread, olive, pesto
Cuisines : Breakfast, Fish & Chips, Fast Food, German, Diner, Belgian, European, Sandwiches, Vegetarian
Whom & When: date, weekend, our, husband, evening, night
Sentiment: lovely, yummy, generous, friendly, great, nice
Service & Atmosphere: selection, attitude, atmosphere, ambiance, pretentious

Table 3: Lexical Cues & Examples - Hygienic (clean)



Health Inspections

(Kang et al., 2013)

Meta-data

Hygienic:

Cooking Method & Garnish: brew, frosting, grill, crush, crust, taco, burrito, toast

Healthy or Fancier Ingredients: celery, calamity, wine, broccoli, salad, flatbread, olive, pesto

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Table 3: Lexical Cues & Examples - Hygienic (clean)

Hygienic gross, mess, sticky, smell, restroom, dirty

Basic Ingredients: beef, pork, noodle, egg, soy, ramen, pho,

Cuisines Vietnamese, Dim Sum, Thai, Mexican, Japanese, Chinese, American, Pizza, Sushi, Indian, Italian, Asian

Sentiment: cheap, never,

Service & Atmosphere cash, worth, district, delivery, think, really, thing, parking, always, usually, definitely
- door: "The wait is always out the *door* when I actually want to go there",

- sticker: "I had *sticker* shock when I saw the prices.",

- student: "heap, large portions and tasty = the perfect *student* food!",

- the size: "i was pretty astonished at *the size* of all the plates for the money.",

- was dry: "The beef *was dry*, the sweet soy and anise-like sauce was TOO salty (almost inedible).",

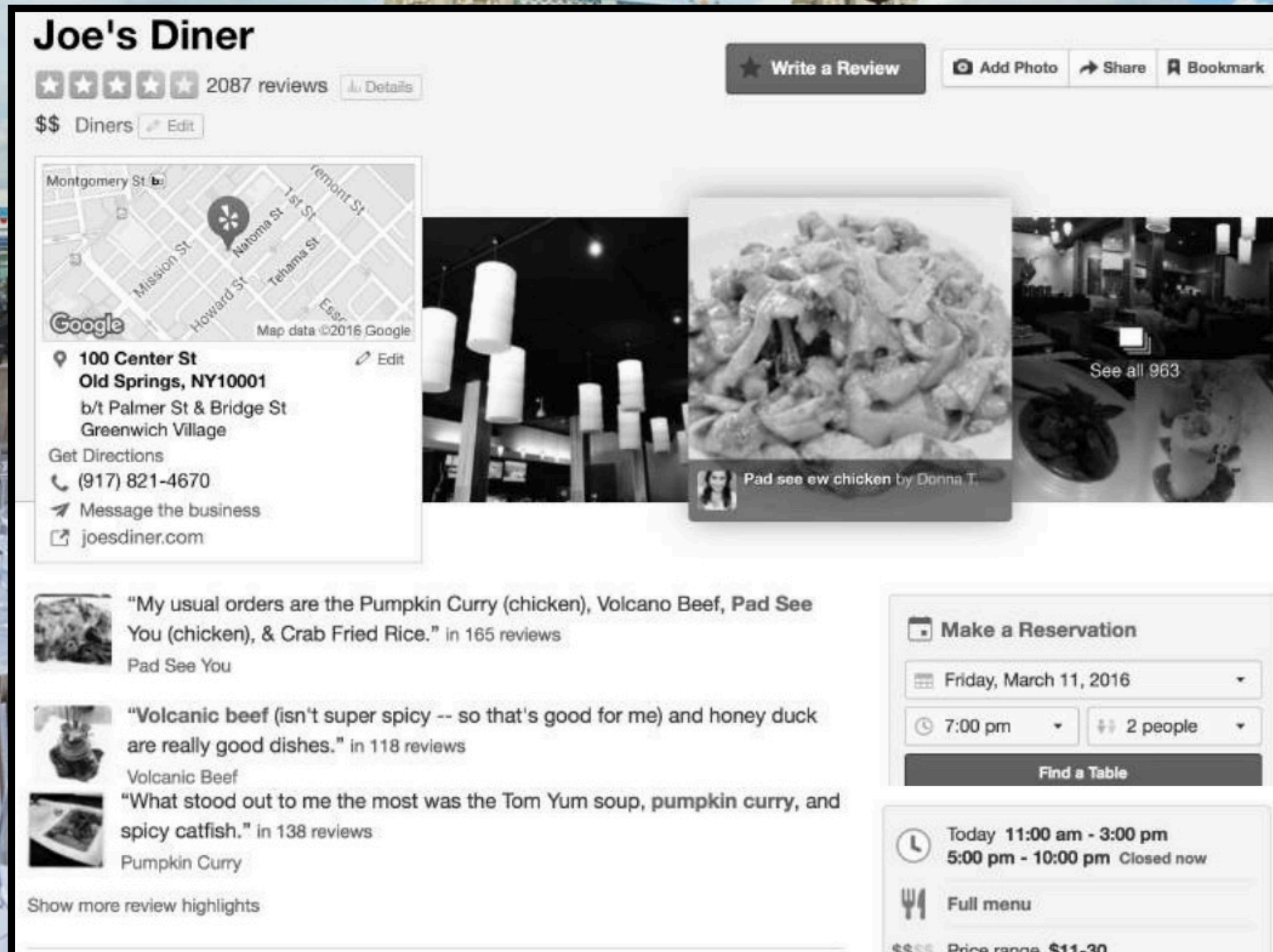
- pool: "There are *pool* tables, TV airing soccer games from around the globe and of course - great drinks!"

Table 2: Lexical Cues & Examples - Unhygienic (dirty)

Health Inspections

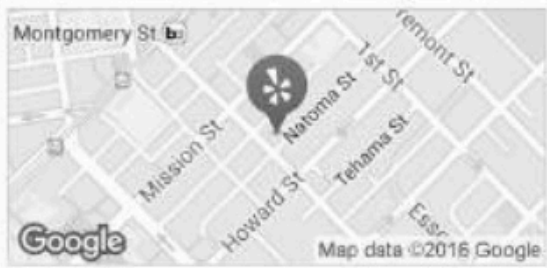
(Kang et al., 2013)

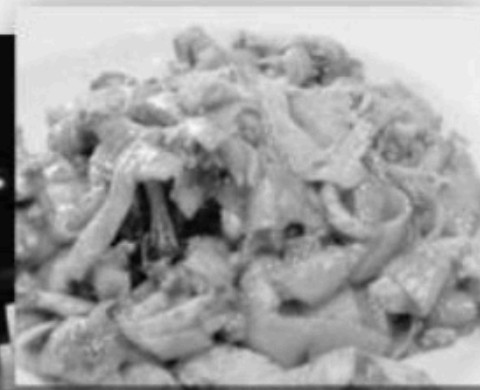
Meta-data



Joe's Diner
★ ★ ★ ★ ★ 2087 reviews [Details](#)
\$\$ Diners [Edit](#)

[Write a Review](#) [Add Photo](#) [Share](#) [Bookmark](#)


100 Center St
Old Springs, NY 10001
b/t Palmer St & Bridge St
Greenwich Village
Get Directions
(917) 821-4670
[Message the business](#)
[joesdiner.com](#)


Pad see ew chicken by Donna T.


"My usual orders are the Pumpkin Curry (chicken), Volcano Beef, Pad See You (chicken), & Crab Fried Rice." in 165 reviews
Pad See You


"Volcanic beef (isn't super spicy -- so that's good for me) and honey duck are really good dishes." in 118 reviews
Volcanic Beef


"What stood out to me the most was the Tom Yum soup, pumpkin curry, and spicy catfish." in 138 reviews
Pumpkin Curry

[Show more review highlights](#)

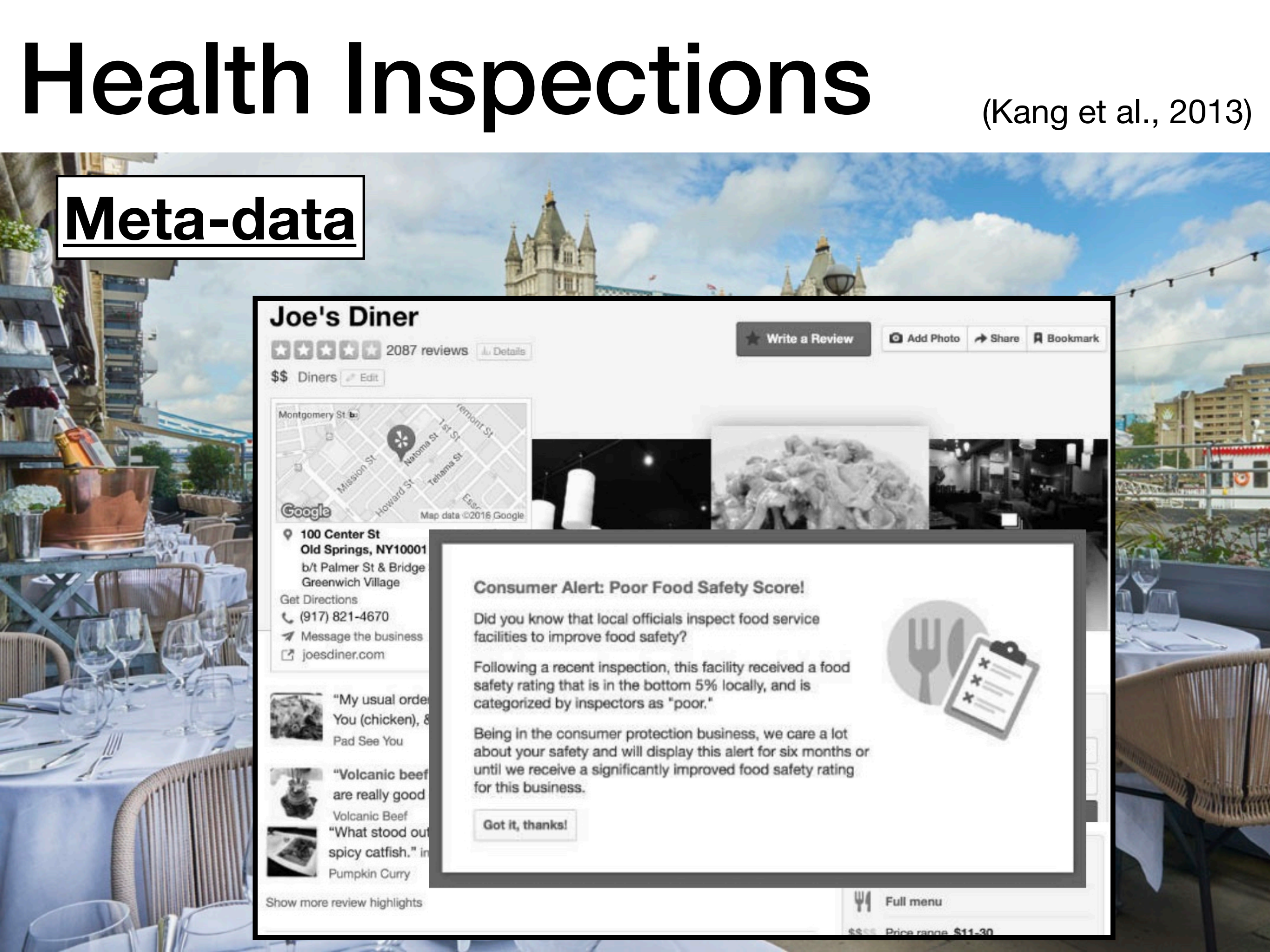
Make a Reservation
Friday, March 11, 2016
7:00 pm 2 people
[Find a Table](#)

Today 11:00 am - 3:00 pm
5:00 pm - 10:00 pm Closed now
[Full menu](#)
Price range: \$11-30

Health Inspections

(Kang et al., 2013)

Meta-data



Joe's Diner

★★★★★ 2087 reviews [Details](#)

\$\$ Diners [Edit](#)

[Write a Review](#) [Add Photo](#) [Share](#) [Bookmark](#)

100 Center St
Old Springs, NY 10001
b/t Palmer St & Bridge
Greenwich Village

Get Directions
(917) 821-4670
Message the business
joesdiner.com

Consumer Alert: Poor Food Safety Score!

Did you know that local officials inspect food service facilities to improve food safety?

Following a recent inspection, this facility received a food safety rating that is in the bottom 5% locally, and is categorized by inspectors as "poor."

Being in the consumer protection business, we care a lot about your safety and will display this alert for six months or until we receive a significantly improved food safety rating for this business.

[Got it, thanks!](#)

Show more review highlights

[Full menu](#)

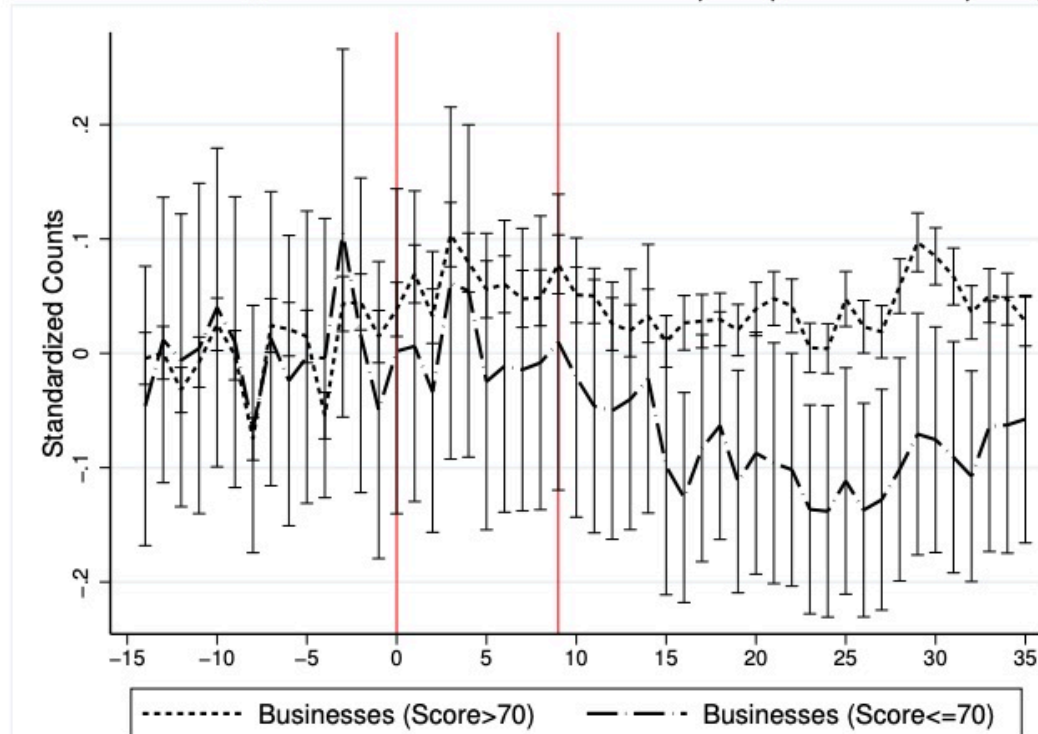
Price range: \$11-30

Health Inspections

(Kang et al., 2013)

Meta-data

Panel A. Weekly consumer leads: 10/11/2012 - 9/25/2013

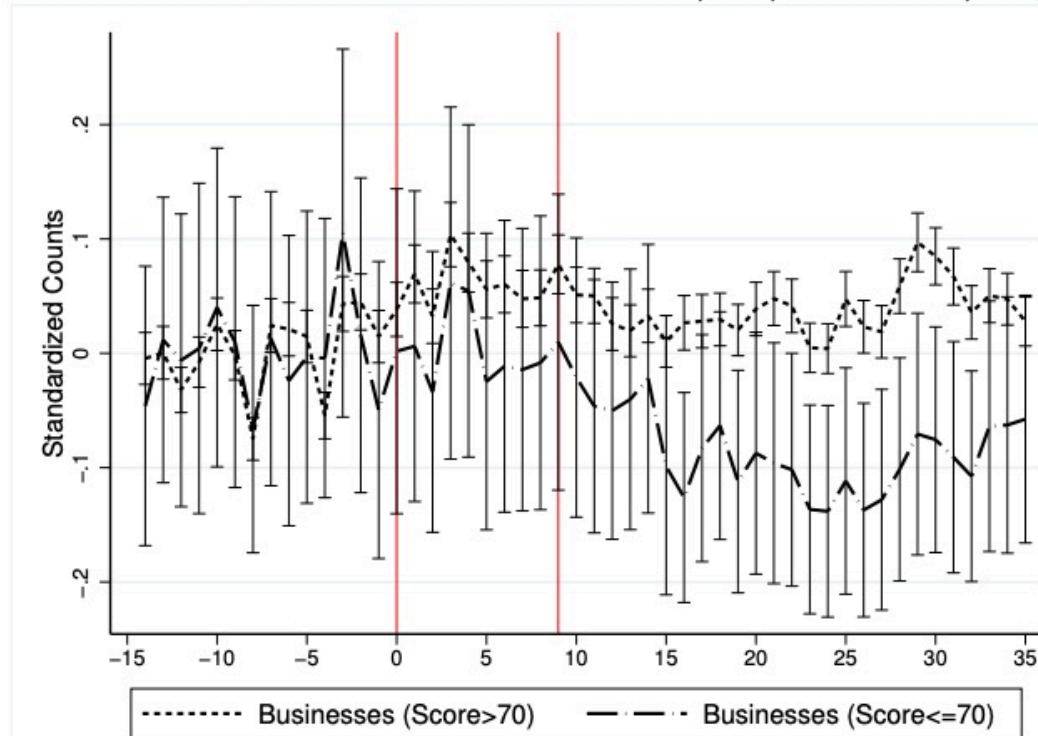


Health Inspections

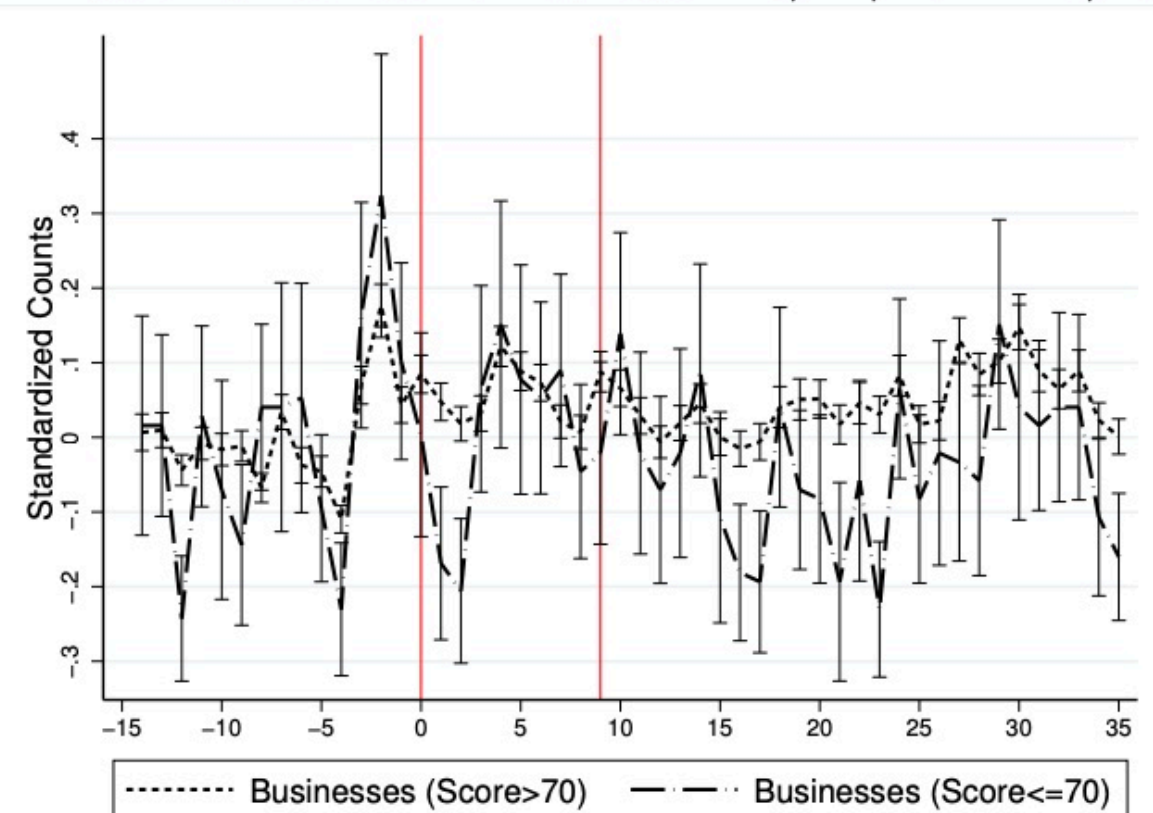
(Kang et al., 2013)

Meta-data

Panel A. Weekly consumer leads: 10/11/2012 - 9/25/2013



Panel B. Weekly number of reviews: 10/11/2012 - 9/25/2013



Dimension Reduction

Sparse Representation

"This is a sentence"

"This is also one sentence"

"This is a document"

"This is not a document"

Dimension Reduction

Sparse Representation

"This is a sentence"

"This is also one sentence"

"This is a document"

"This is not a document"

This	is	a	sentence	also	one	document	not	wolf	star	...
1	1	1	1	0	0	0	0	0	0	0
1	1	0	1	1	1	0	0	0	0	0
1	1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	1	1	0	0	0

Dimension Reduction

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This	is	a	sentence	also	one	document	not	wolf	star	...
1	1	1	1	0	0	0	0	0	0	0
1	1	0	1	1	1	0	0	0	0	0
1	1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	1	1	0	0	0

Dense Representations

Dimension Reduction

Sparse Representation

"This is a sentence"

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This	is	a	sentence	also	one	document	not	wolf	star	...
1	1	1	1	0	0	0	0	0	0	0
1	1	0	1	1	1	0	0	0	0	0
1	1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	1	1	0	0	0

Dense Representations

Price	ft ²	Bedrooms	zone	Parking?
1800	440	3	3	yes
1400	610	1	1	No
1100	575	2	4	No
2200	800	2	2	Yes

Dimension Reduction

Sparse Representation

"This is a sentence"

"This is also one sentence"

"This is a document"

"This is not a document"

This	is	a	sentence	also	one	document	not	wolf	star	...
1	1	1	1	0	0	0	0	0	0	0
1	1	0	1	1	1	0	0	0	0	0
1	1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	1	1	0	0	0

Dense Representations

Lower dimensionality -> easier to estimate

Dimension Reduction

Sparse Representation

"This is a sentence"

"This is also one sentence"

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This	is	a	sentence	also	one	document	not	wolf	star	...
1	1	1	1	0	0	0	0	0	0	0
1	1	0	1	1	1	0	0	0	0	0
1	1	1	0	0	0	1	0	0	0	0
1	1	1	0	0	0	1	1	0	0	0

Dense Representations

Lower dimensionality -> easier to estimate

Captures “synonymy”

- different words have similar meanings

Dimension Reduction

What words are similar to one another?

Similarity is high-dimensional!

Dimension Reduction

What words are similar to one another?

Similarity is high-dimensional!

Synonyms: same meaning in all senses

Probably no true examples....

Dimension Reduction

What words are similar to one another?

Similarity is high-dimensional!

Synonyms: same meaning in all senses

Probably no true examples....

Big/large

Dimension Reduction

What words are similar to one another?

Similarity is high-dimensional!

Synonyms: same meaning in all senses

Probably no true examples....

Big/large - my large sister != my big sister

Dimension Reduction

What words are similar to one another?

Similarity is high-dimensional!

Synonyms: same meaning in all senses

Probably no true examples....

Big/large - my large sister != my big sister

Antonyms: opposite along one dimension, otherwise similar

Hot/cold, in/out, tall/short, long/short

Dimension Reduction

What words are similar to one another?

Similarity is high-dimensional!

Synonyms: same meaning in all senses

Probably no true examples....

Big/large - my large sister != my big sister

Antonyms: opposite along one dimension, otherwise similar

Hot/cold, in/out, tall/short, long/short

Homonyms: different meanings for same character string

River bank / National bank

Dimension Reduction

Theory-Driven - *“We’ll tell you what a word means”*

Dimension Reduction

Theory-Driven - *“We’ll tell you what a word means”*
Dictionaries

Dimension Reduction

Theory-Driven - *“We’ll tell you what a word means”*
Dictionaries

Data-Driven - *“You will know a word by the company it keeps”*

Dimension Reduction

Theory-Driven - *“We’ll tell you what a word means”*
Dictionaries

Data-Driven - *“You will know a word by the company it keeps”*

Two senses of similarity:

Dimension Reduction

Theory-Driven - *“We’ll tell you what a word means”*
Dictionaries

Data-Driven - *“You will know a word by the company it keeps”*

Two senses of similarity:

Syntagmatic - often found in the same documents

“GDP”, “inflation”, “economy”, “currency”

e.g. topic models (LDA), Latent Semantic Analysis

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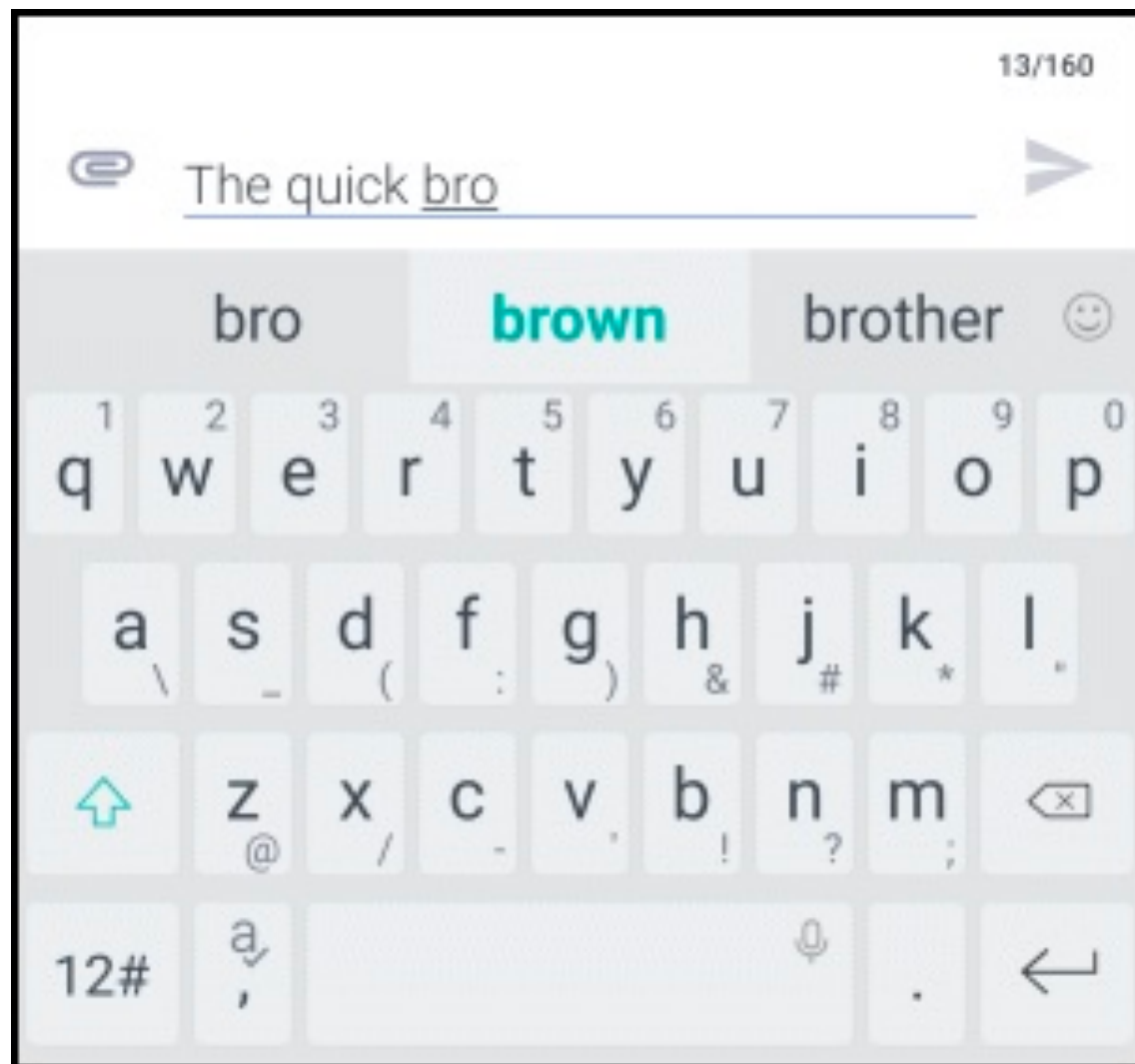
Paradigmatic - often have the same neighbours

“wrote”, “remarked”, “said”, “added”

e.g. word2vec, GloVe, BERT, GPT-1/2/3

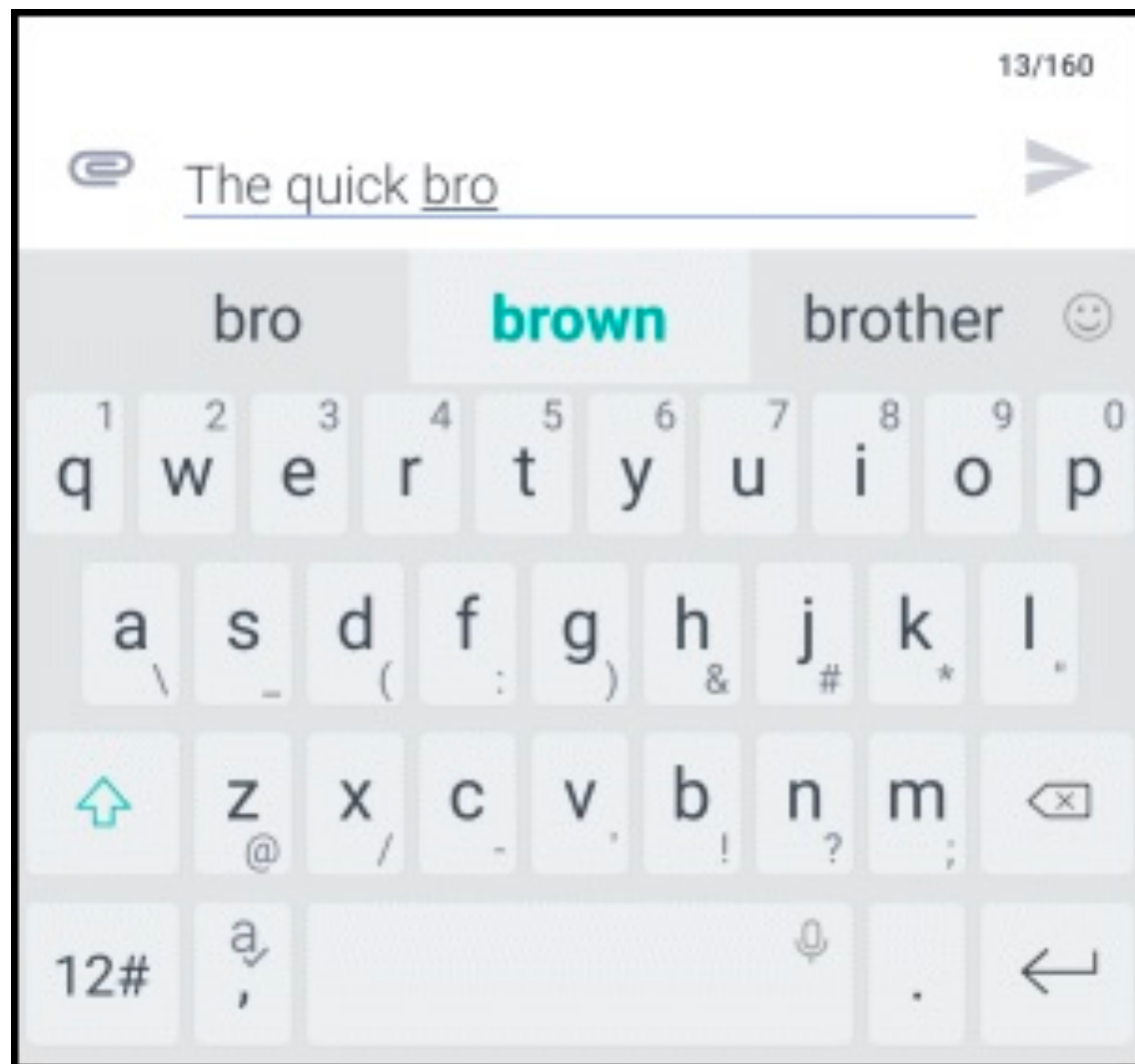
Language Modeling Task

Predictive Text



Language Modeling Task

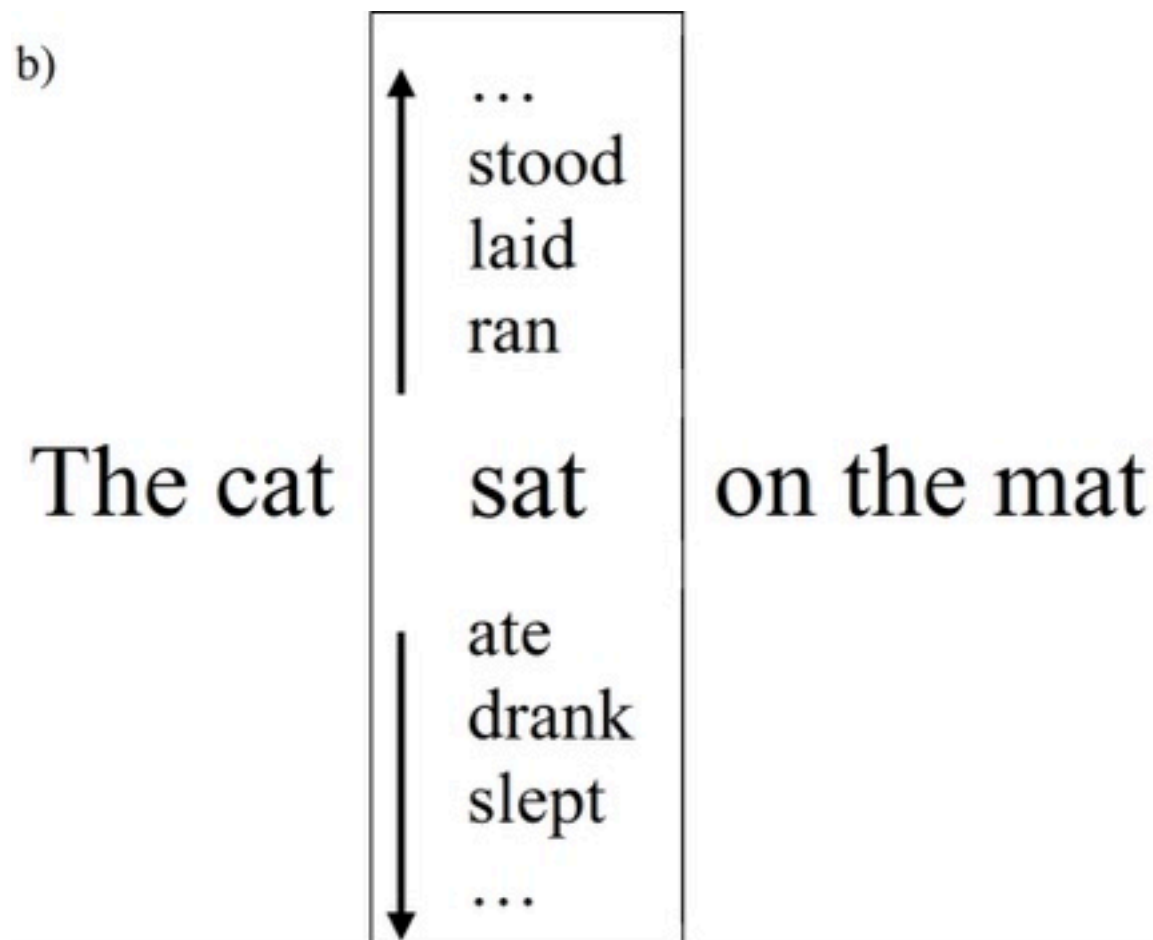
Predictive Text



- what is the next
- what is the next **holiday**
- what is the next **marvel movie**
- what is the next **warriors game**
- what is the next **full moon**
- what is the next **holiday coming up**
- what is the next **generation called**
- what is the next **marvel show**
- what is the next **covid variant**
- what is the next **bitcoin**
- what is the next **spiderman movie**

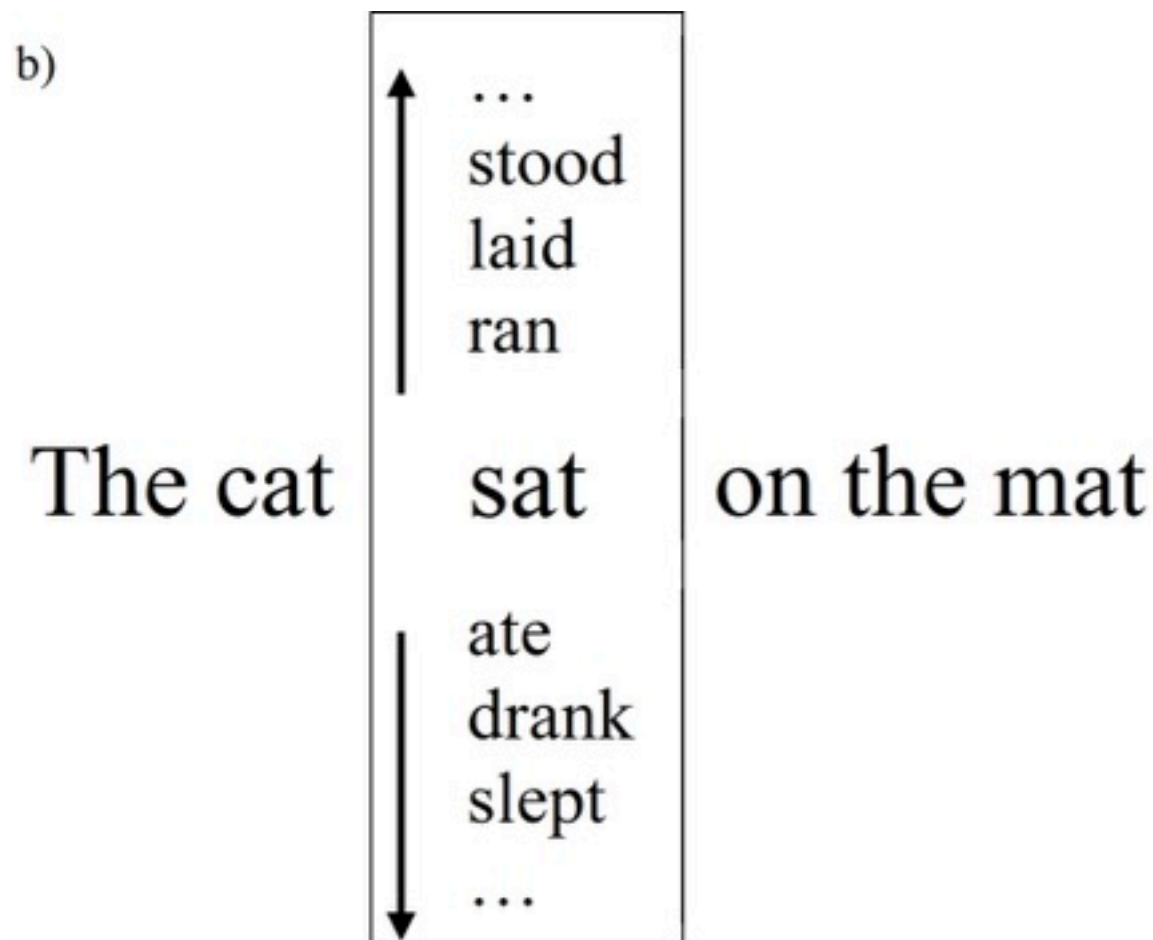
Language Modeling Task

Continuous Bag of Words



Language Modeling Task

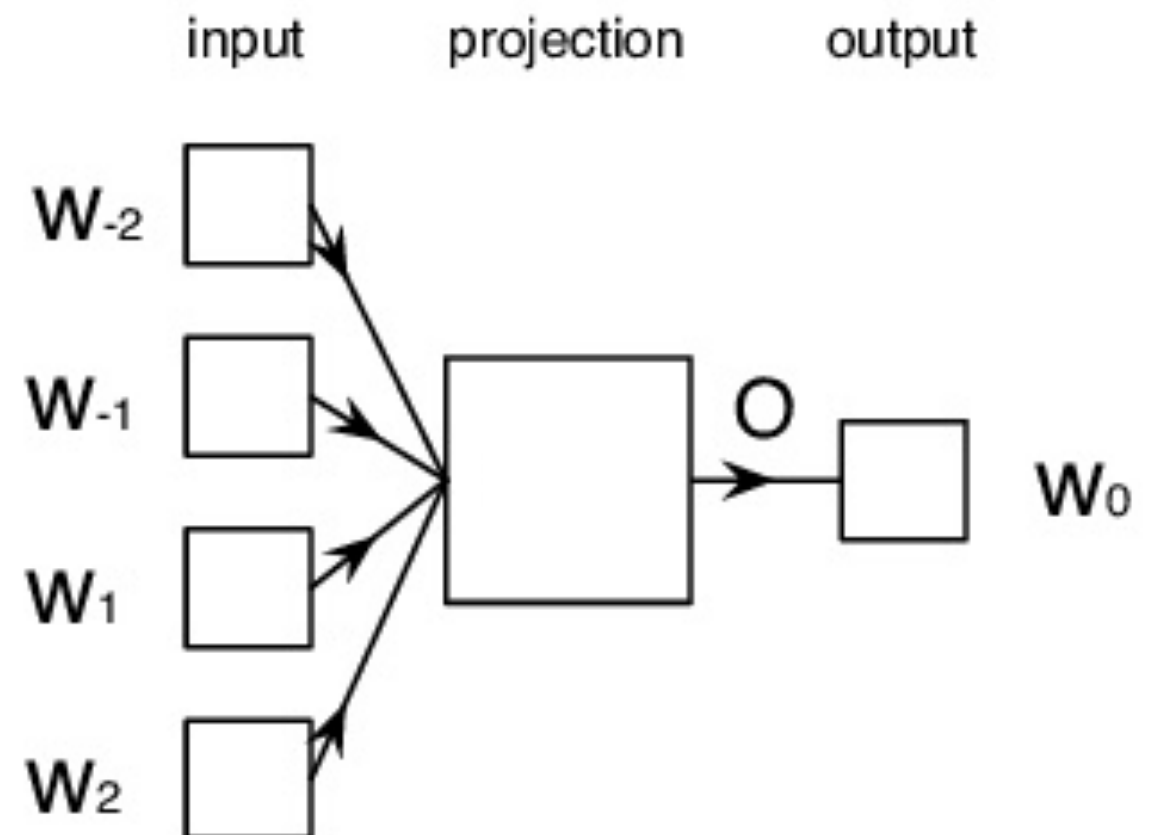
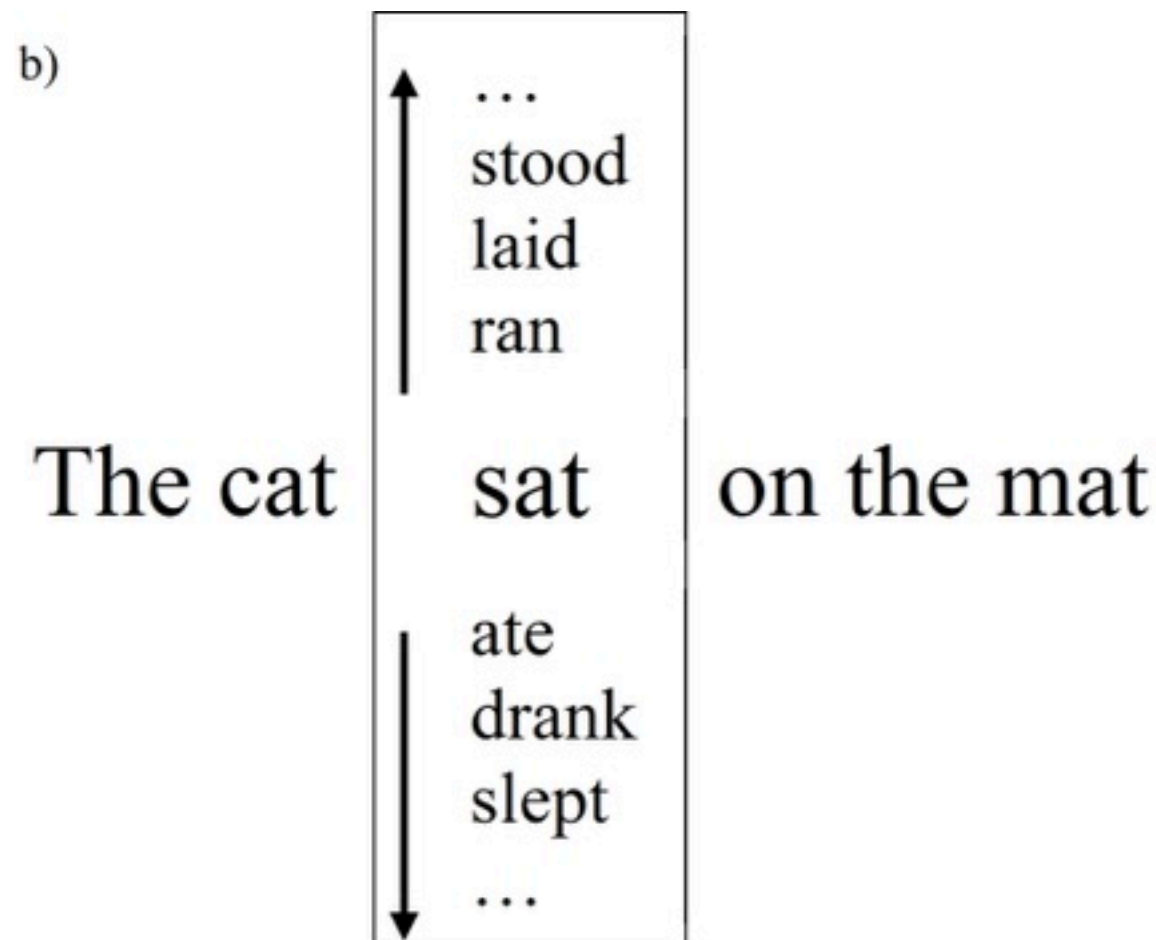
Continuous Bag of Words



Predict the masked word using the context words
Estimate mapping of words onto hidden layer

Language Modeling Task

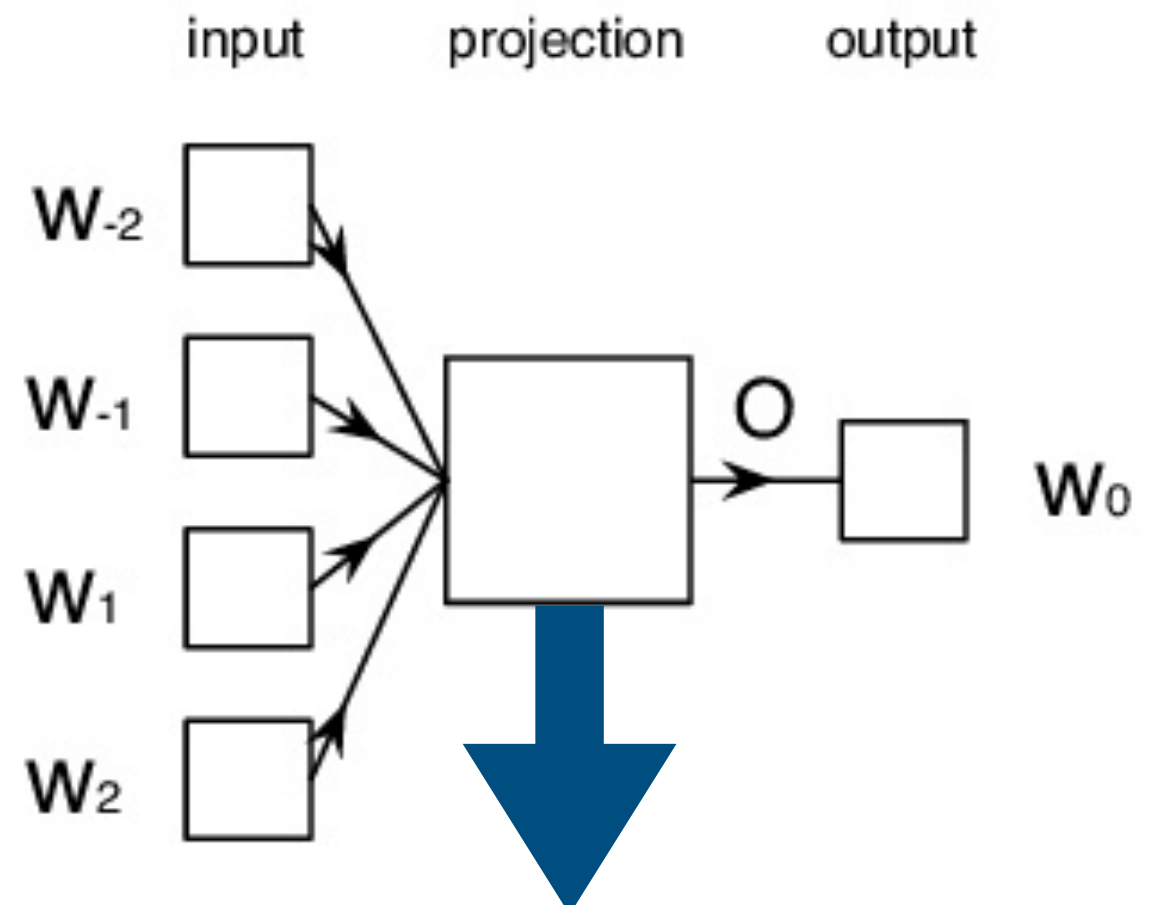
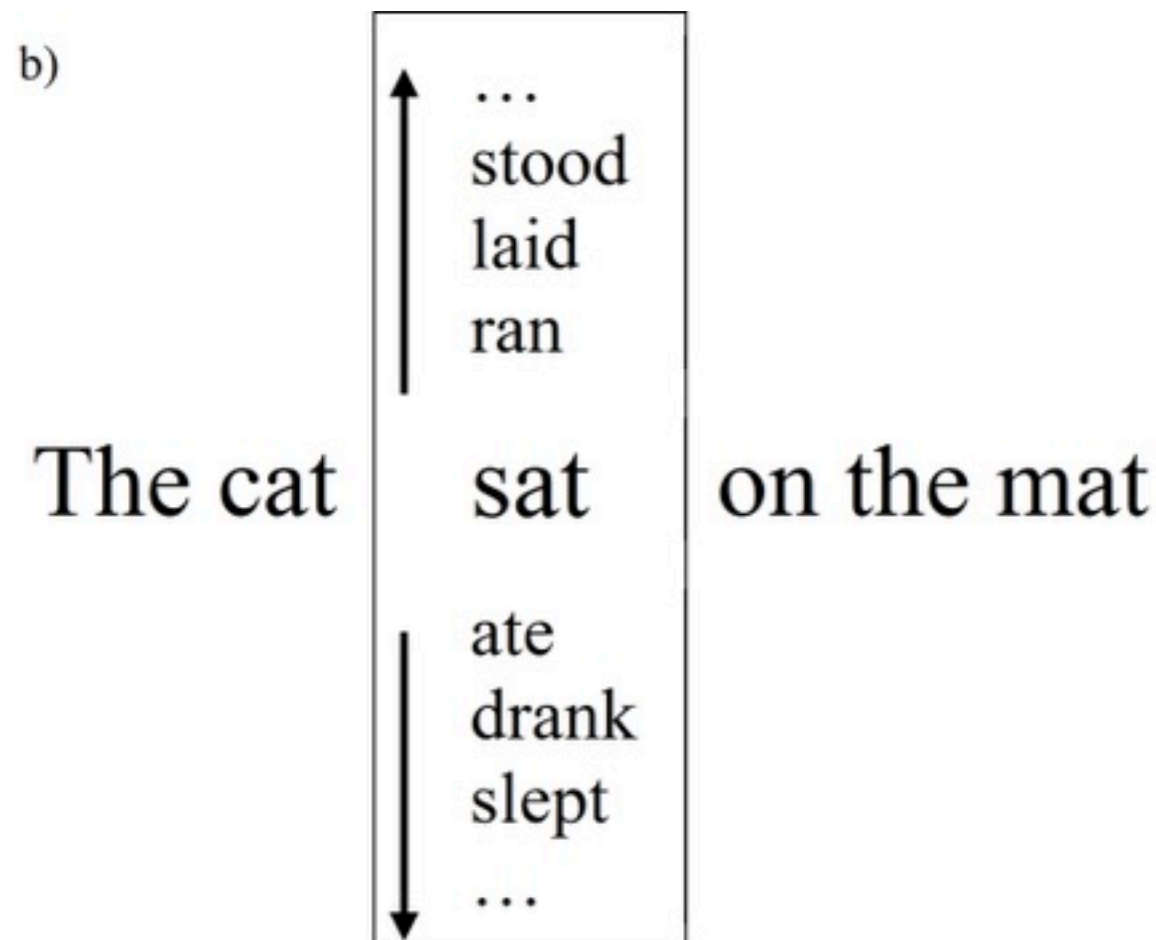
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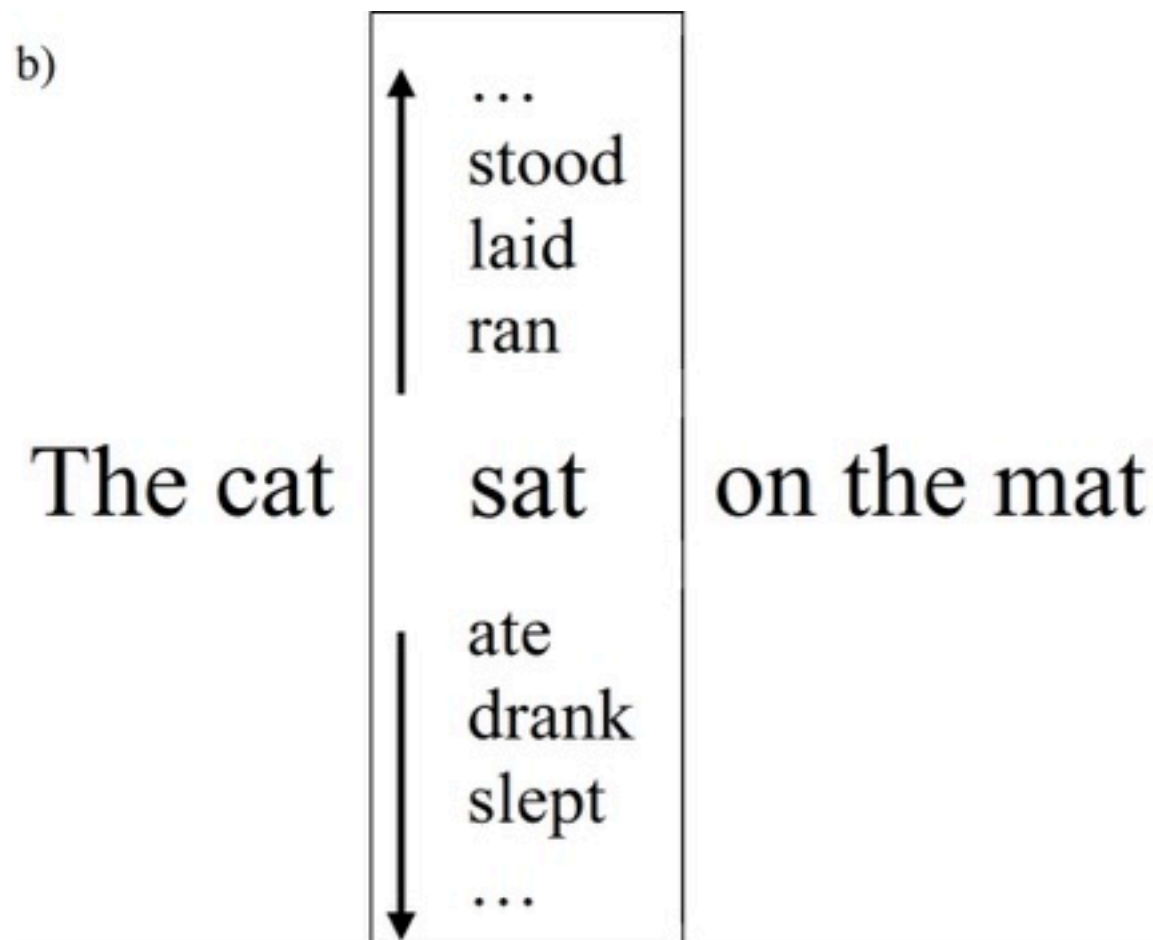
	vec1	vec2	vec3	vec4	vec5	vec6
aardvark	-0.52	-0.49	-0.70	-0.02	0.11	0.37
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able	-0.46	-0.85	-0.35	0.33	-0.13	0.57

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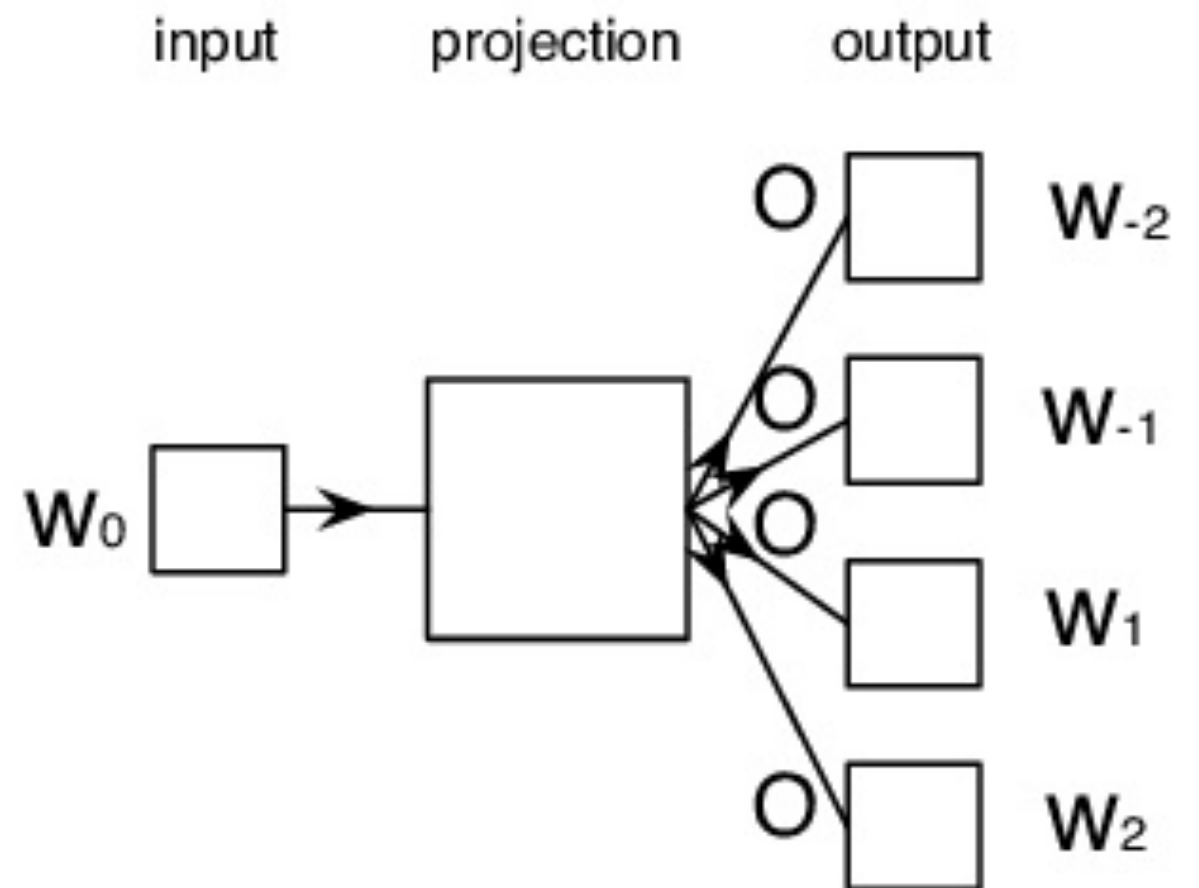
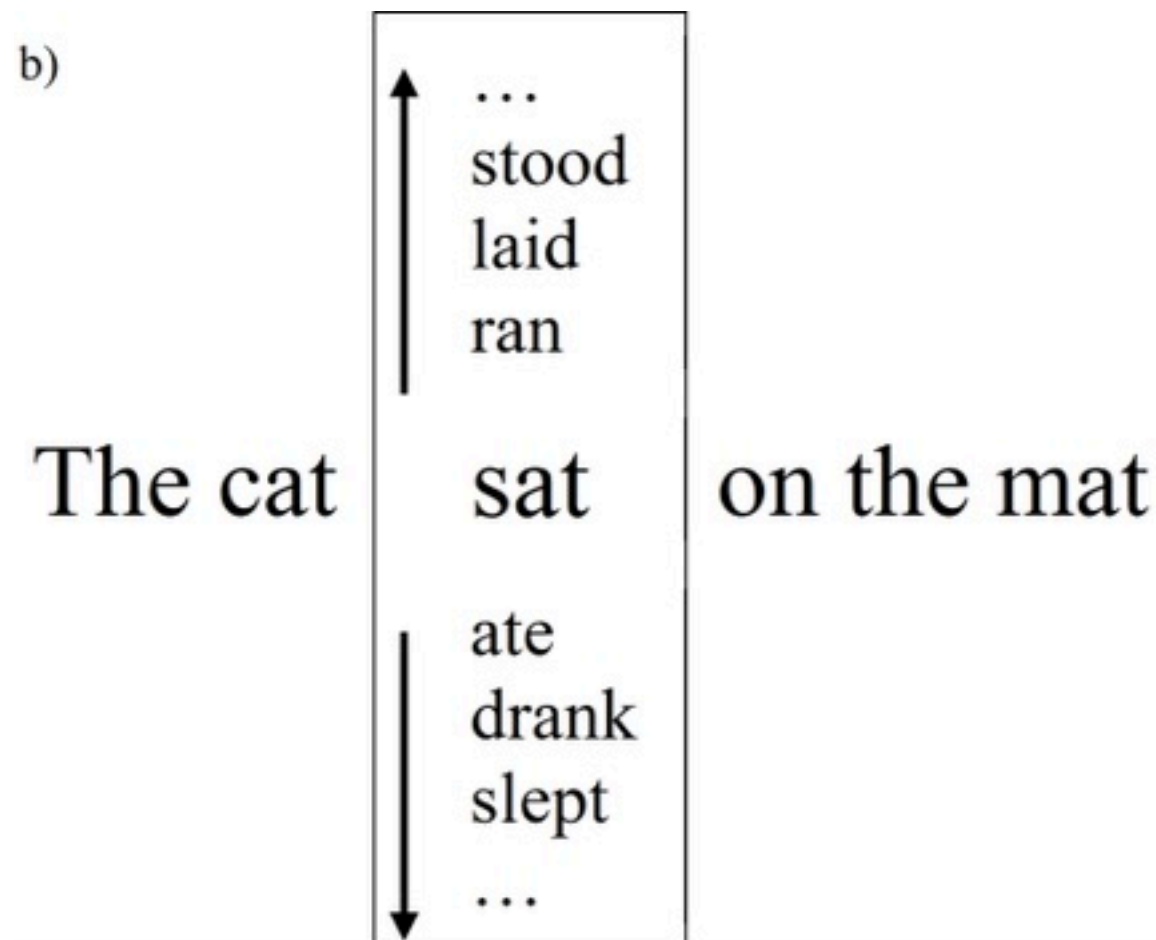
Skip-grams



Predict the context words using the masked word
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Language Modeling Task

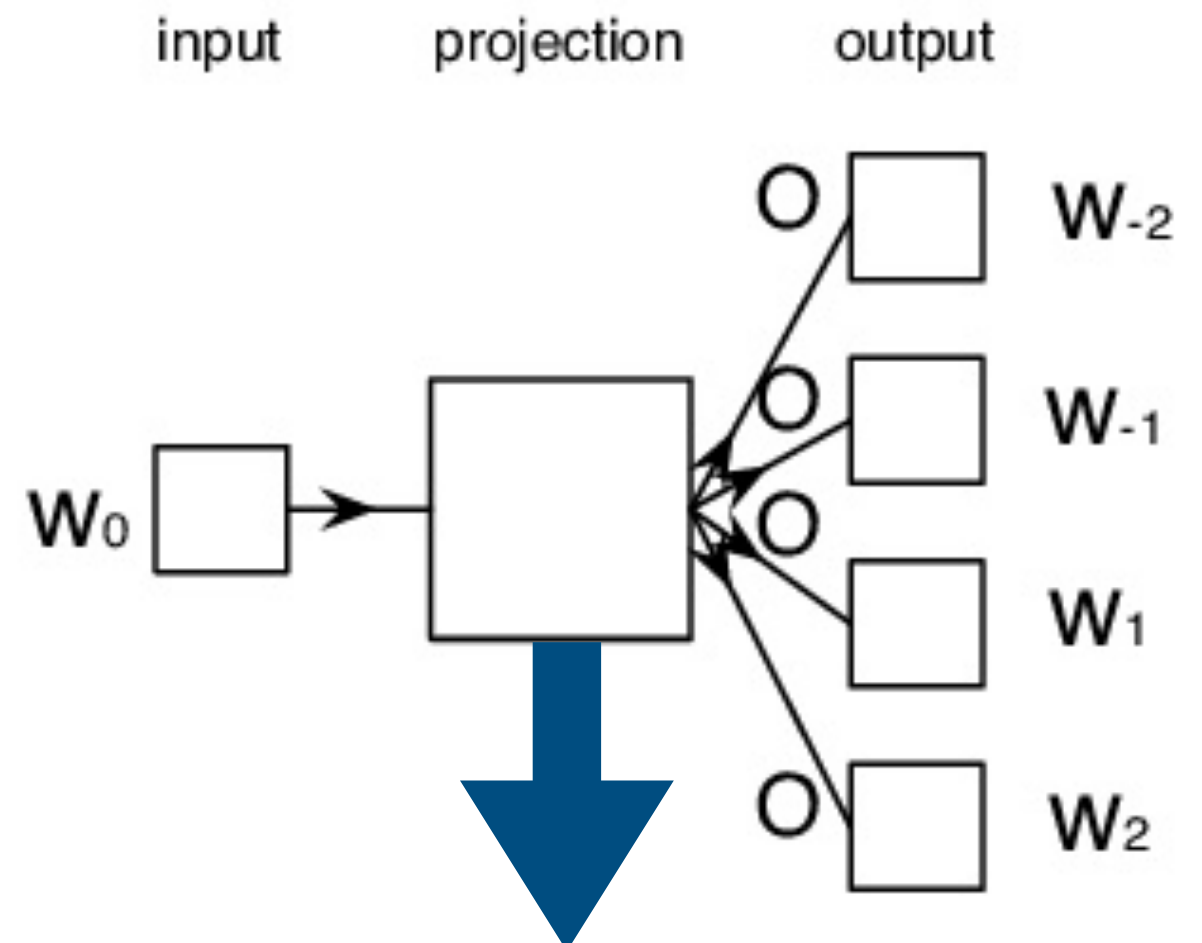
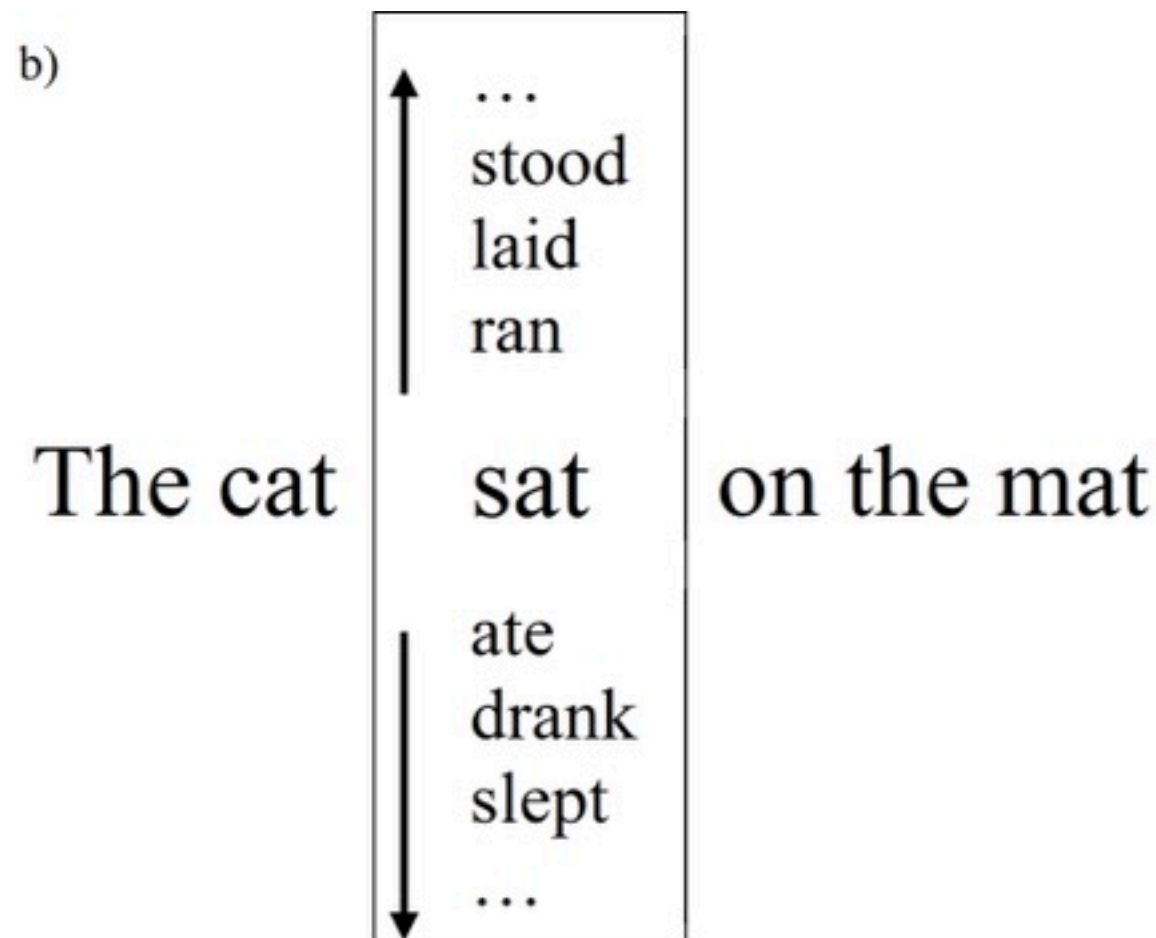
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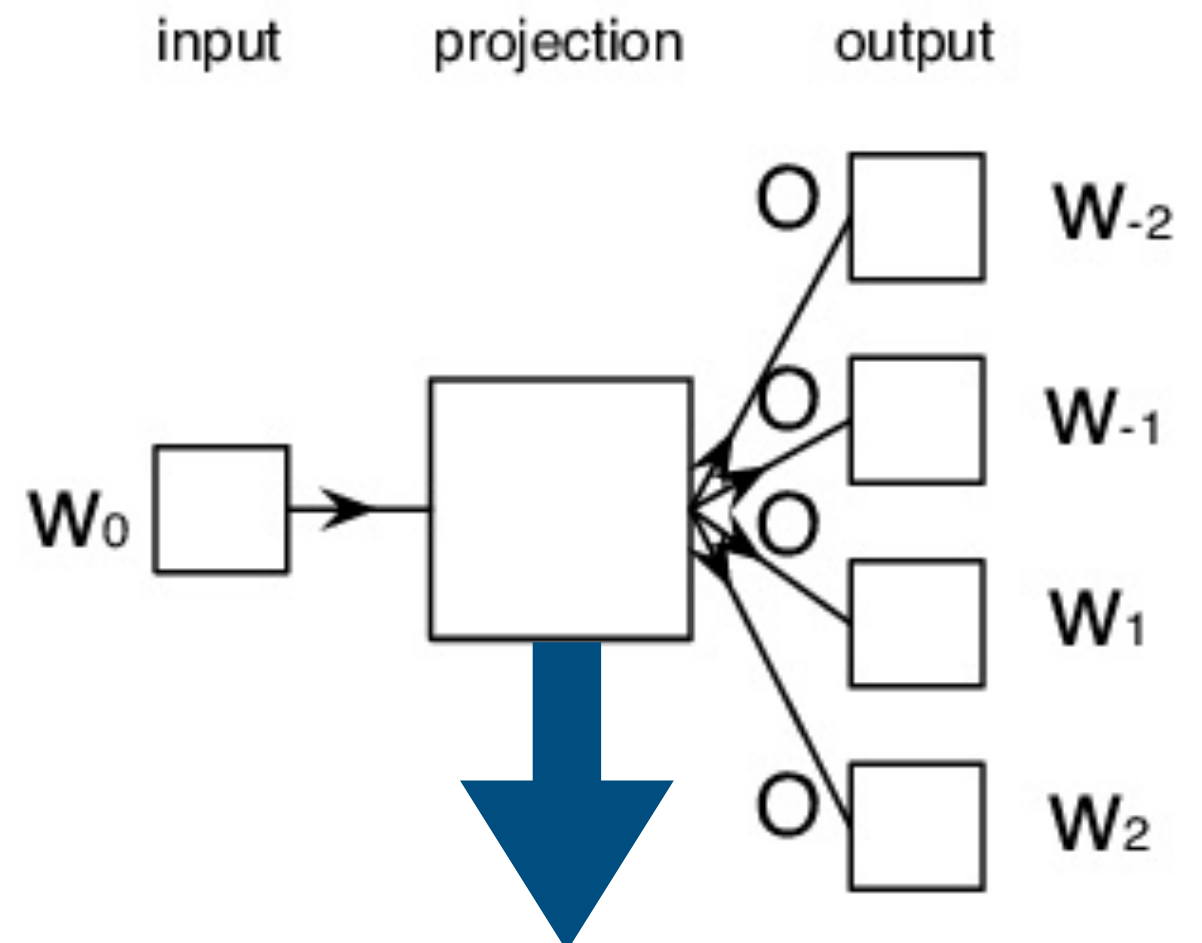
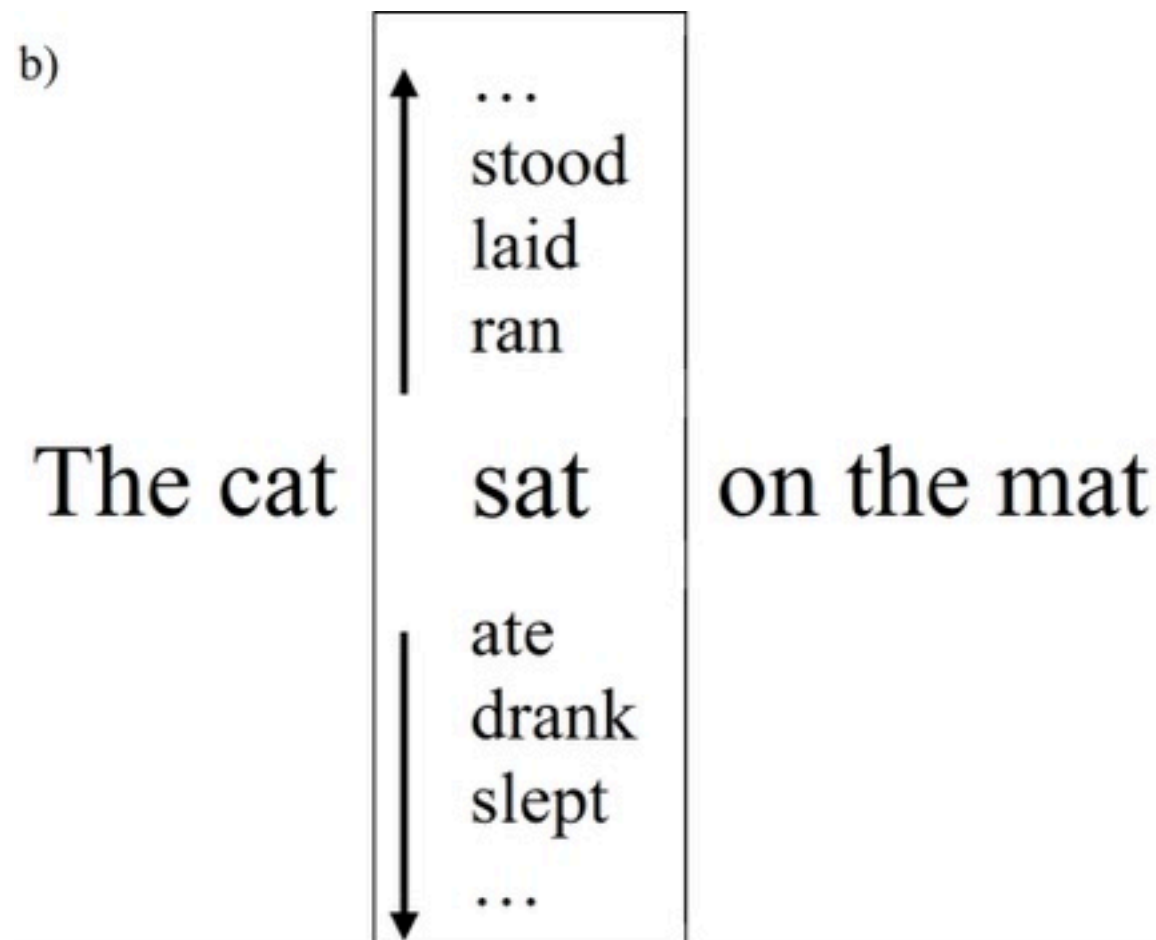


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Predict the context words using the masked word

"Self-Supervised Learning"

Negative Sampling

The cat [sat] on the mat

Masked Word	Context Word	In data
sat	the	1
sat	cat	1
sat	on	1
sat	the	1

Negative Sampling

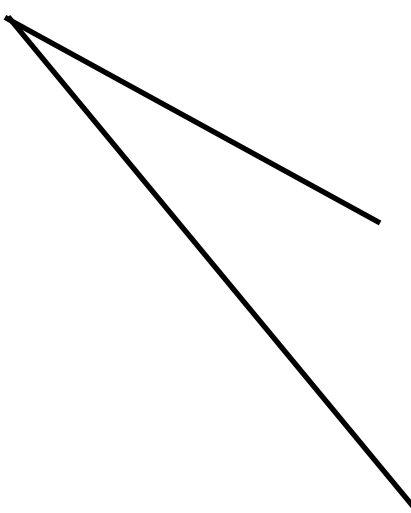
The cat [sat] on the mat

Masked Word	Context Word	In data
sat	the	1
sat	cat	1
sat	on	1
sat	the	1
sat	horse	0
sat	flower	0
sat	running	0
sat	nice	0

Negative Sampling

The cat [sat] on the mat

random other words
not from context

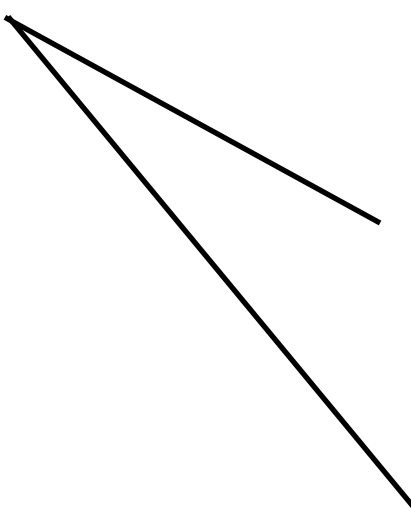


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sat	nice	0

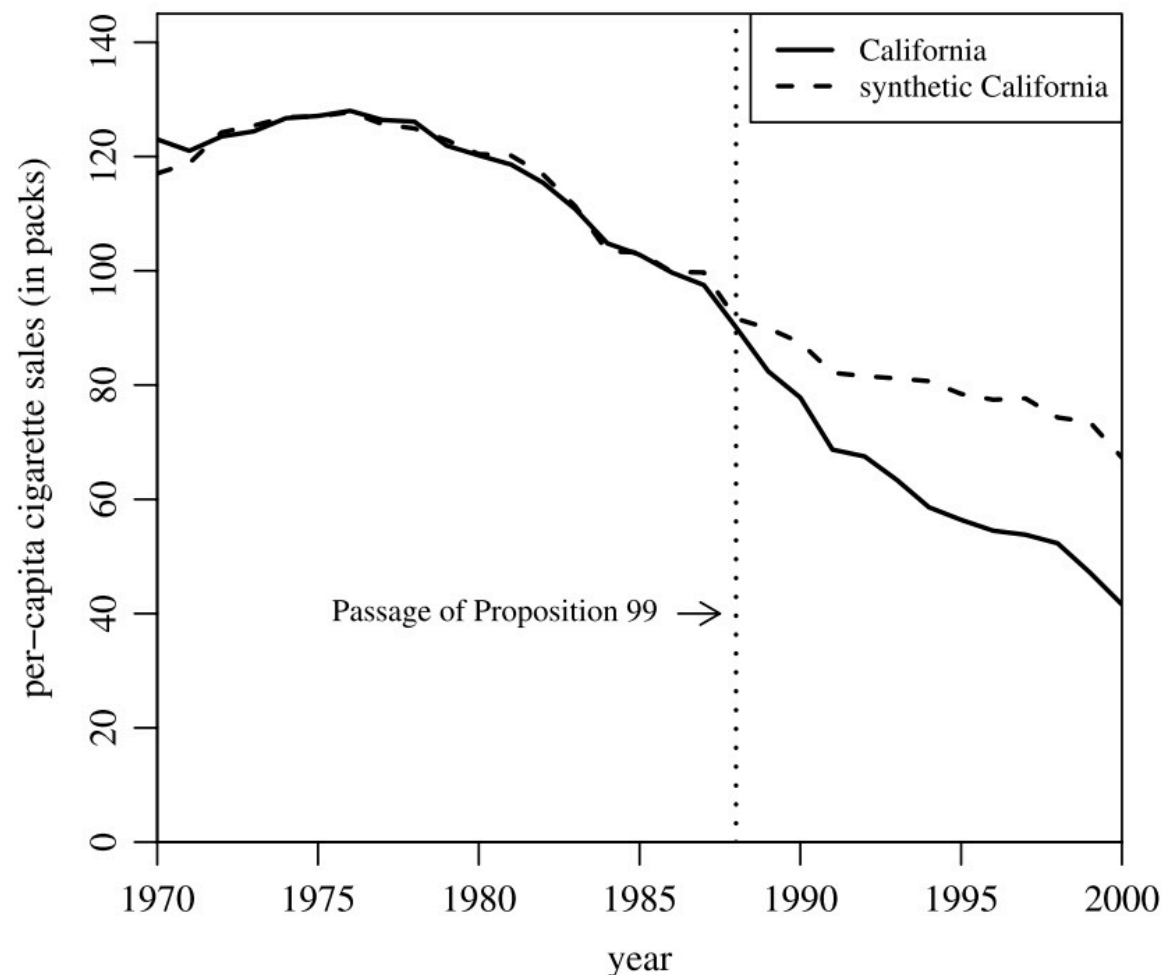
Similar to synthetic control
- generate comparison set

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Language Modeling Task

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Similar to synthetic control

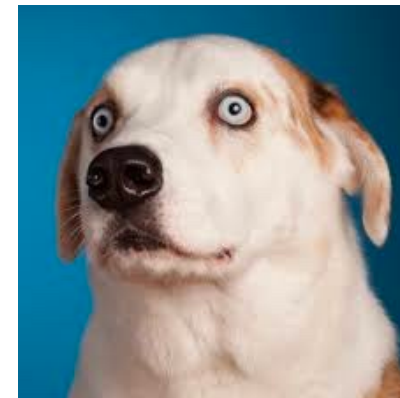
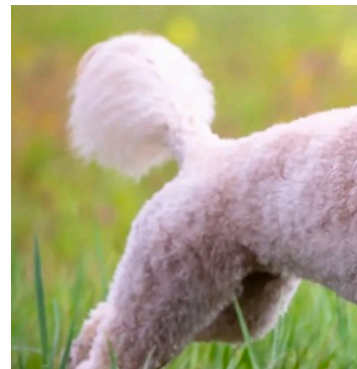
- generate comparison set and test

Self-Supervised Image Learning



Chen et al., 2020

Self-Supervised Image Learning



Chen et al., 2020

Self-Supervised Image Learning



(a) Original



(b) Crop and resize



(c) Crop, resize (and flip)



(d) Color distort. (drop)



(e) Color distort. (jitter)



(f) Rotate $\{90^\circ, 180^\circ, 270^\circ\}$



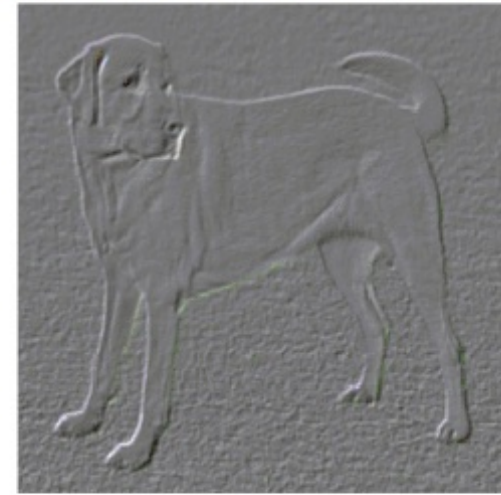
(g) Cutout



(h) Gaussian noise



(i) Gaussian blur

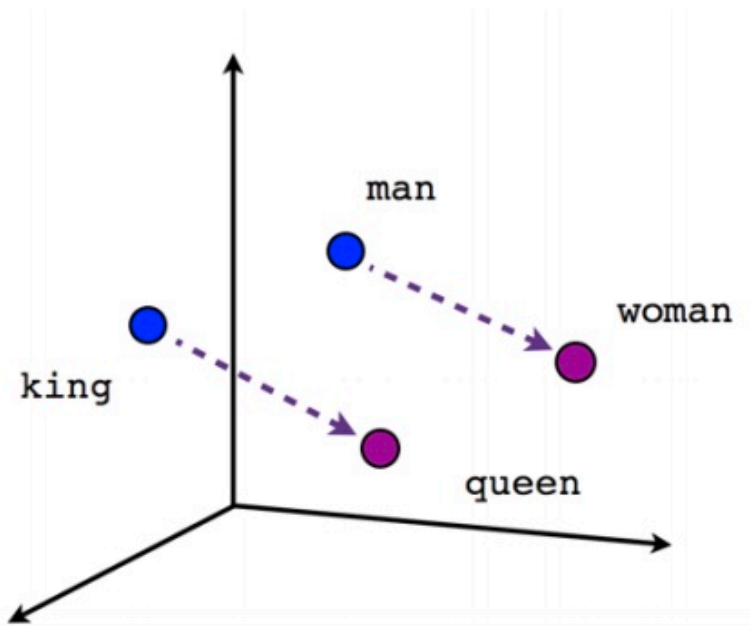


(j) Sobel filtering

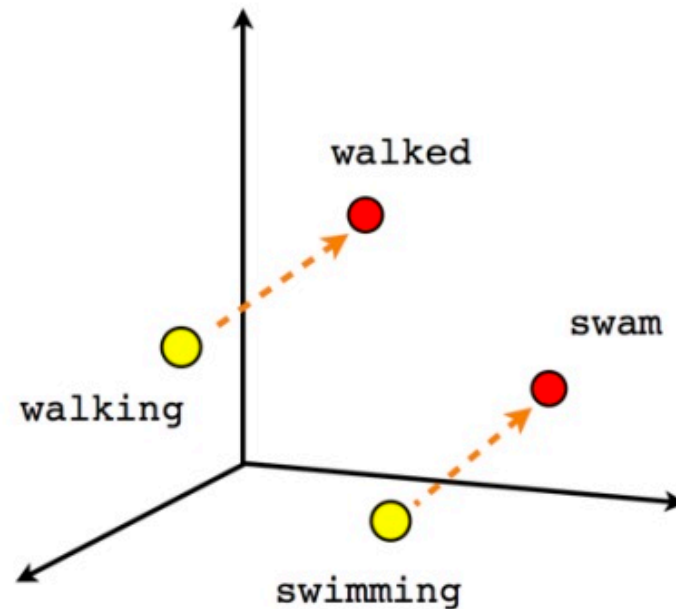
Chen et al., 2020

Embedding Spaces

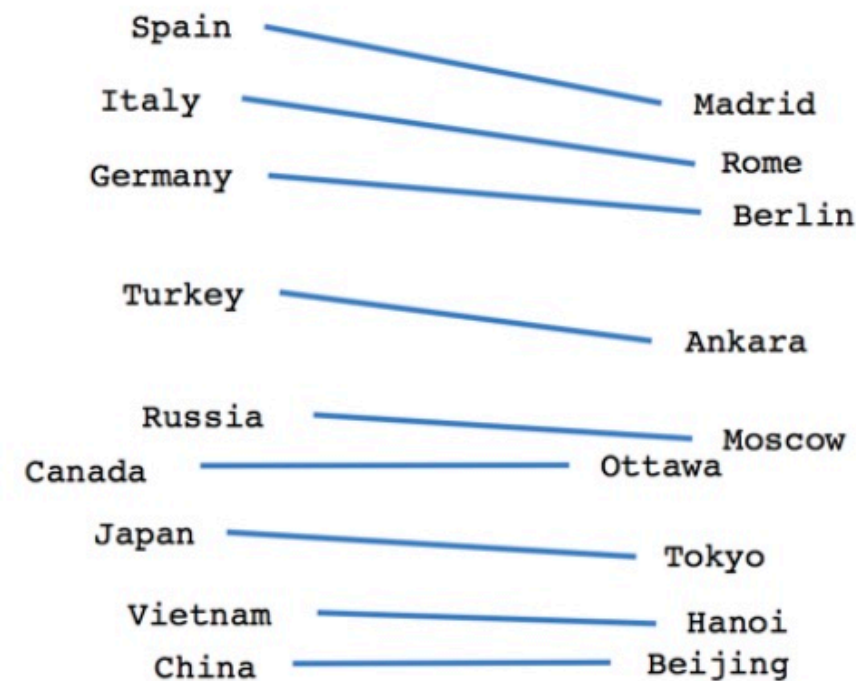
What does meaning look like?



Male-Female



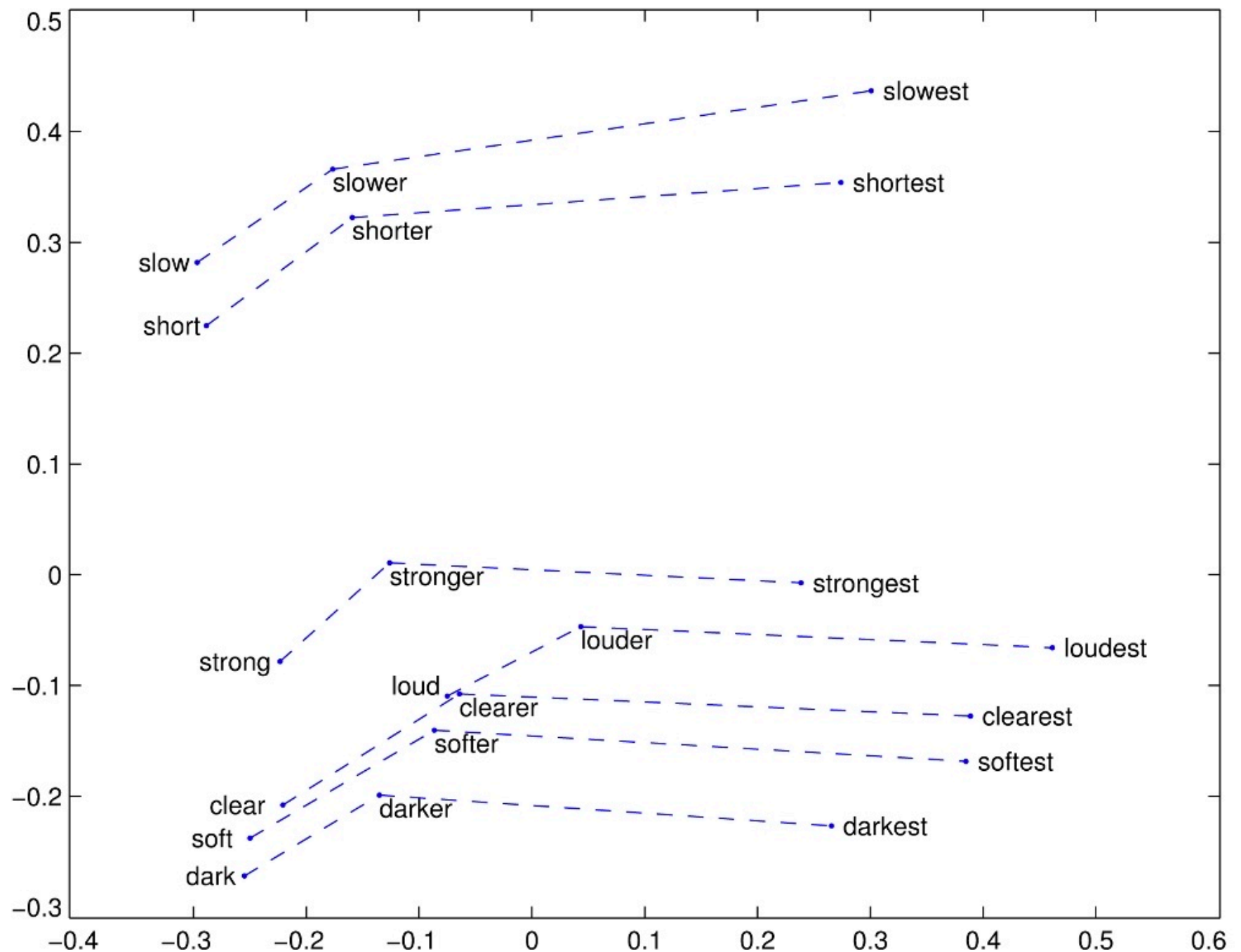
Verb tense



Country-Capital

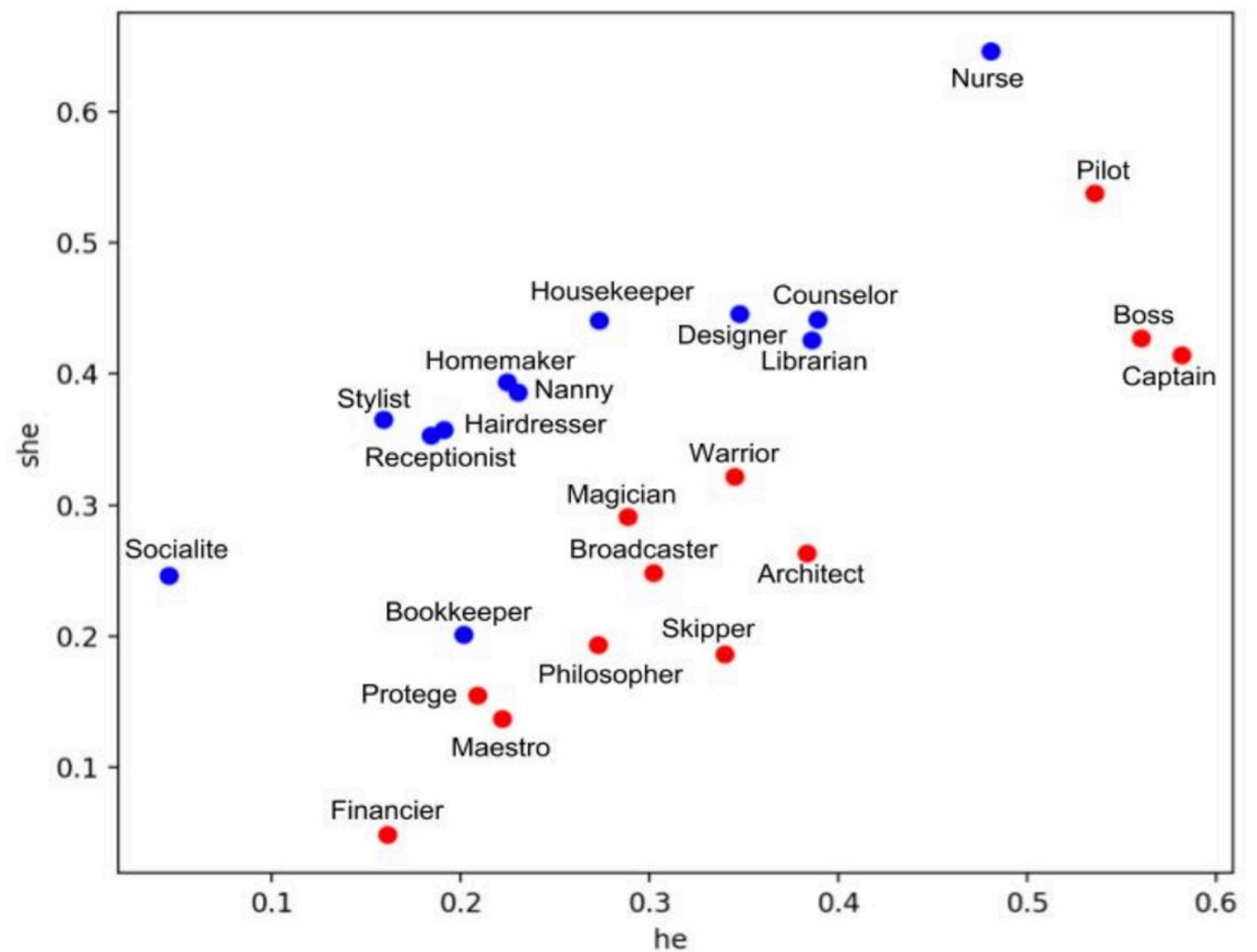
Embedding Spaces

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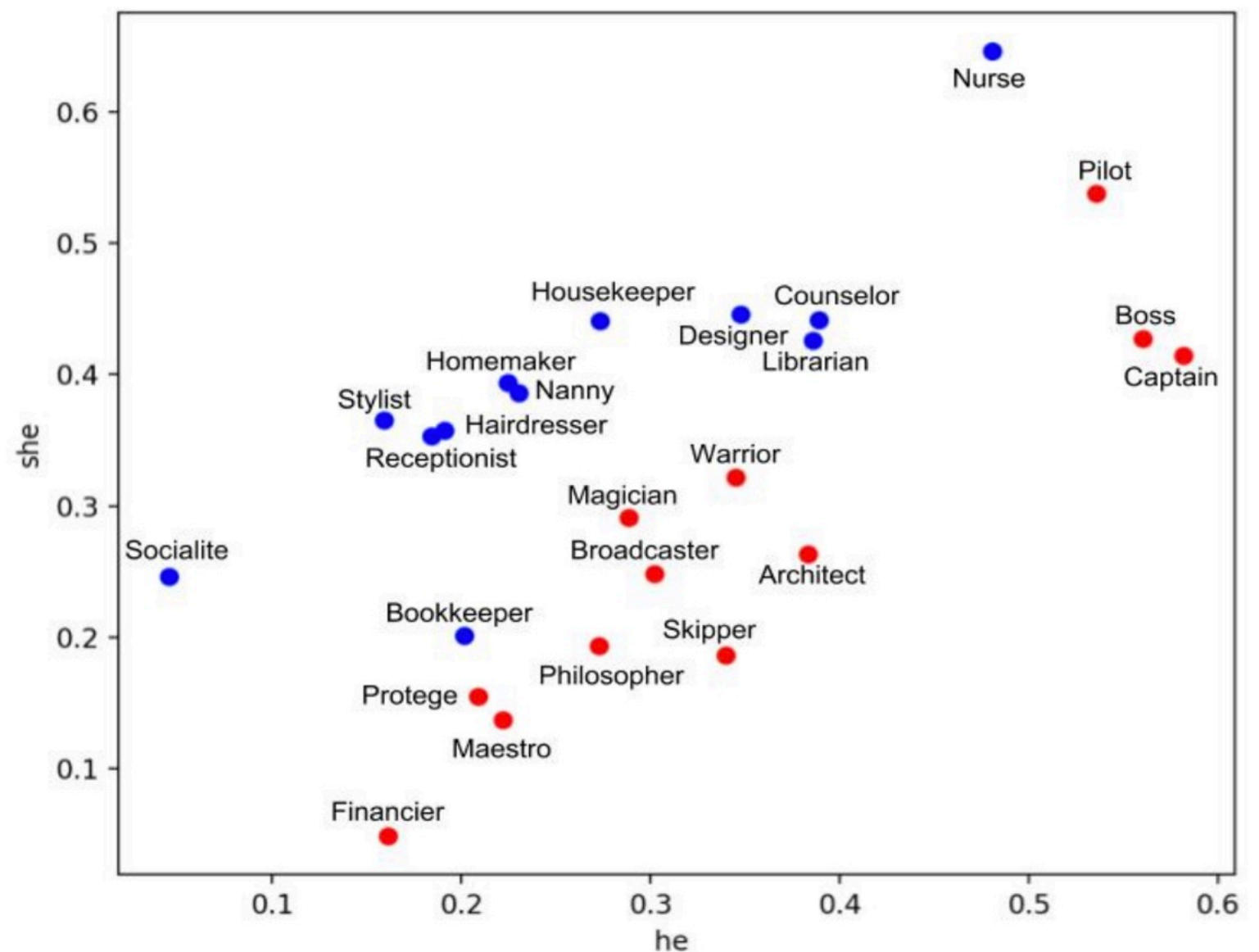


Caliskan, Bryson & Narayanan, 2017

Embedding Spaces

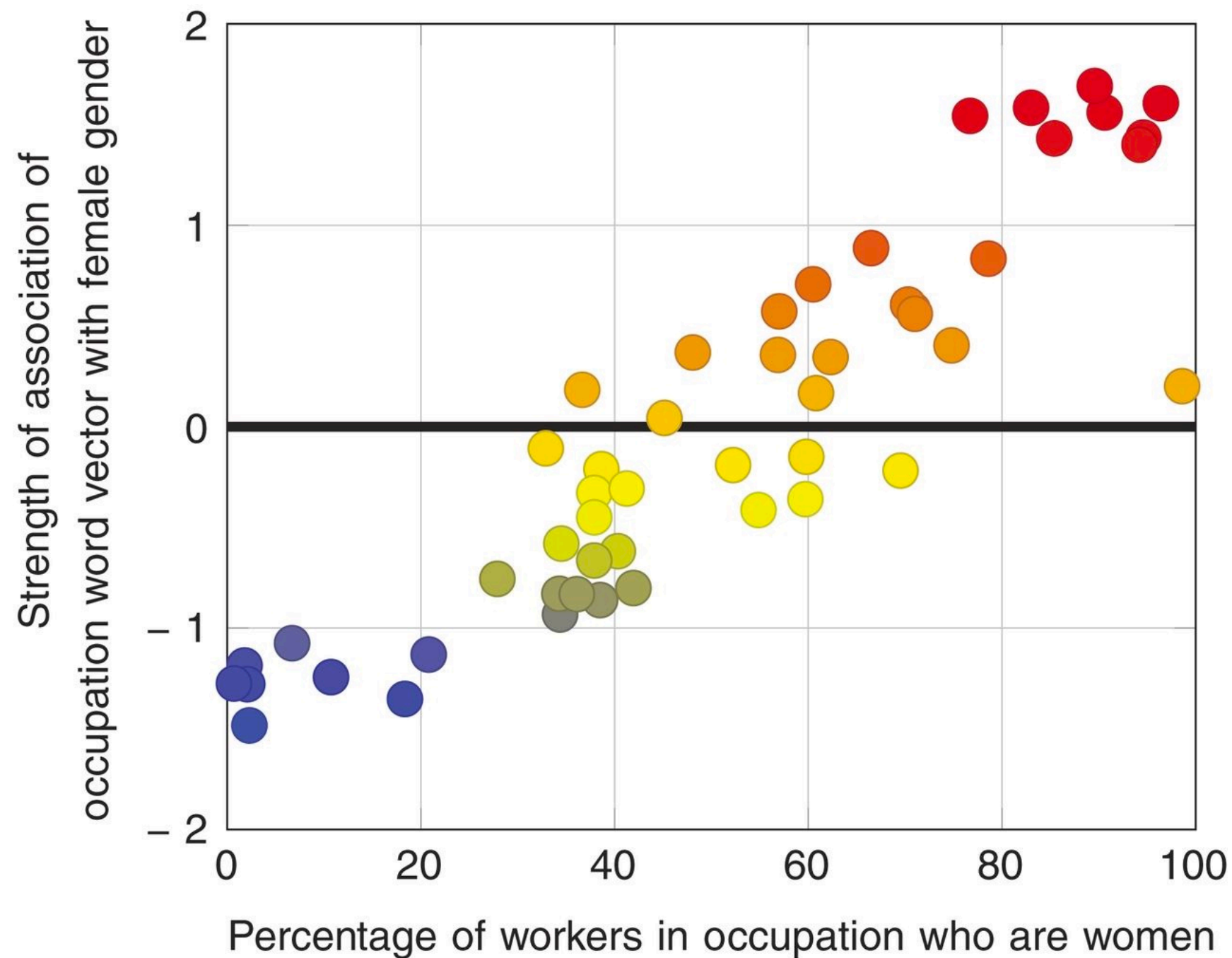
What does meaning look like?

Where is the bias?



Caliskan, Bryson & Narayanan, 2017

Applications of Word Similarity



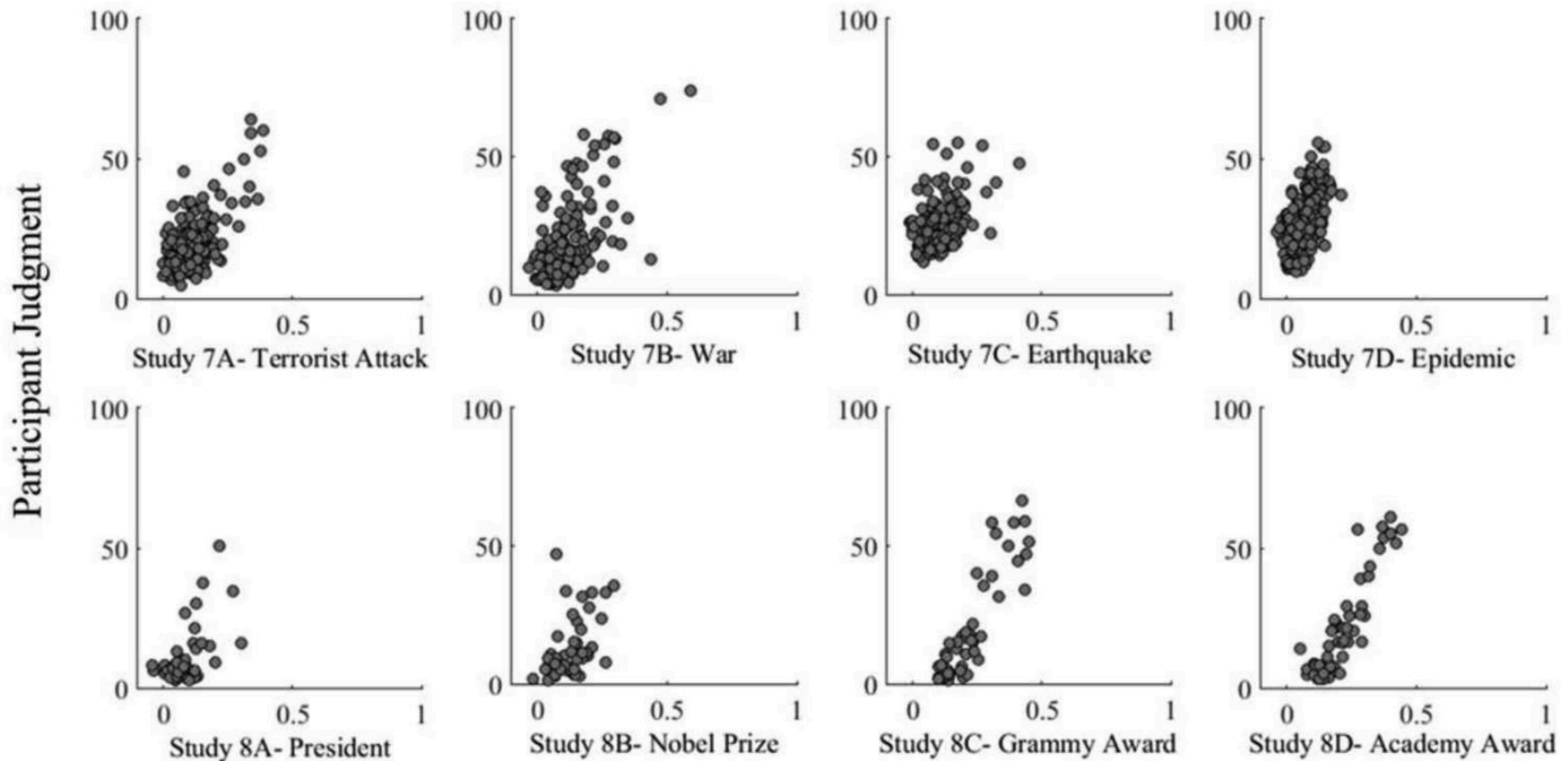
Caliskan, Bryson & Narayanan, 2017

Applications of Word Similarity



Garg et al., 2018

Applications of Word Similarity



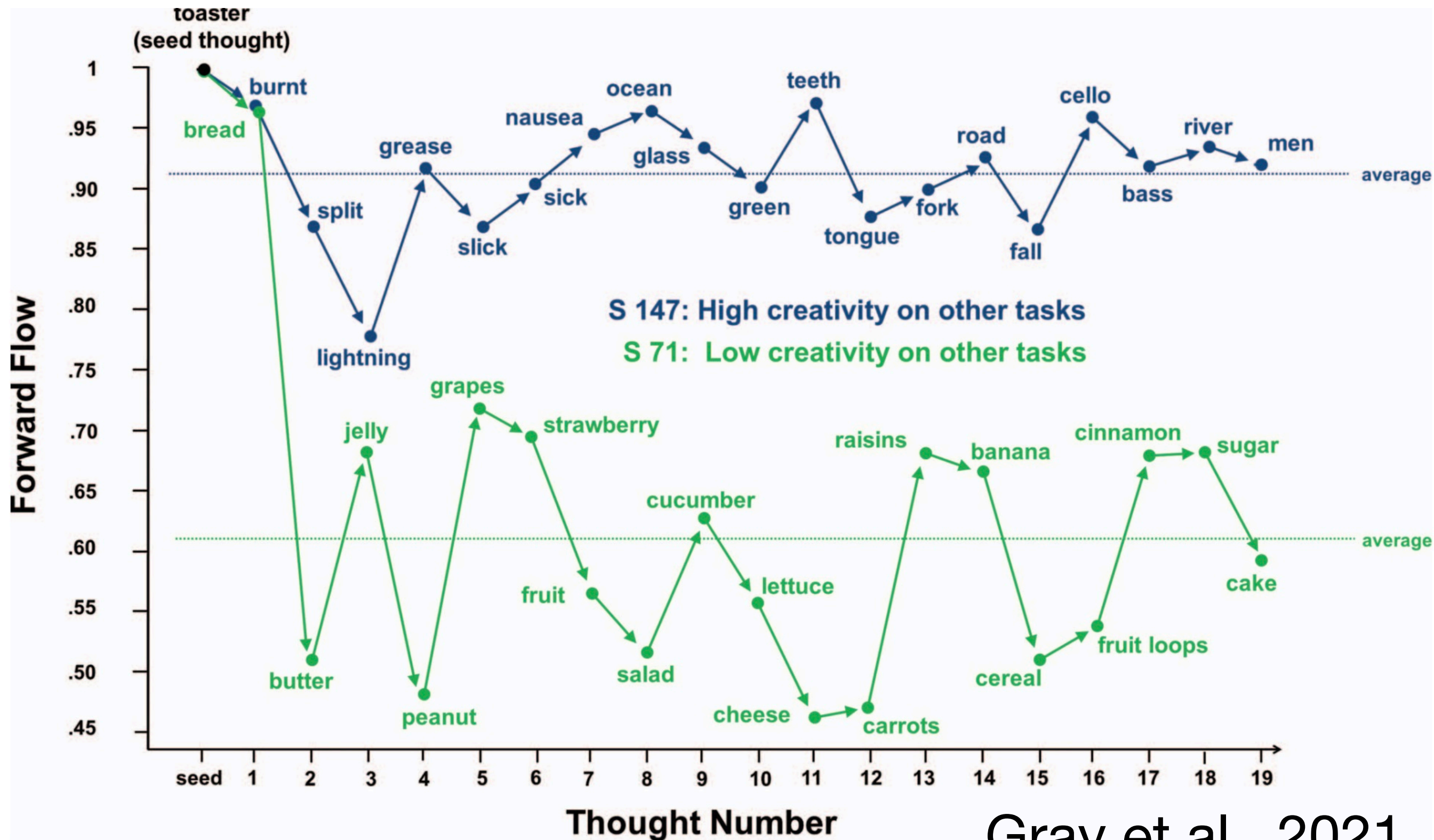
Applications of Word Similarity

<https://semantle.novalis.org/>

Guess the hidden word with
similarity of previous guesses

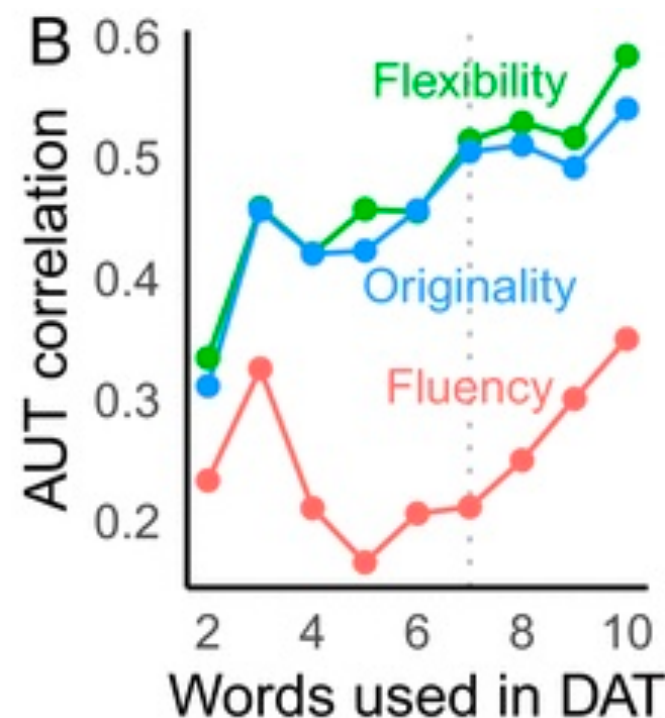
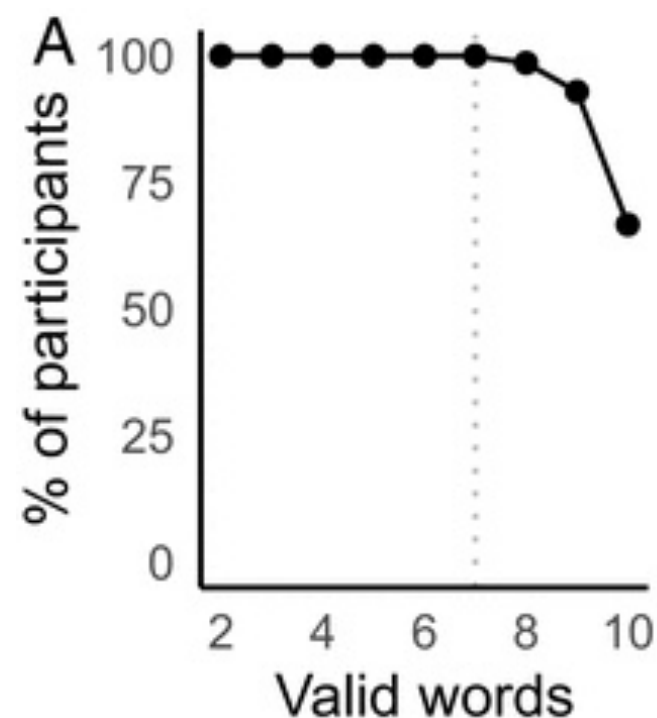
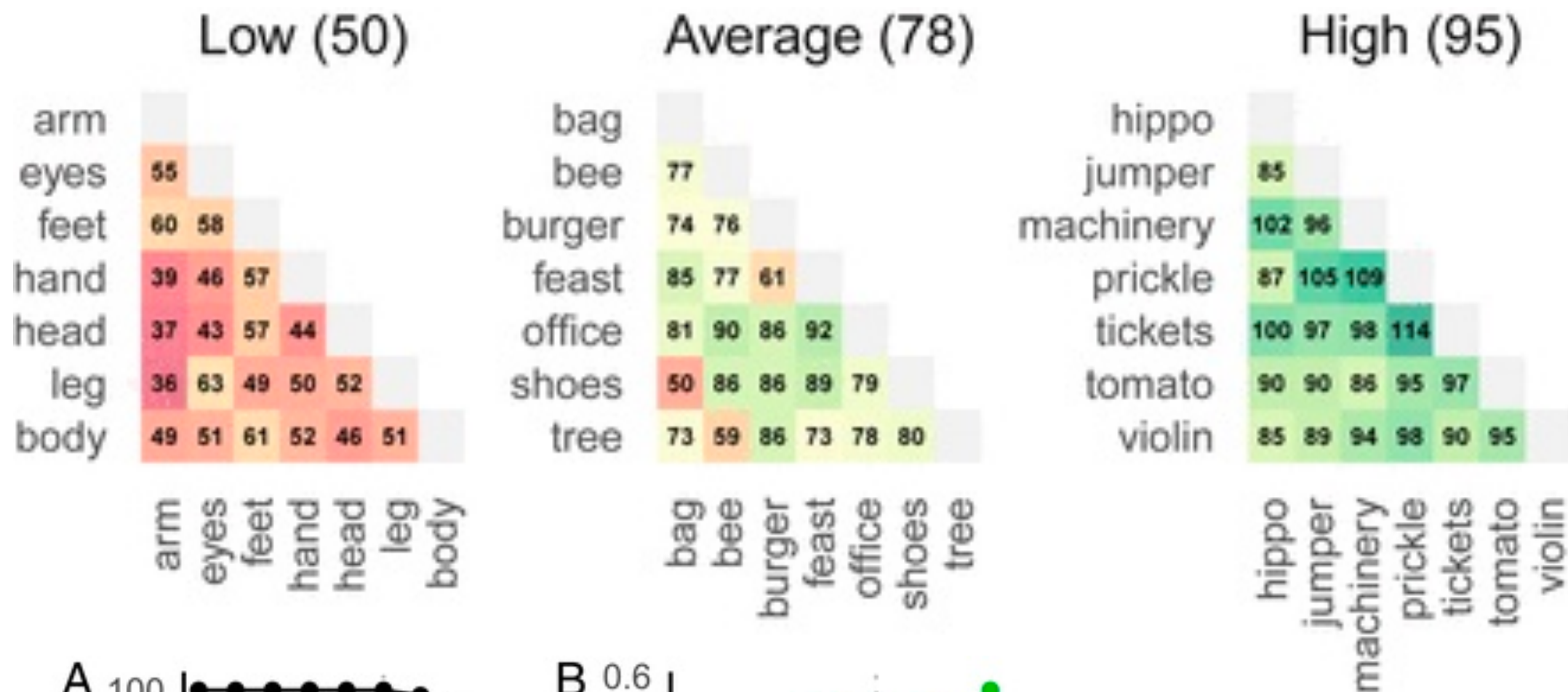
Verizon 8:47 PM 100%			
Guess			
You found it in 89! The secret word is surgeon. Feel free to keep entering words if you are curious about the similarity to other words. Share and play again tomorrow.			
#	Guess	Similarity	Getting close
89	monthly	100.00	FOUND!
44	rent	37.71	973/1000
68	stipend	36.25	964/1000
73	pay	35.34	961/1000
52	fee	35.08	959/1000
69	allowance	34.52	956/1000
61	payment	34.46	955/1000
55	rate	31.37	923/1000
74	dues	30.92	912/1000
57	rental	26.97	784/1000
64	salary	25.56	709/1000
semantle.novalis.org			

Applications of Word Similarity



Gray et al., 2021

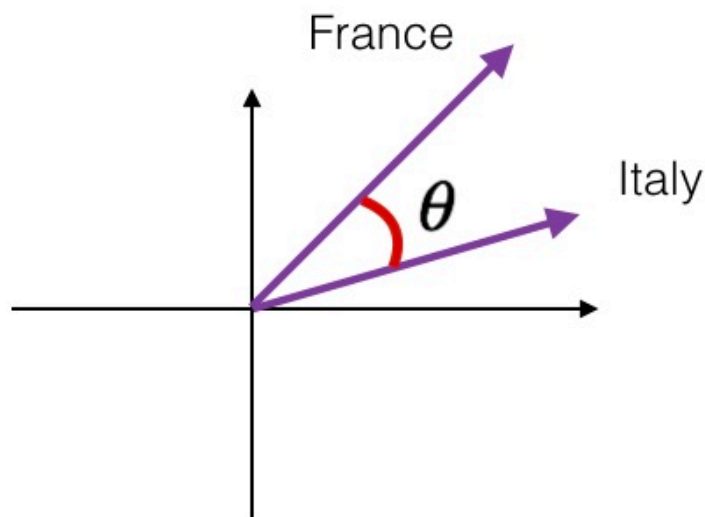
Applications of Word Similarity



Vector Space Model of Text

Cosine similarity

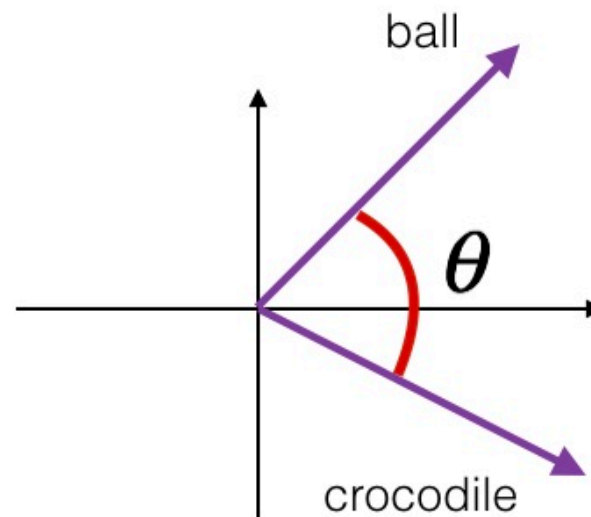
Indifferent to document length



France and Italy are quite similar

θ is close to 0°

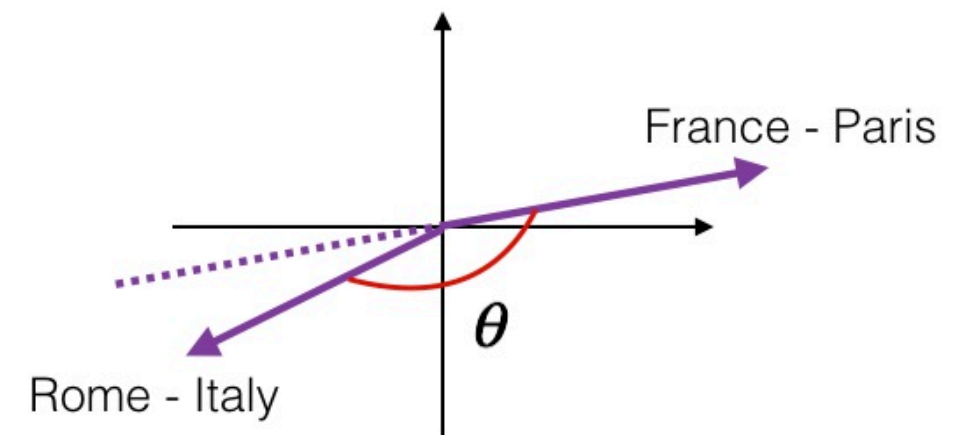
$$\cos(\theta) \approx 1$$



ball and crocodile are not similar

θ is close to 90°

$$\cos(\theta) \approx 0$$



the two vectors are similar but opposite
the first one encodes (city - country)
while the second one encodes (country - city)

θ is close to 180°

$$\cos(\theta) \approx -1$$

Embedding Models

Calculating similarity

Consider three sentences

“I am very sad”, “I am very happy”, “I am thrilled”

Which of the first two is more similar to the third?

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According to
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Embedding Models

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According to
bag of words

	i	am	veri	sad	happi	thrill
text1	1	1	1	1	0	0
text2	1	1	1	0	1	0
text3	1	1	0	0	0	1

Embedding Models

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	i	am	veri	sad	happi	thrill
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text2	1	1	1	0	1	0
text3	1	1	0	0	0	1

$$\cos(t1, t3) = .57$$

$$\cos(t2, t3) = .57$$

Embedding Models

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According to
word2vec

Embedding Models

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According to
word2vec

	vec1	vec2	vec3	vec4	vec5	vec6
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Embedding Models

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$$\cos(t1, t3) = .81$$

$$\cos(t2, t3) = .87$$

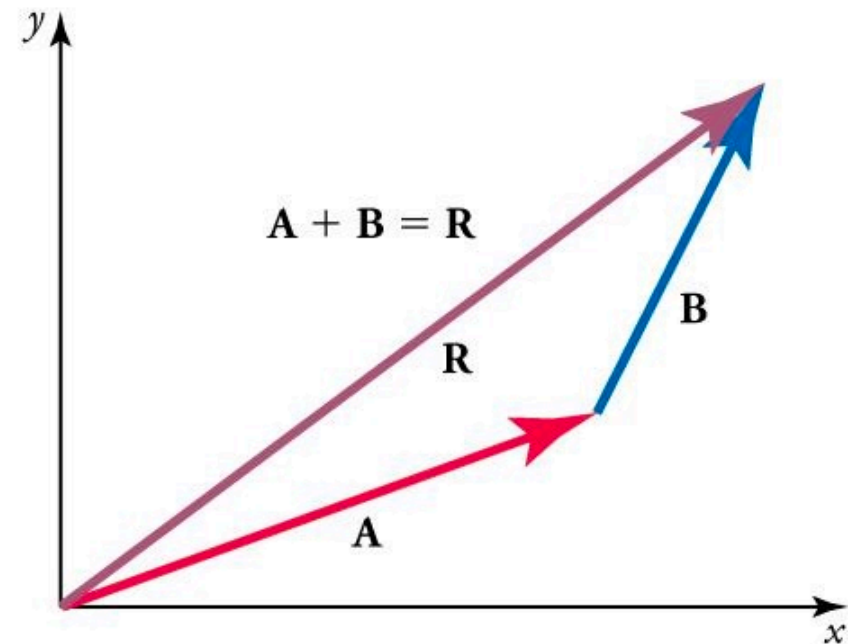
Sentence Embeddings

Simple

Sentence Embeddings

Simple

Add up the vectors of individual words



(Arora et al., 2017)

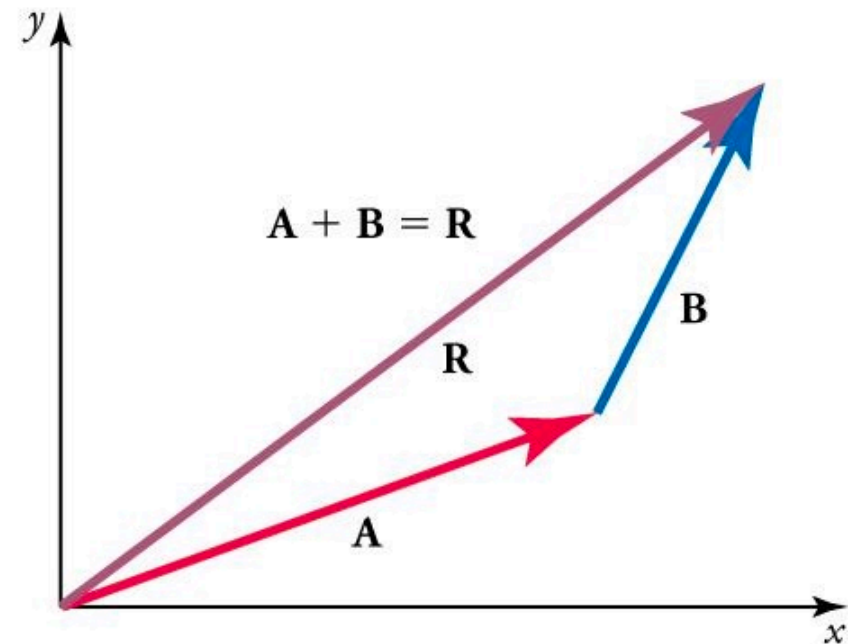
Sentence Embeddings

Simple

Add up the vectors of individual words

How to combine?

- Unweighted average



(Arora et al., 2017)

Sentence Embeddings

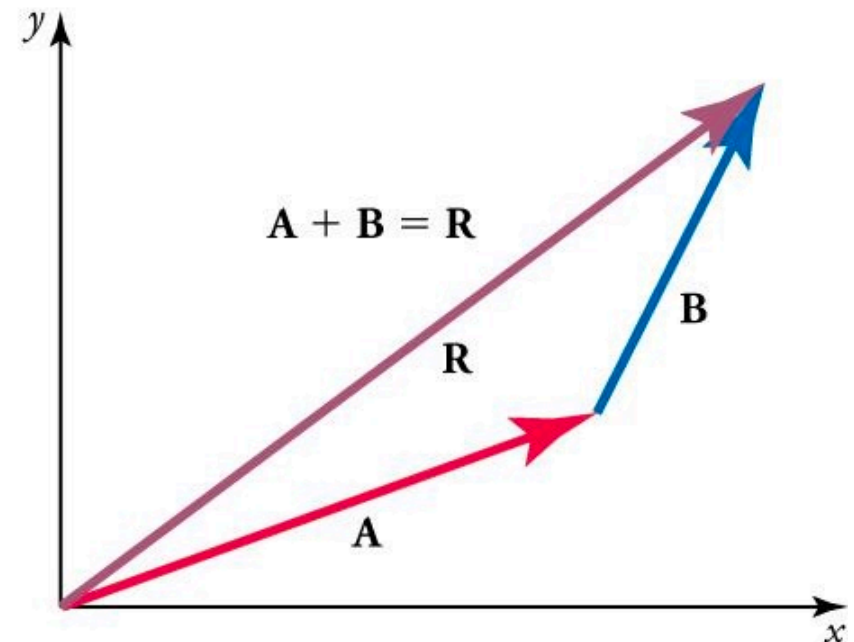
Simple

Add up the vectors of individual words

How to combine?

- Unweighted average

- Weight by inverse word frequency
contextual/universal frequency



(Arora et al., 2017)

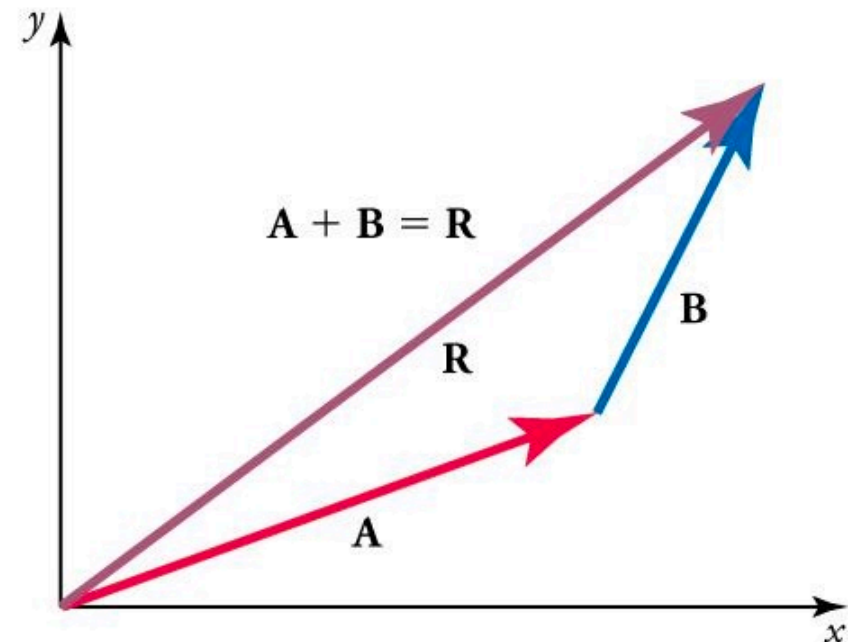
Sentence Embeddings

Simple

Add up the vectors of individual words

How to combine?

- Unweighted average



- Weight by inverse word frequency
contextual/universal frequency

- Remove first principal component
(driven by in-context topics)

(Arora et al., 2017)

Sentence Embeddings

More complex

Train embeddings for whole sentences
(BERT, GPT-2, etc.)

Sentence Embeddings

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Many more layers, and attention to whole sentence

Sentence Embeddings

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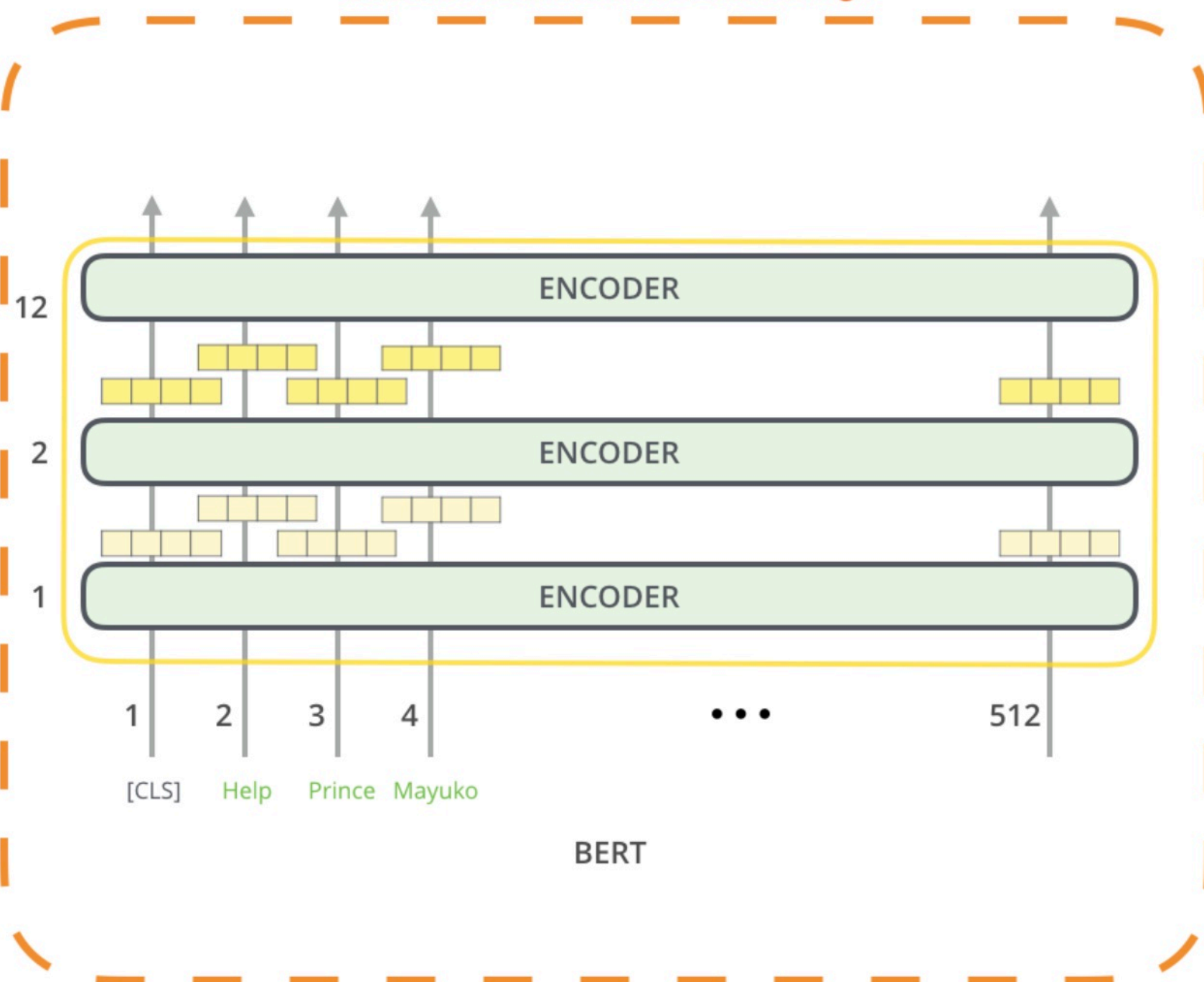
Many more layers, and attention to whole sentence

Two objectives:

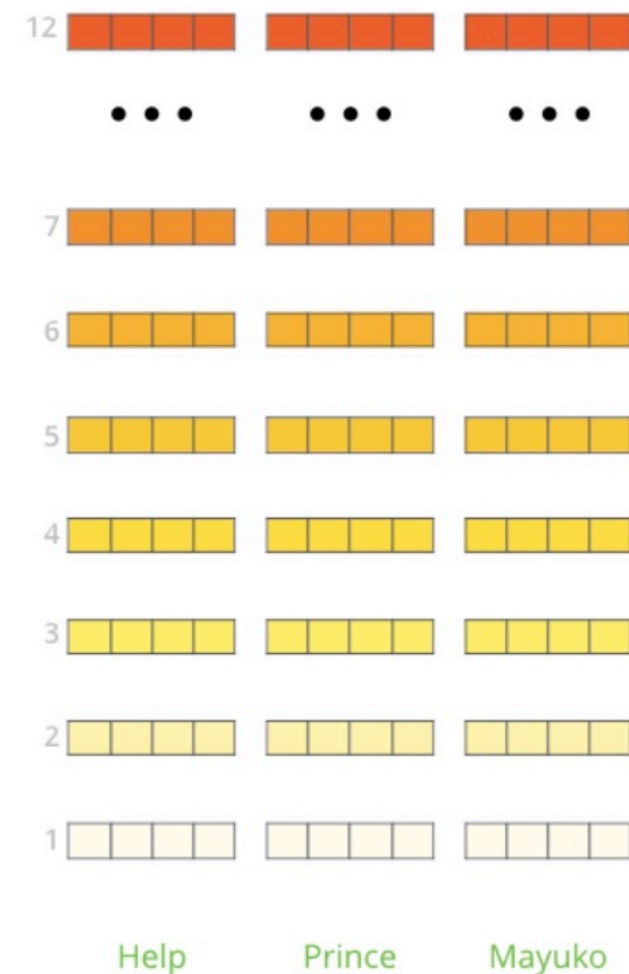
- Masked word prediction
- Two-sentence pairs: does one follow the other?

Sentence Embeddings

Generate Contextualized Embeddings



The output of each encoder layer along each token's path can be used as a feature representing that token.



But which one should we use?

Sentence Embeddings

What is the best contextualized embedding for “Help” in that context?
For named-entity recognition task CoNLL-2003 NER

Dev F1 Score

12 

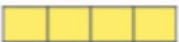
• • •

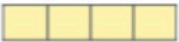
7 

6 

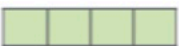
5 

4 

3 

2 

1 



Help

First Layer

Embedding 

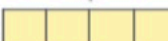
91.0

Last Hidden Layer

12 

94.9

Sum All 12
Layers

12 
+
• • •
+
2 
+
1 
=


95.5

Second-to-Last
Hidden Layer

11 

95.6

Sum Last Four
Hidden

12 
+
11 
+
10 
+
9 
=


95.9

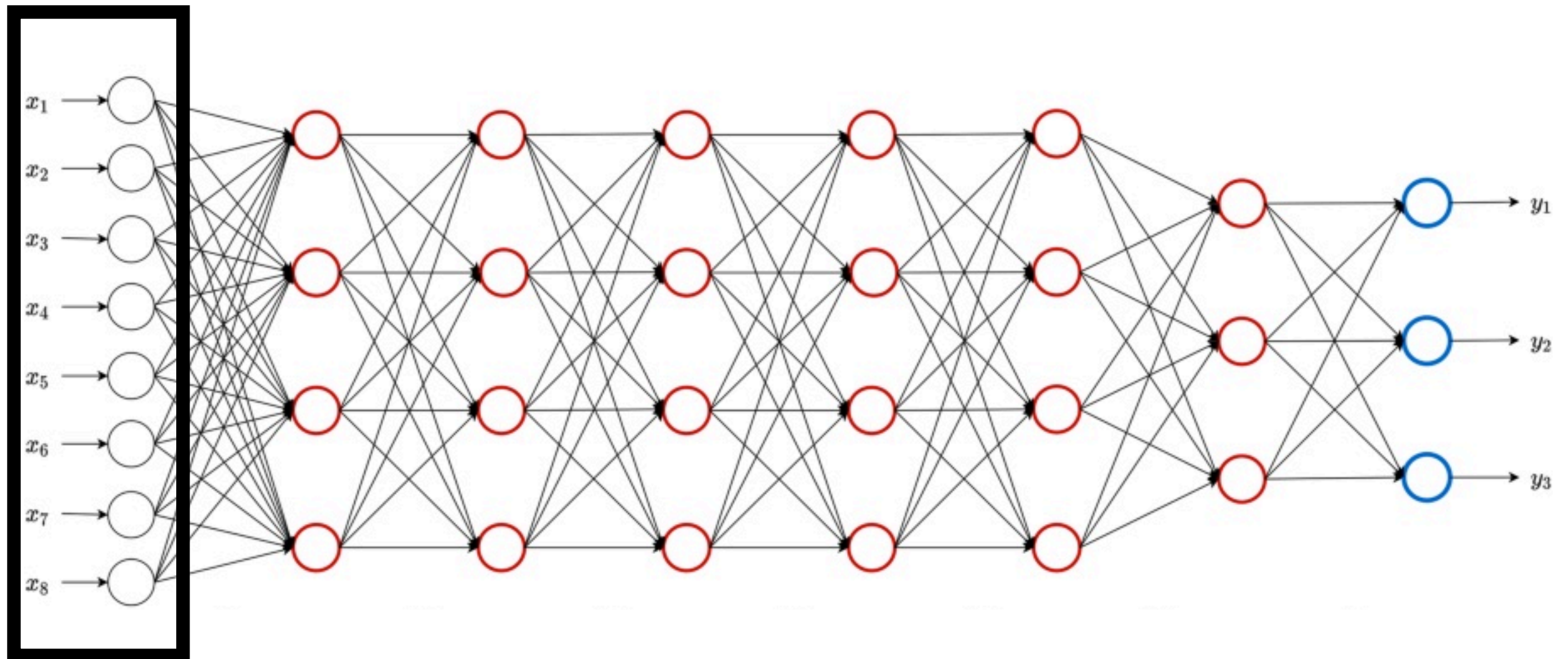
Concat Last
Four Hidden

9 10 11 12


96.1

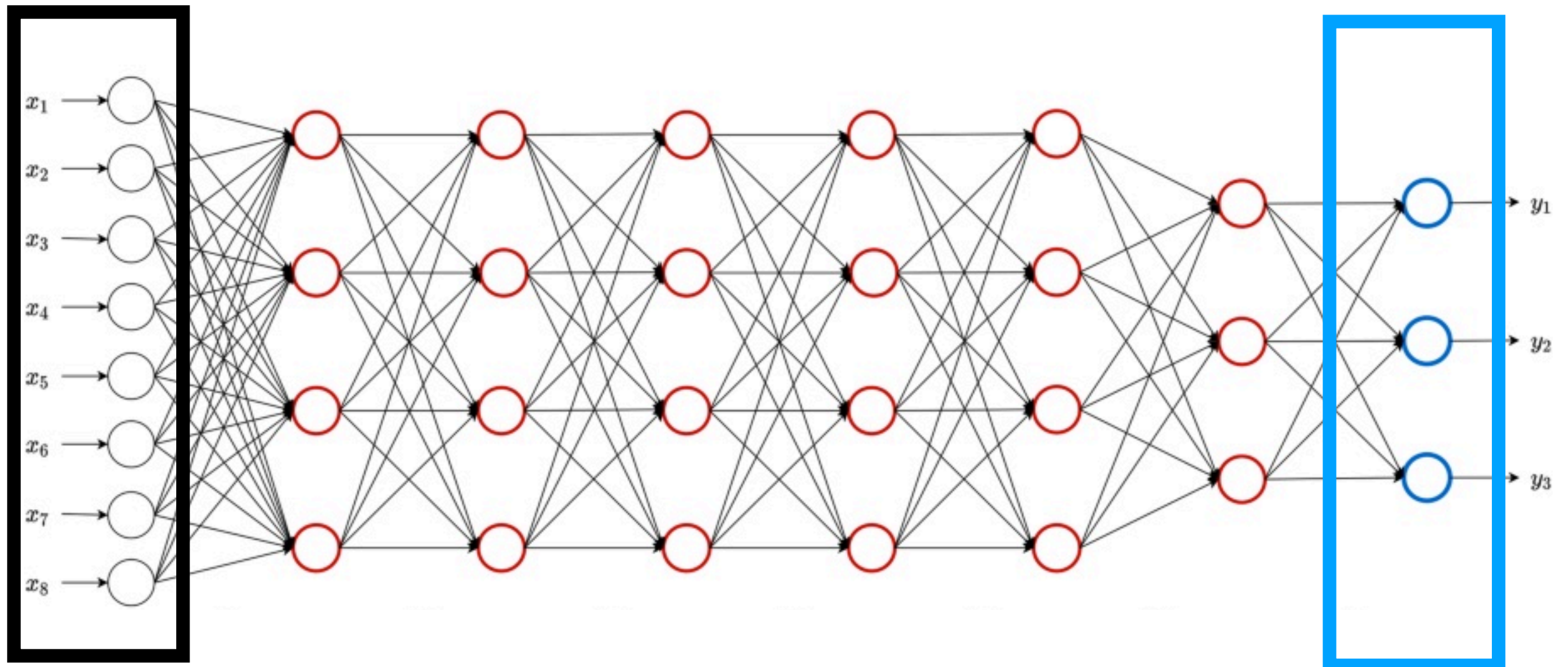
Pre-Trained Models

**Words in
documents**



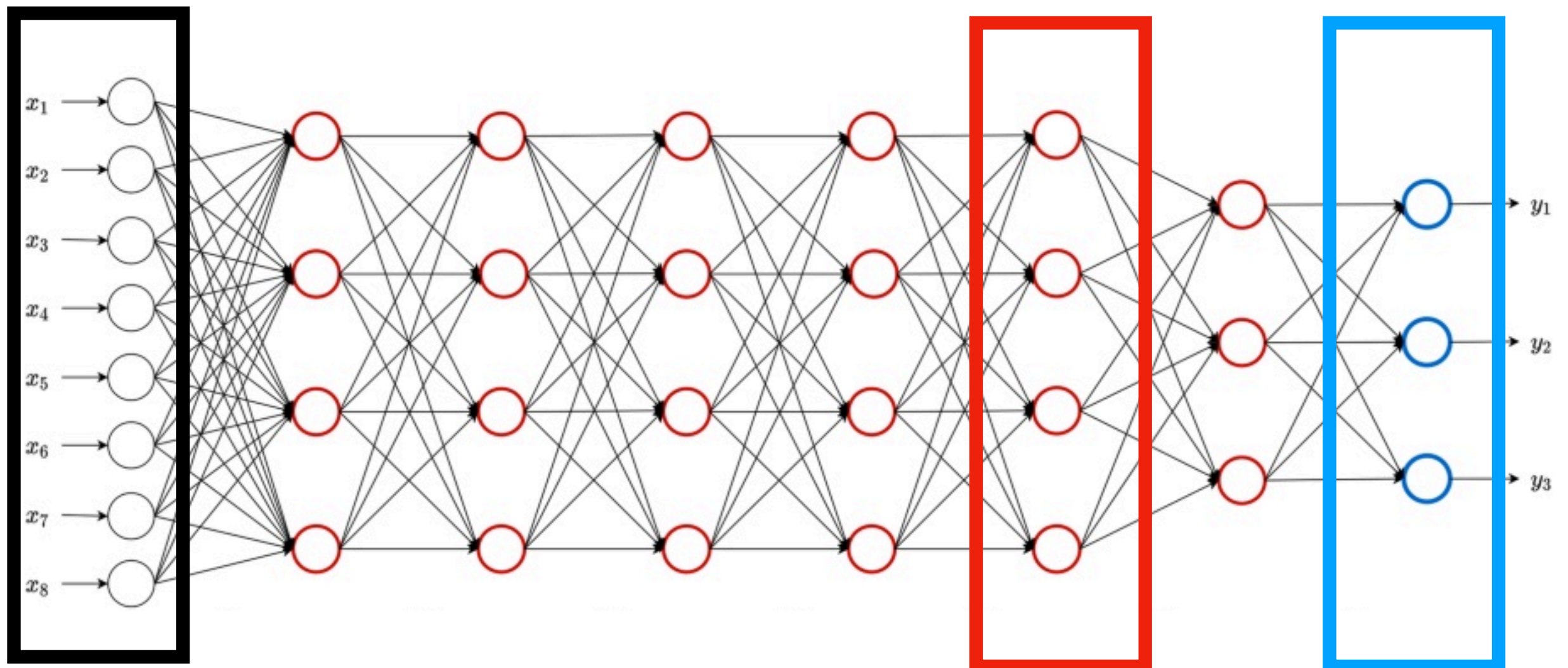
Pre-Trained Models

**Masked word
Prediction**



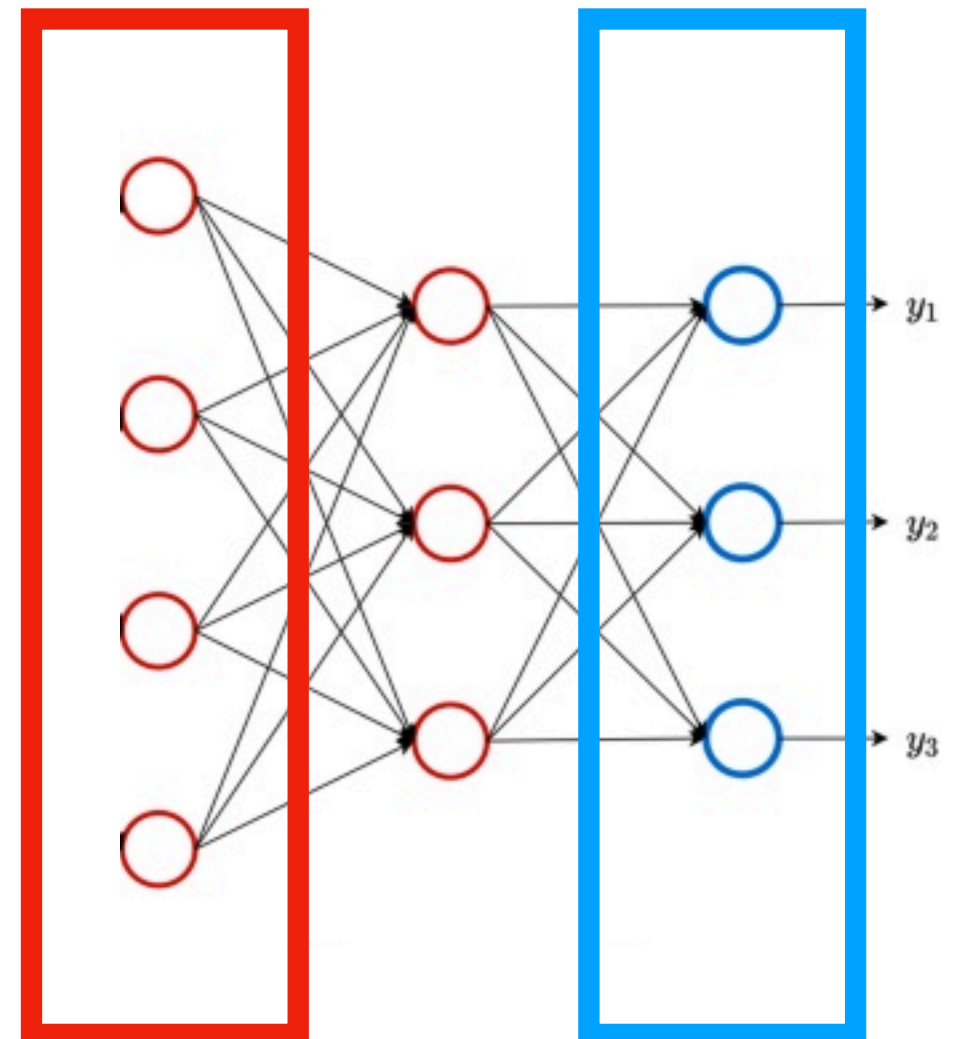
Pre-Trained Models

**The vector
representation**



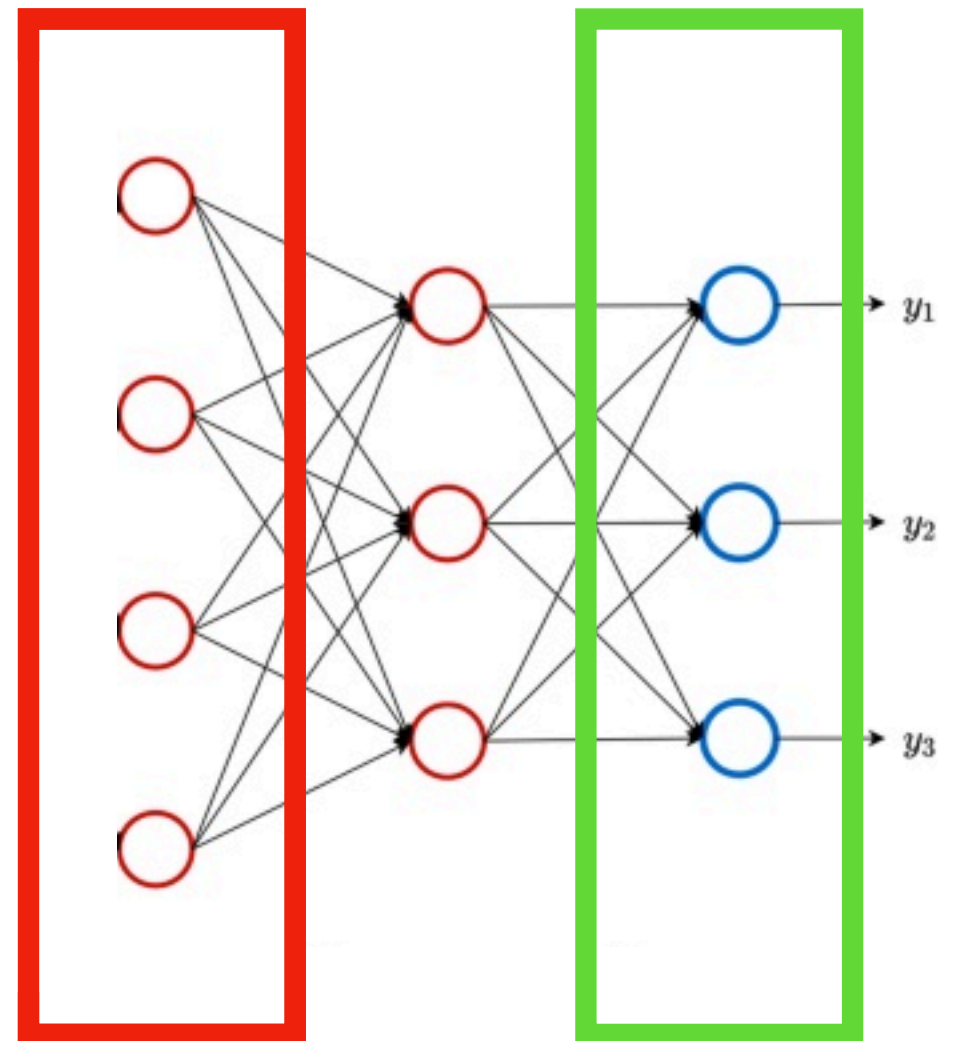
Pre-Trained Models

The vector representation

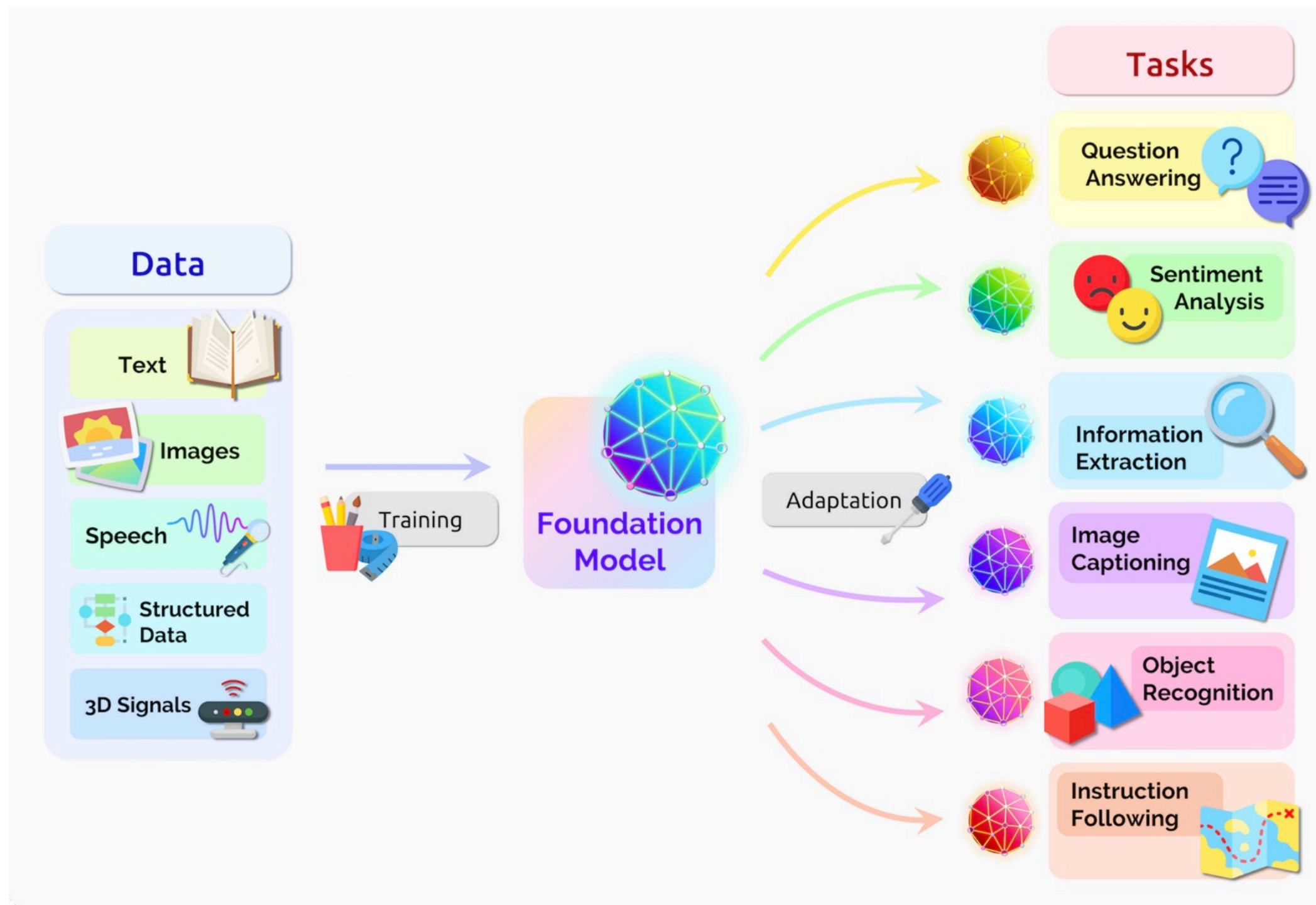


Pre-Trained Models

**New quantity
of interest**

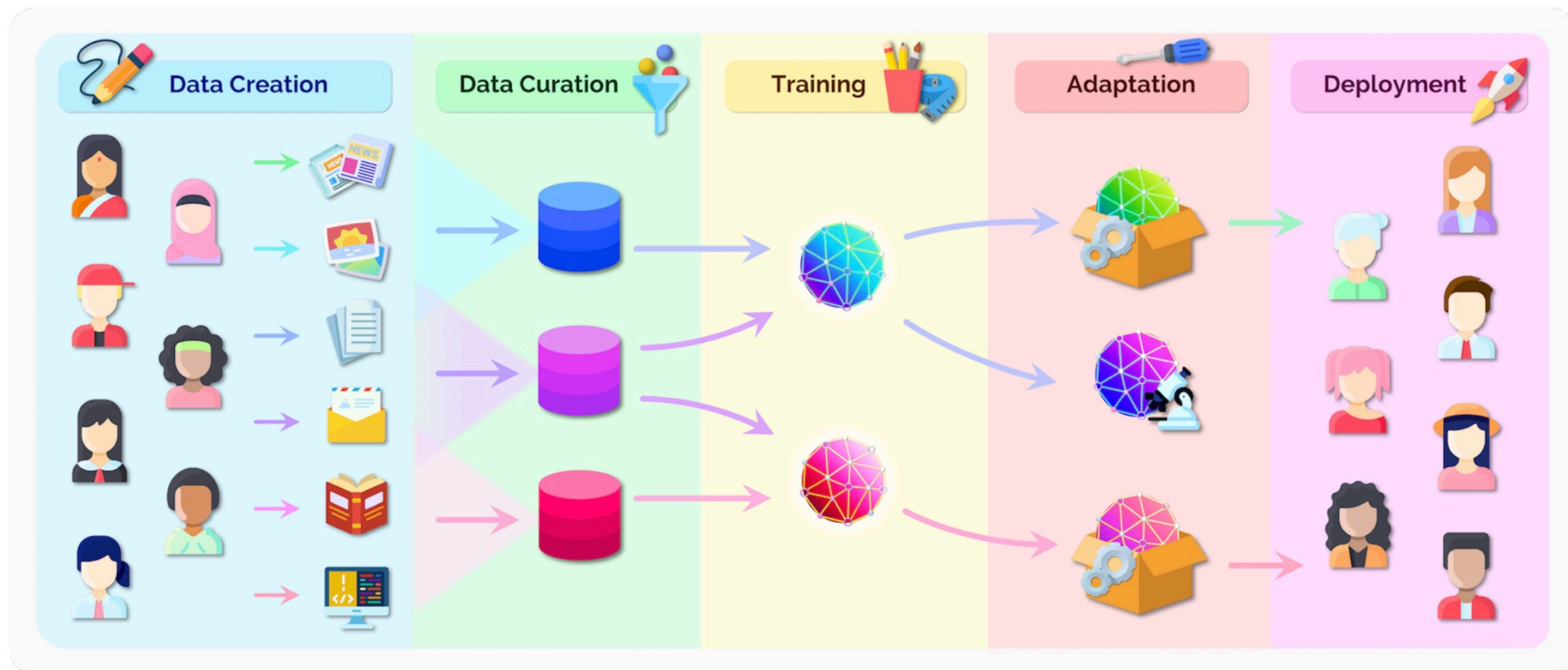


Pre-Trained Models



(Bommasani et al., 2021)

Pre-Trained Models



(Bommasani et al., 2021)

Next Steps in Meaning

Next Steps in Meaning

Benchmark Tasks

Next Steps in Meaning

Benchmark Tasks

Natural Language Inference

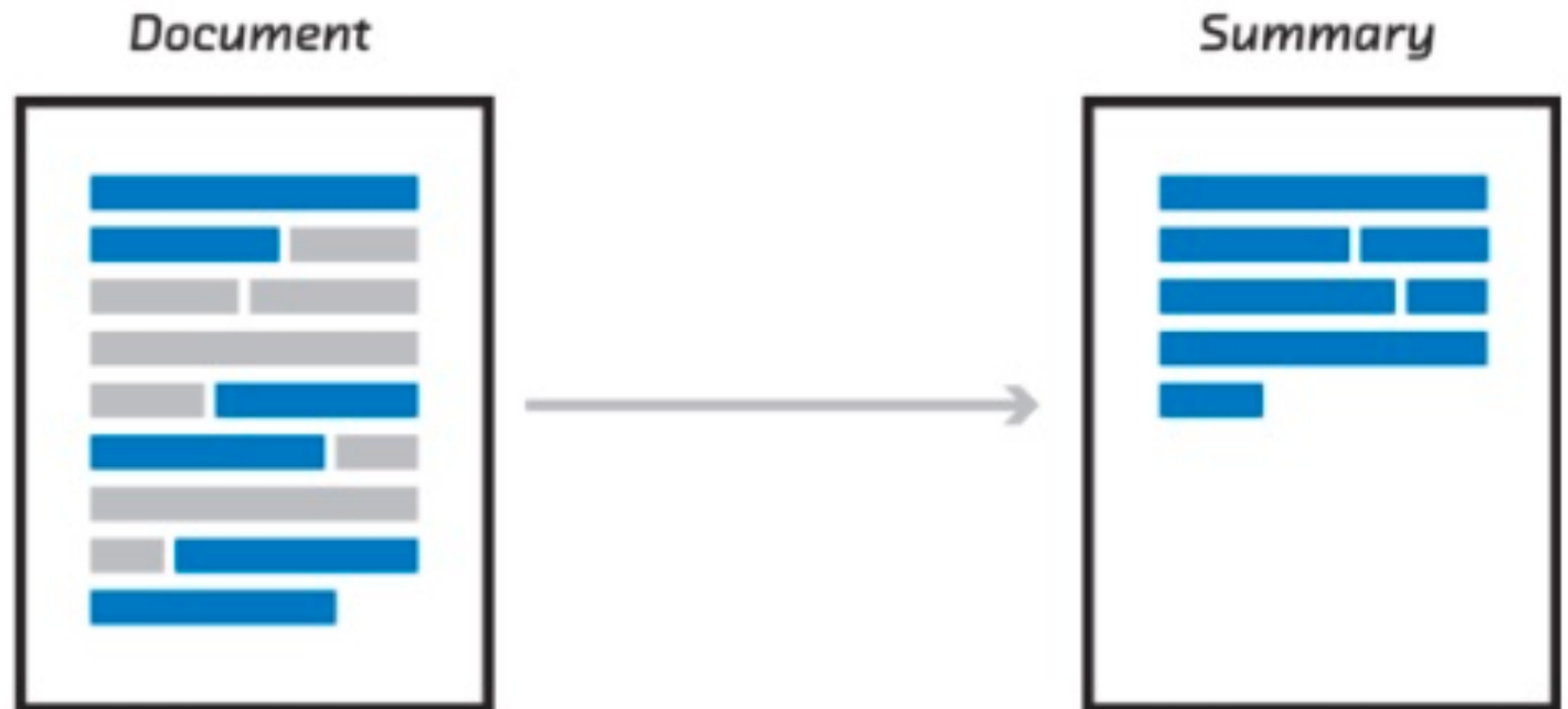
P^a	A senior is waiting at the window of a restaurant that serves sandwiches.	Relationship
H^b	A person waits to be served his food.	Entailment
	A man is looking to order a grilled cheese sandwich.	Neutral
	A man is waiting in line for the bus.	Contradiction
^a P, Premise. ^b H, Hypothesis.		

Next Steps in Meaning

Benchmark Tasks

Natural Language Inference

Text Summarization



Next Steps in Meaning

Benchmark Tasks

Natural Language Inference

Text Summarization

Co-reference Resolution

1 A horse walks into a bar . 0 The shocked bartender points a finger 0 his way in alarm and yells , “ Hey ! ”

1 The horse says , “ 0 You read 1 my mind , buddy . ”

Next Steps in Meaning

Benchmark Tasks

Natural Language Inference

Text Summarization

Co-reference Resolution

Question Answering

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?
graupel

Where do water droplets collide with ice crystals to form precipitation?
within a cloud

Next Steps in Meaning

Benchmark Tasks

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average -
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.8	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	87.4	91.3	45.4	80.0	82.3	56.0	75.1
BERT _{BASE}	84.6/83.4	71.2	90.5	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	92.7	94.9	60.5	86.5	89.3	70.1	82.1

Table 1: GLUE Test results, scored by the evaluation server (<https://gluebenchmark.com/leaderboard>). The number below each task denotes the number of training examples. The “Average” column is slightly different than the official GLUE score, since we exclude the problematic WNLI set.⁸ BERT and OpenAI GPT are single-model, single task. F1 scores are reported for QQP and MRPC, Spearman correlations are reported for STS-B, and accuracy scores are reported for the other tasks. We exclude entries that use BERT as one of their components.

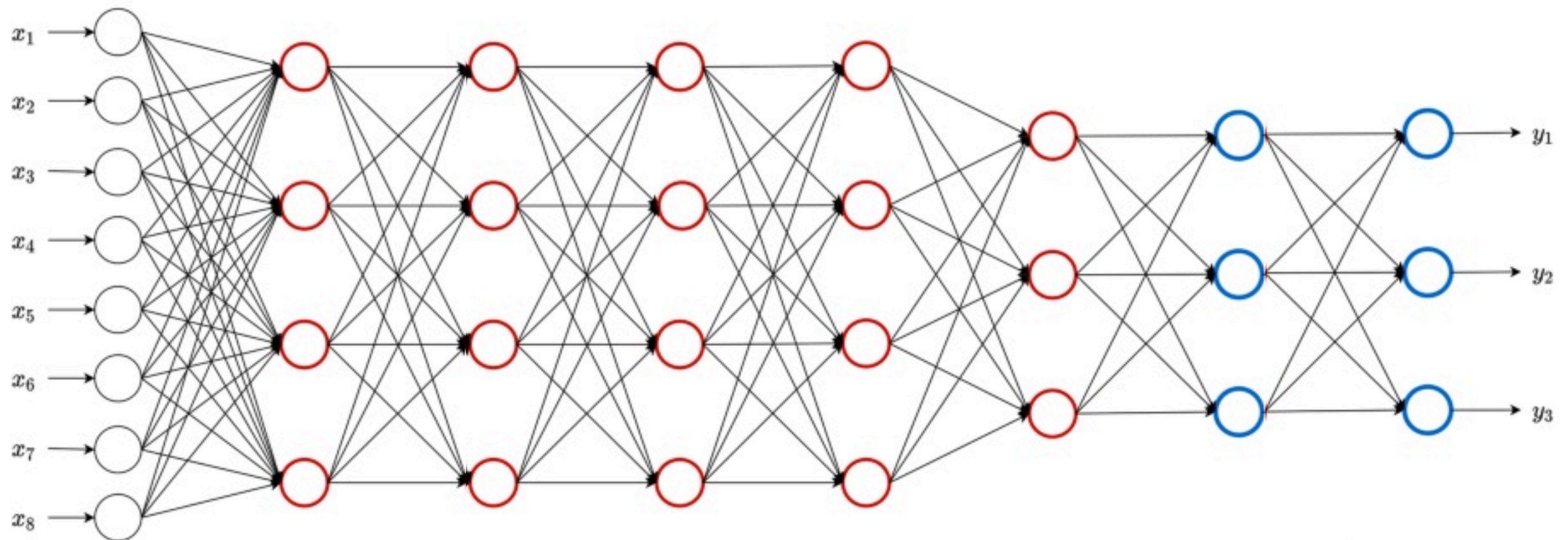
Next Steps in Meaning

Benchmark Tasks

	PP	DAN	RNN	LSTM (no)	LSTM (o.g.)	skip-thought	Ours
similarity (SICK)	84.9	85.96	73.13	85.45	83.41	85.8	86.03
entailment (SICK)	83.1	84.5	76.4	83.2	82.0	-	84.6
sentiment (SST)	79.4	83.4	86.5	86.6	89.2	-	82.2

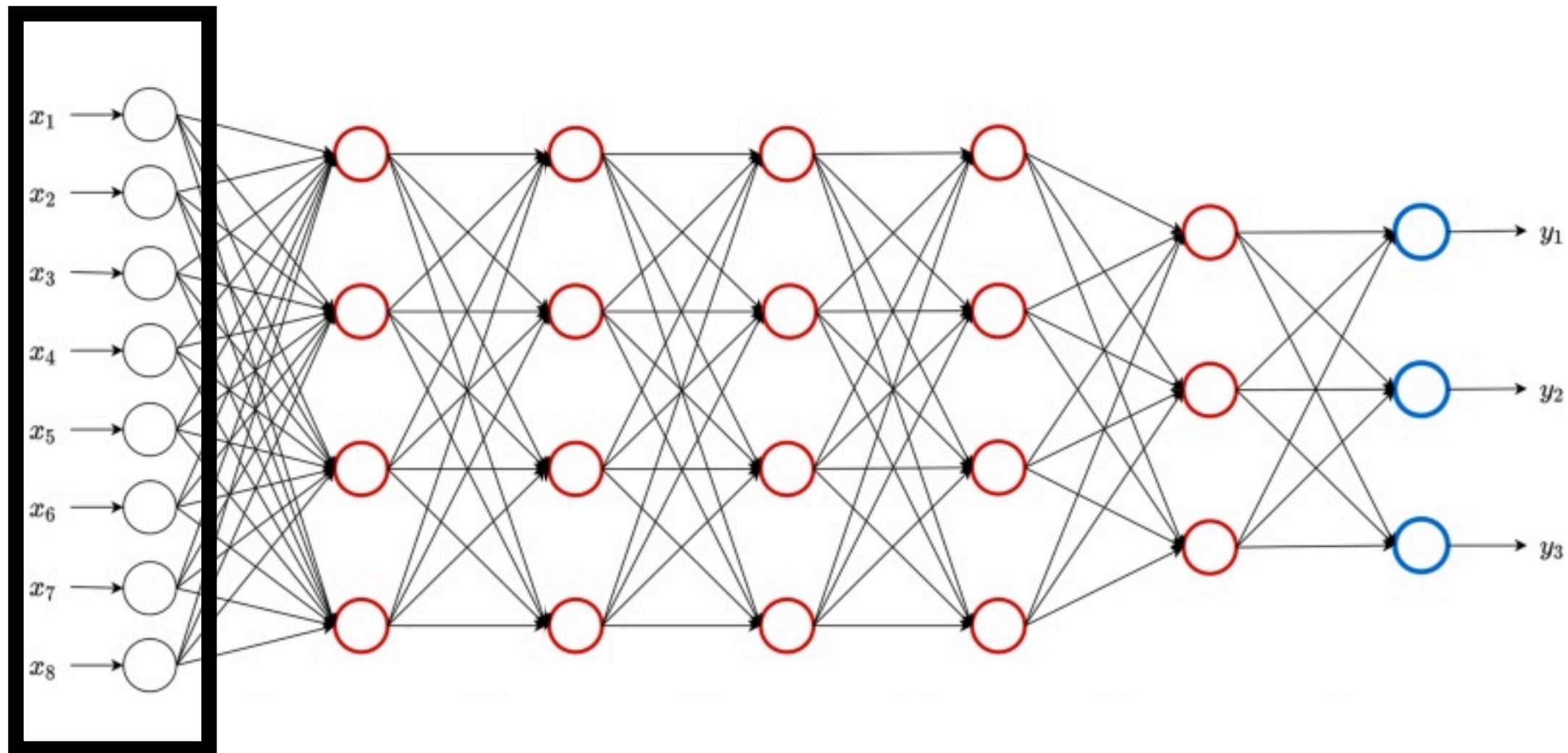
Table 2: Results on similarity, entailment, and sentiment tasks. The sentence embeddings are computed unsupervisedly, and then used as features in downstream supervised tasks. The row for similarity (SICK) shows Pearson’s $r \times 100$ and the last two rows show accuracy. The highest score in each row is in boldface. Results in Column 2 to 6 are collected from (Wieting et al., 2016), and those in Column 7 for skip-thought are from (Lei Ba et al., 2016).

Pre-Trained Models



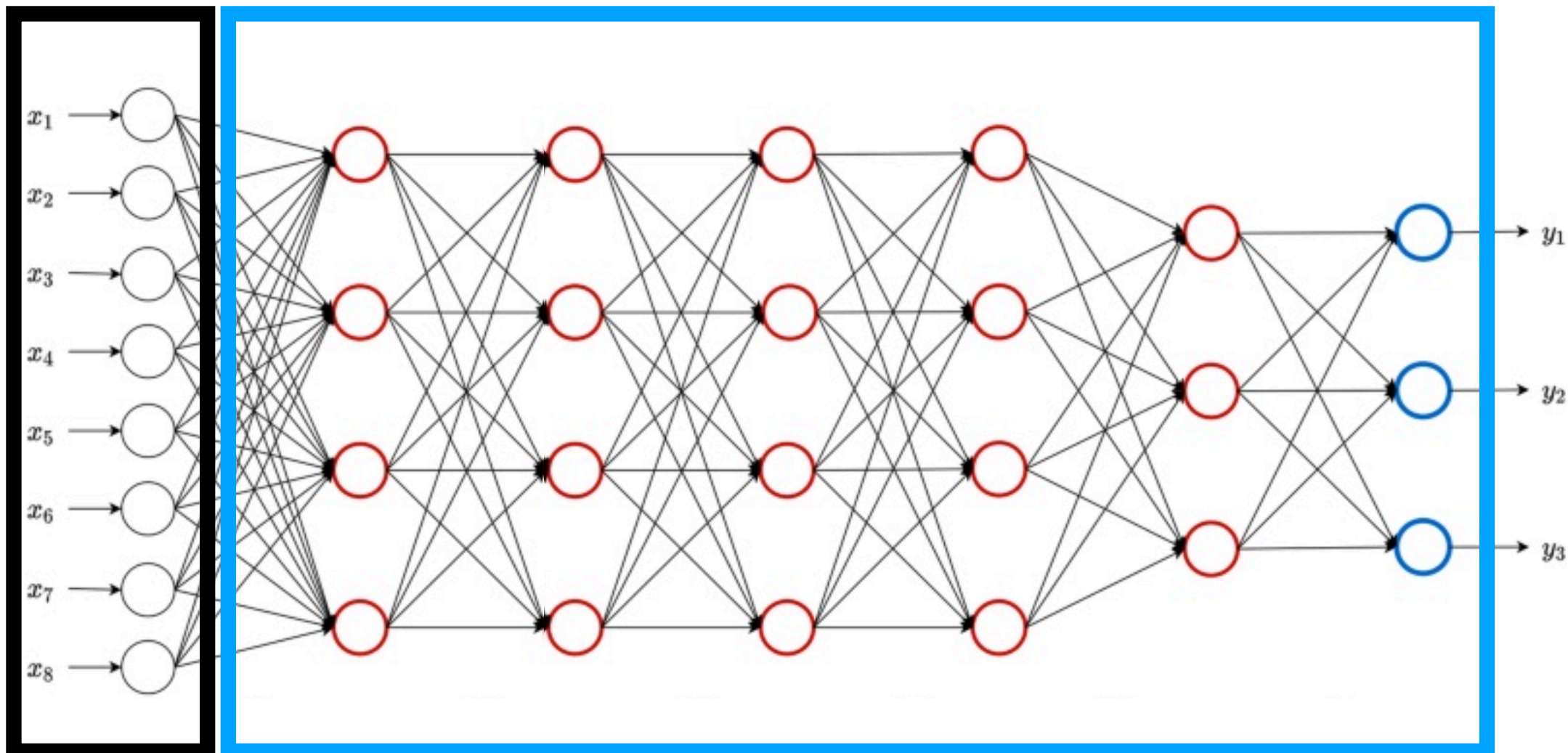
Pre-Trained Models

Words in a Document



Pre-Trained Models

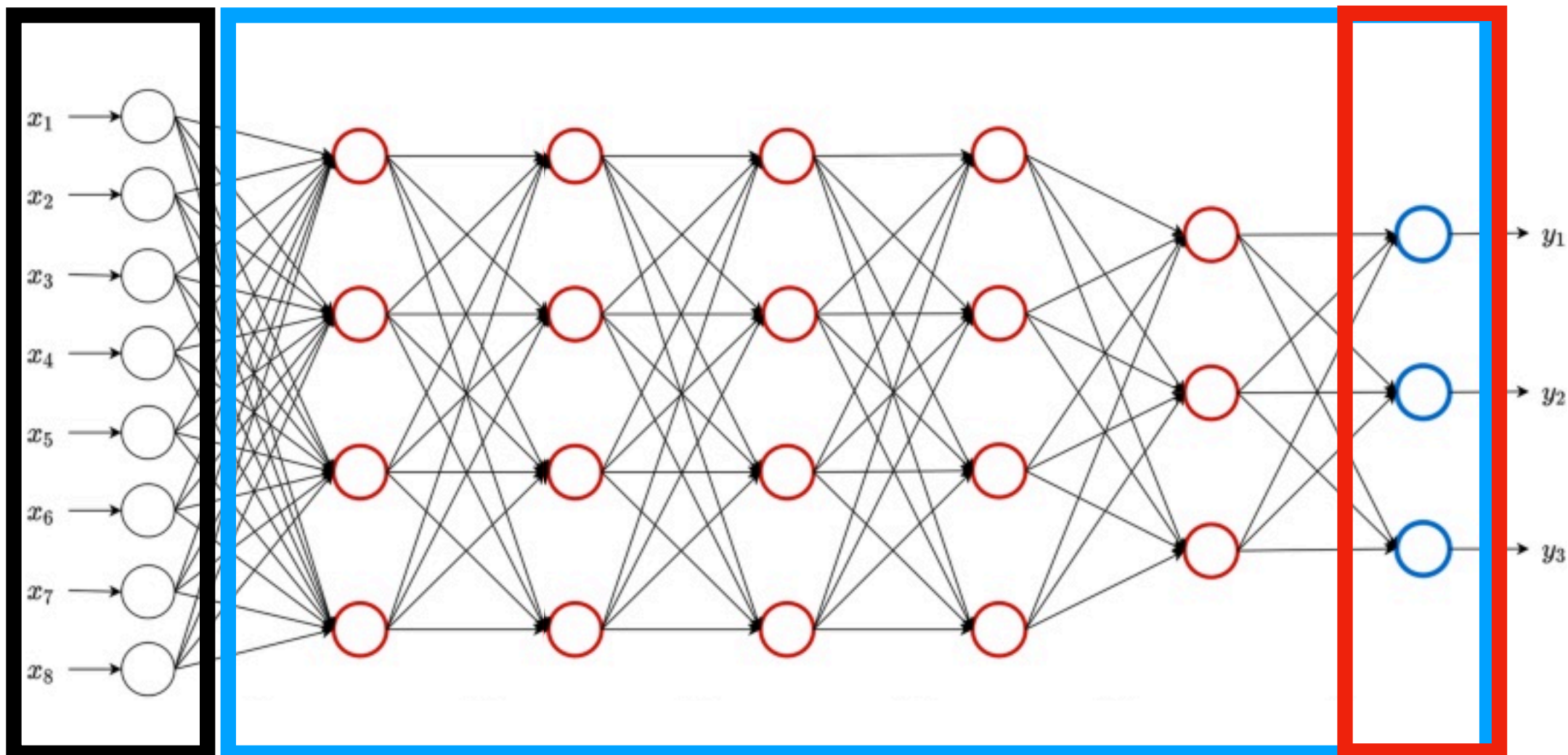
spaCy



Pre-Trained Models

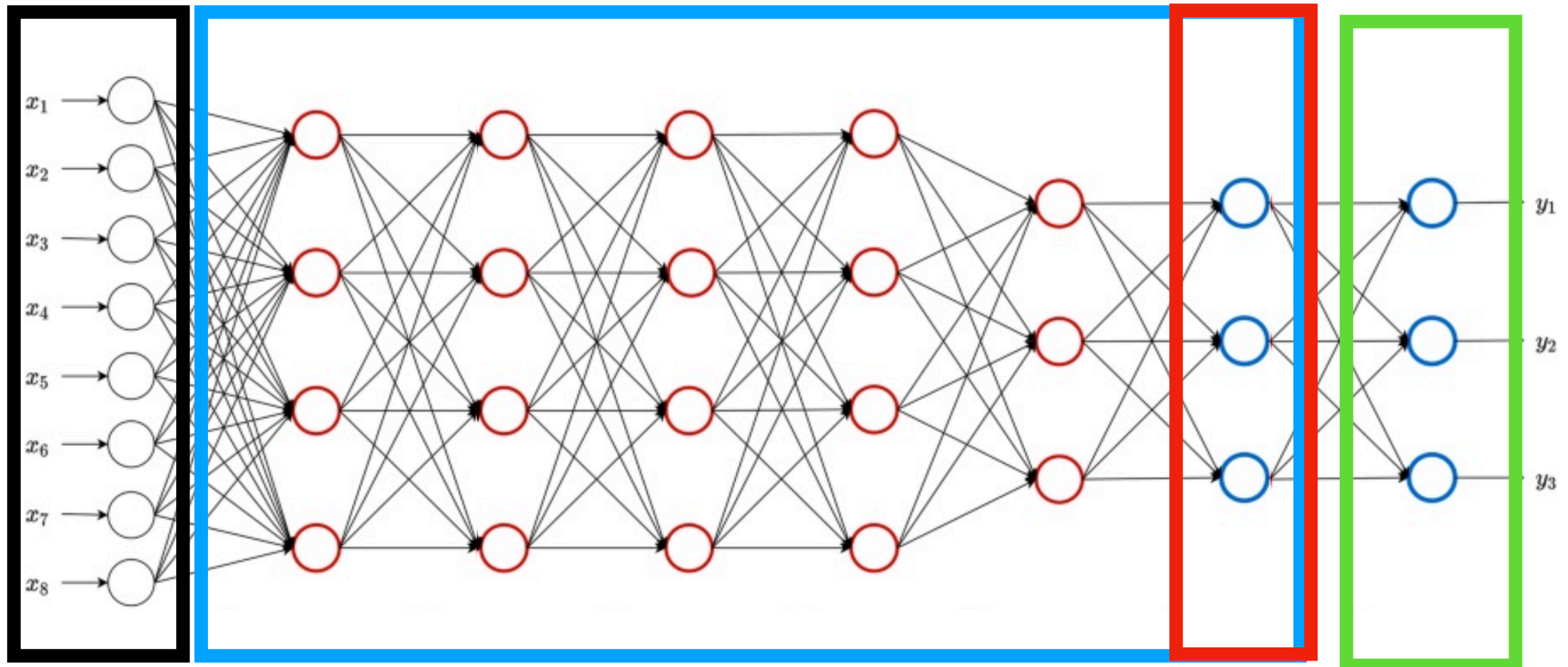
spaCy

**Grammar
Labels**



Pre-Trained Models

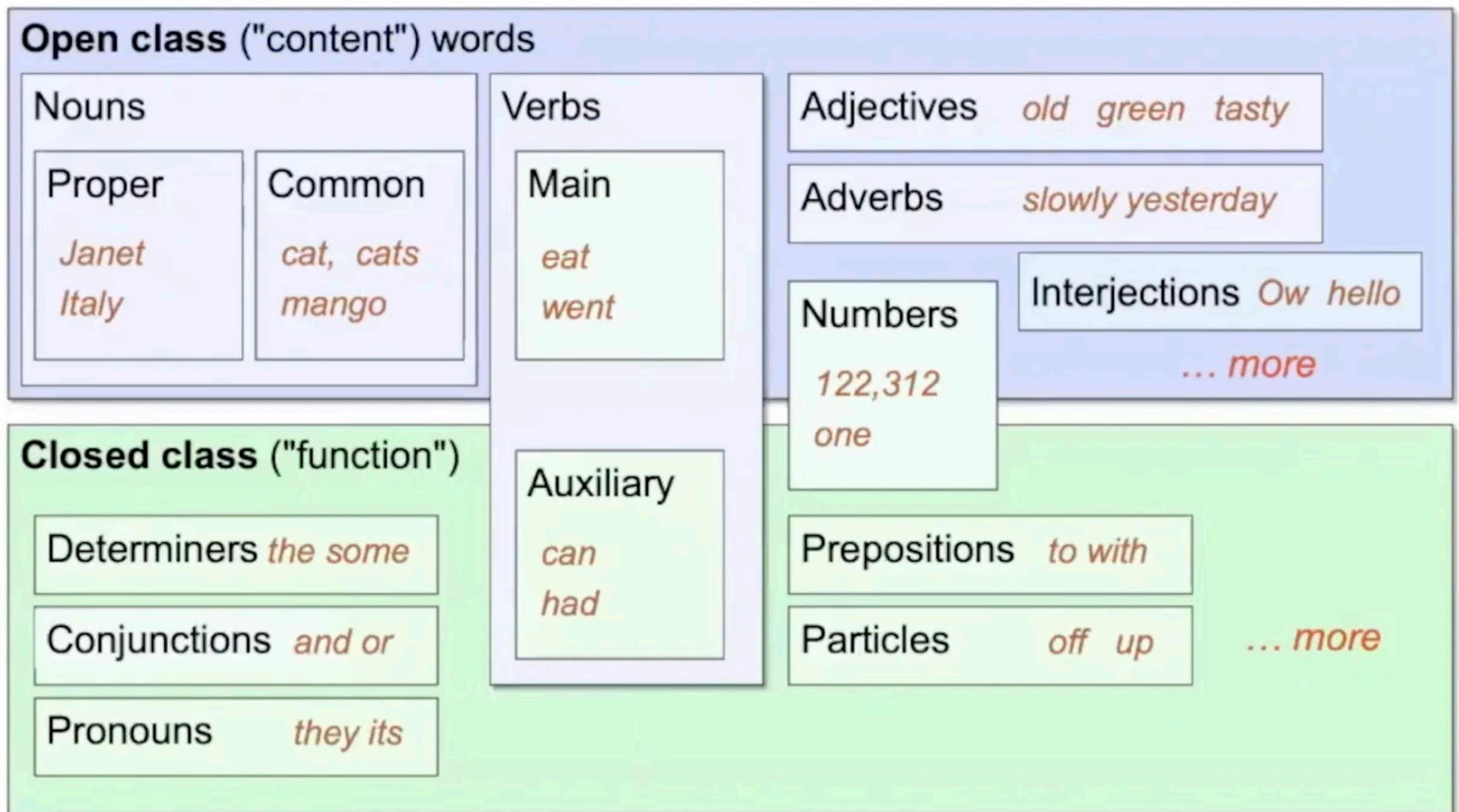
**New quantity
of interest**



Pre-Trained Models

Part of Speech Tagging

spaCy



Pre-Trained Models

Part of Speech Tagging

spaCy

Roughly 15% of word types are ambiguous

- Hence 85% of word types are unambiguous
- *Janet* is always PROP, *hesitantly* is always ADV

Pre-Trained Models

Part of Speech Tagging

spaCy

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But those 15% tend to be very common.

So ~60% of word tokens are ambiguous

Pre-Trained Models

Part of Speech Tagging

spaCy

Roughly 15% of word types are ambiguous

- Hence 85% of word types are unambiguous
- *Janet* is always PROPN, *hesitantly* is always ADV

But those 15% tend to be very common.

So ~60% of word tokens are ambiguous

E.g., *back*

earnings growth took a *back*/ADJ seat

a small building in the *back*/NOUN

a clear majority of senators *back*/VERB the bill

enable the country to buy *back*/PART debt

I was twenty-one *back*/ADV then

Pre-Trained Models

Lemmatization

spaCy

Stemming vs Lemmatization

change
changing
changes
changed
changer

Diagram illustrating stemming: The words 'change', 'changing', 'changes', 'changed', and 'changer' are listed on the left. Arrows from each word point to the word 'chang' on the right, which is colored blue. This represents the process of reducing words to their root form by removing suffixes.

change
changing
changes
changed
changer

Diagram illustrating lemmatization: The words 'change', 'changing', 'changes', 'changed', and 'changer' are listed on the left. Arrows from each word point to the word 'change' on the right, which is colored green. This represents the process of reducing words to their base or dictionary form, considering the word's context.

Pre-Trained Models

Lemmatization

spaCy

Stemming

adjustable → adjust
formality → formaliti
formaliti → formal
airliner → airlin

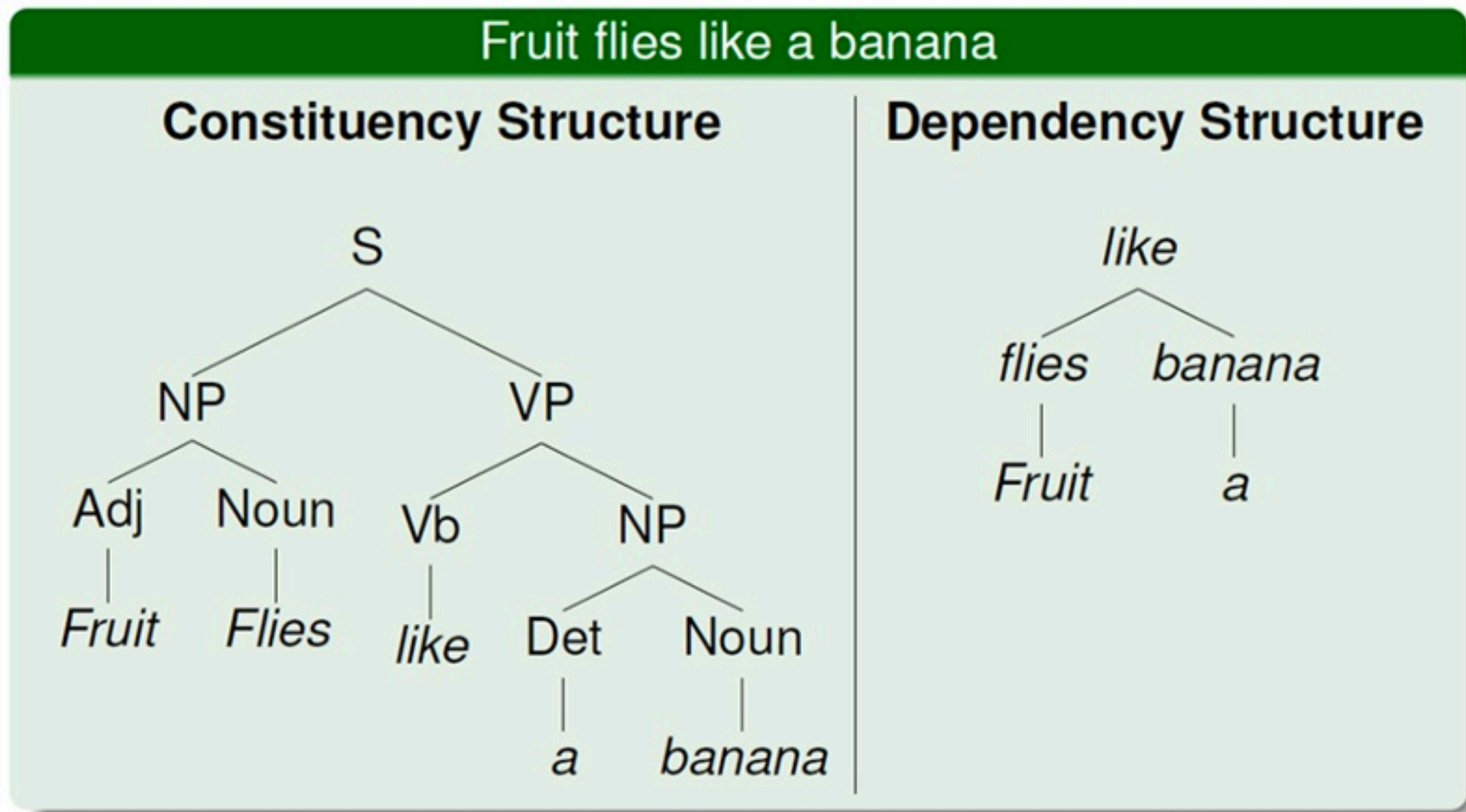
Lemmatization

was → (to) be
better → good
meeting → meeting

Pre-Trained Models

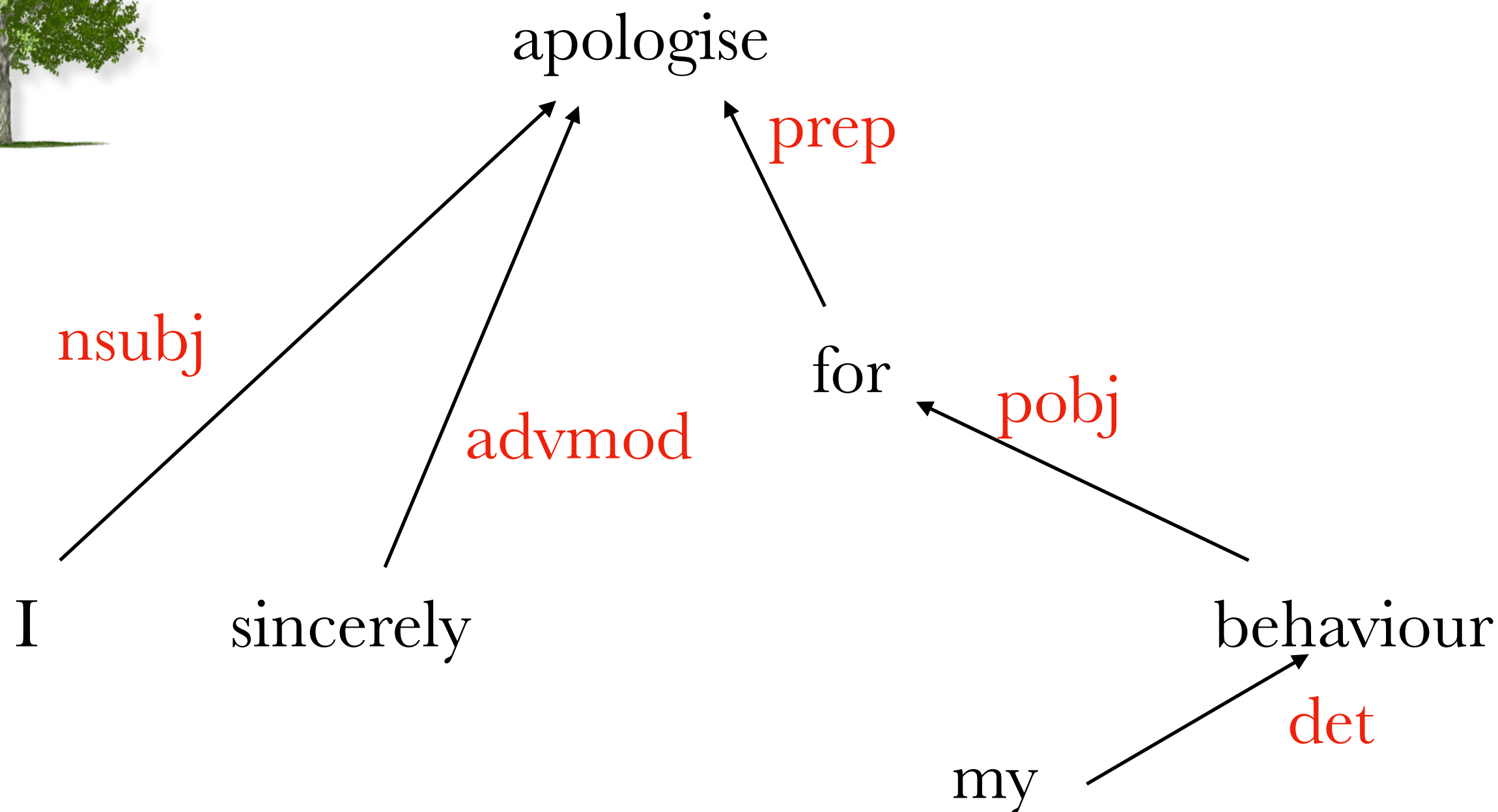
Two Types of Structure

spaCy





Dependency Trees

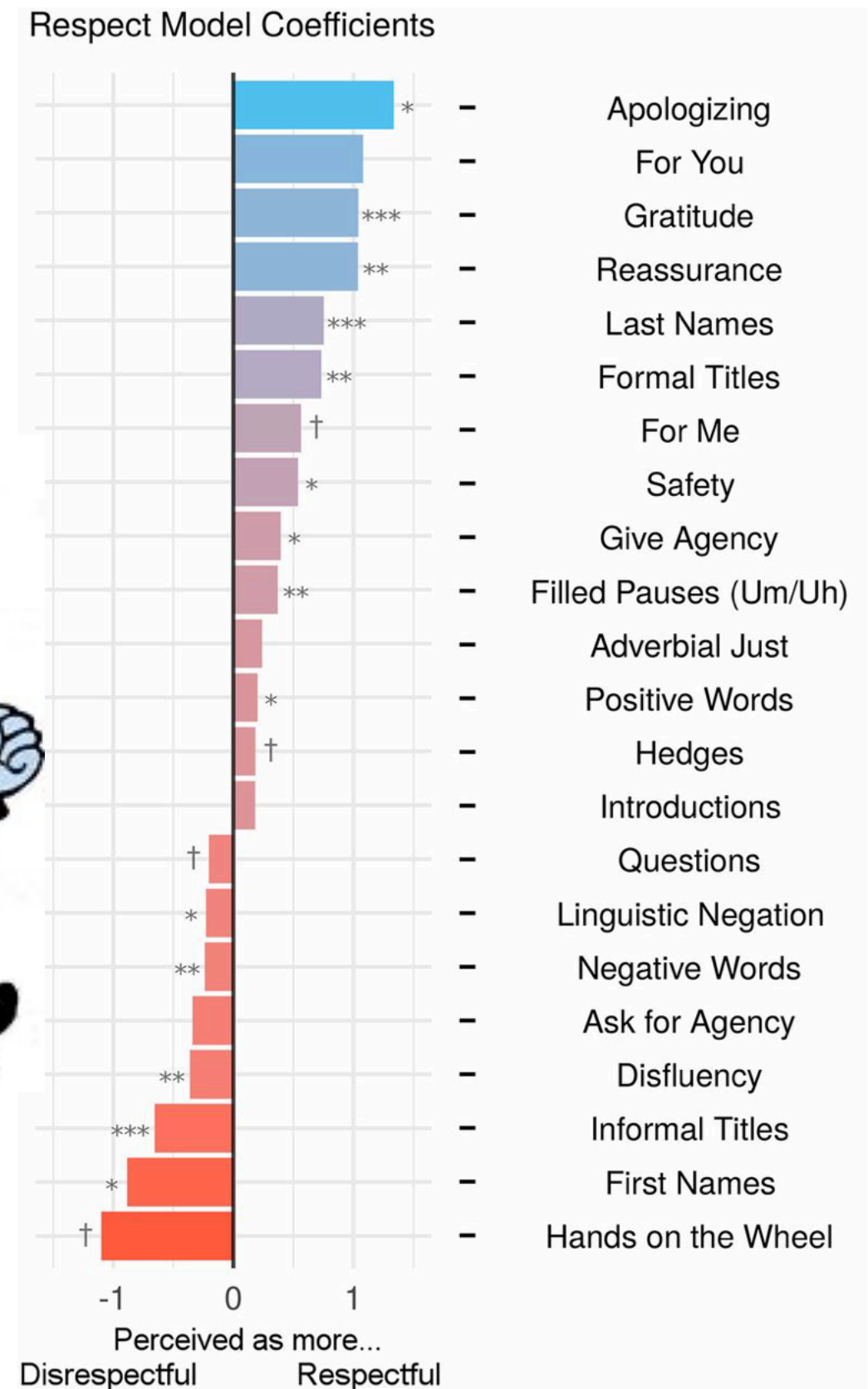


Markers of Politeness

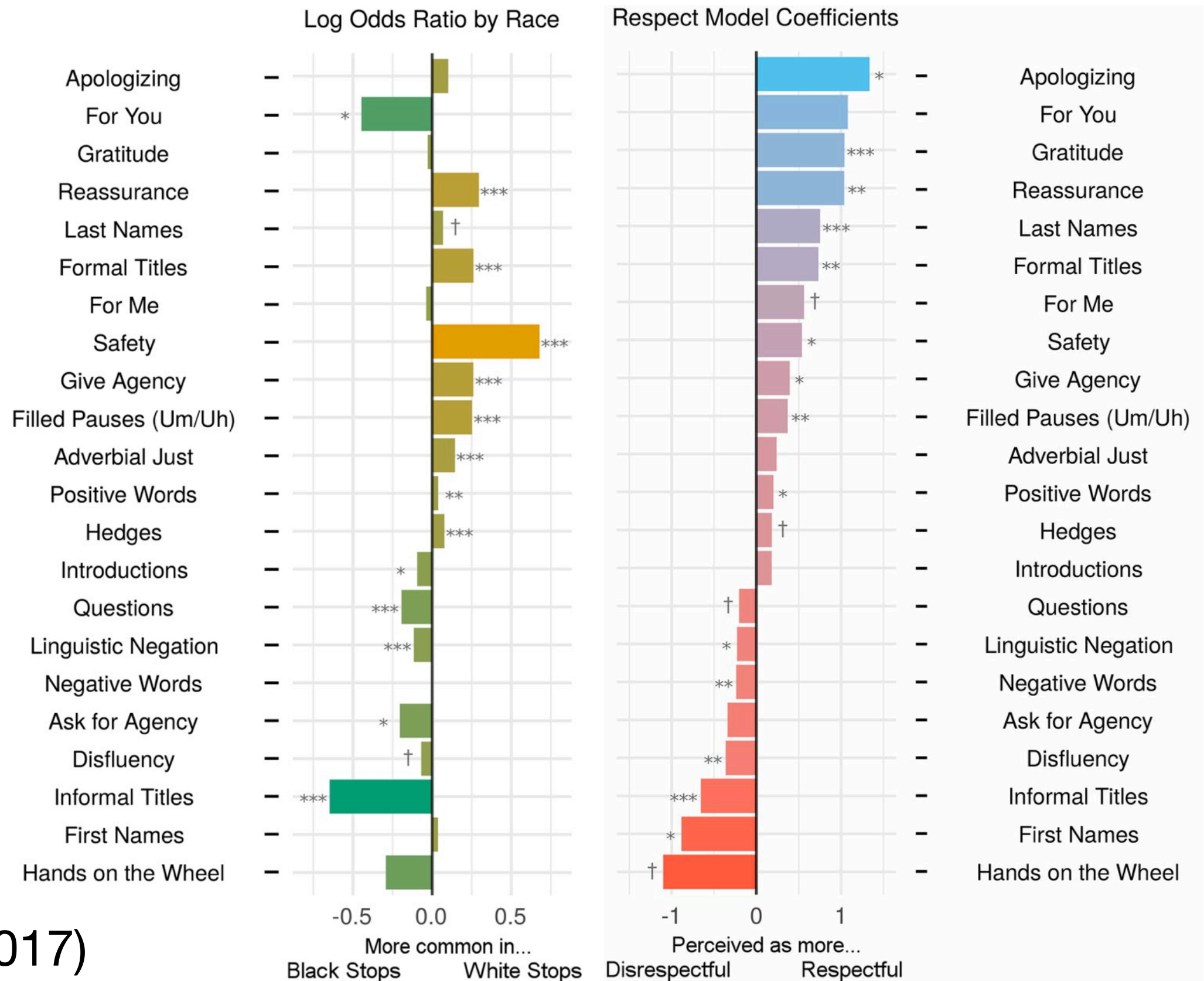
Language from Police Body Camera Footage shows Racial Disparities in Officer Respect



(Voigt, Camp, Prabakharan, Hamilton, Hetey, Griffiths, Jurgens, Jurafsky & Eberhart, 2017)



Markers of Politeness



(Voigt et al., 2017)

Markers of Politeness

EXAMPLE	RESPECT SCORE
FIRST NAME ASK FOR AGENCY QUESTIONS [name], can I see that driver's license again? It- it's showing suspended. Is that- that's you? ↑ ↑ ↑ DISFLUENCY NEGATIVE WORD DISFLUENCY	-1.07
INFORMAL TITLE ASK FOR AGENCY ADVERBIAL "JUST" All right, my man. Do me a favor. Just keep your hands on the steering wheel real quick. ↑ "HANDS ON THE WHEEL"	-0.51

(Voigt et al., 2017)

Markers of Politeness

APOLOGY	INTRODUCTION	LAST NAME	
↓	↓	↓	
Sorry to stop you. My name's Officer [name] with the Police Department.			0.84

FORMAL TITLE	SAFETY	PLEASE	
↓	↓	↓	
There you go, ma'am. Drive safe, please.			1.21

ADVERBIAL "JUST"	FILLED PAUSE	REASSURANCE	
↓	↓	↓	
It just says that, uh, you've fixed it. No problem. Thank you very much, sir.			2.07
	↑	↑	
GRATITUDE	FORMAL TITLE		

(Voigt et al., 2017)

Pre-Trained Models

Named Entity Recognition

spaCy

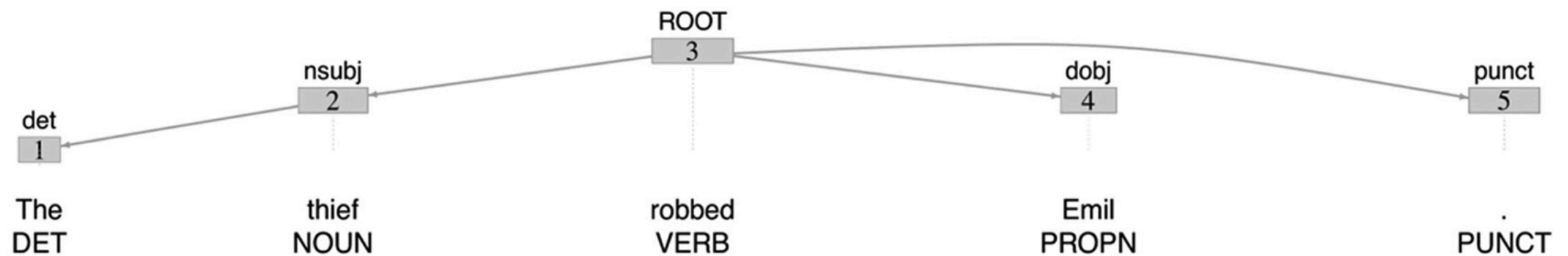
Barack Obama Person the 44th President of the United States Title, was born in Honolulu, Hawaii Location. He graduated from Columbia University Org and Harvard Law School Org. In 2009 Date, Obama was elected as the first African American Ethnicity President of the United States Location. During his presidency, Obama implemented the Affordable Care Act Law and strengthened diplomatic relations with Cuba Location. He served two terms in office before being succeeded by President Donald Trump Title in 2017 Date.

Pre-Trained Models

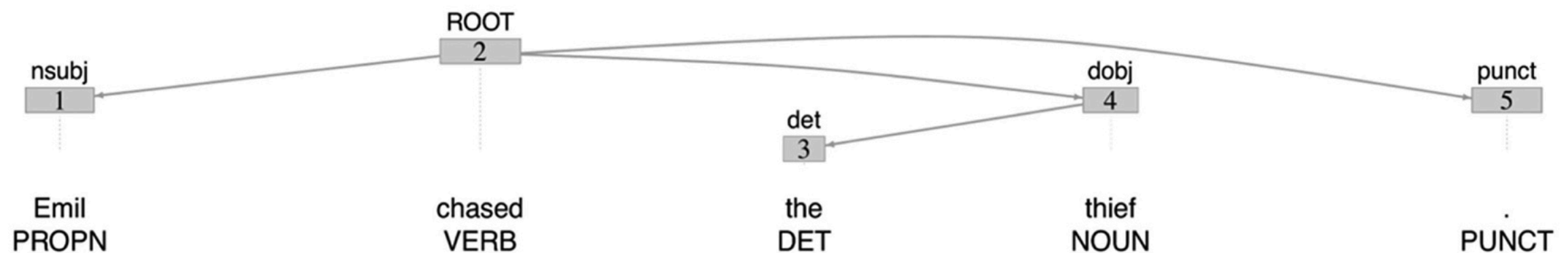
Knowledge Extraction

spaCy

A



B



Stuhler, 2022

Pre-Trained Models



Knowledge Extraction

actions of an entity

treatments of an entity

agents acting upon an entity

patients acted upon by an entity

characterizations of an entity

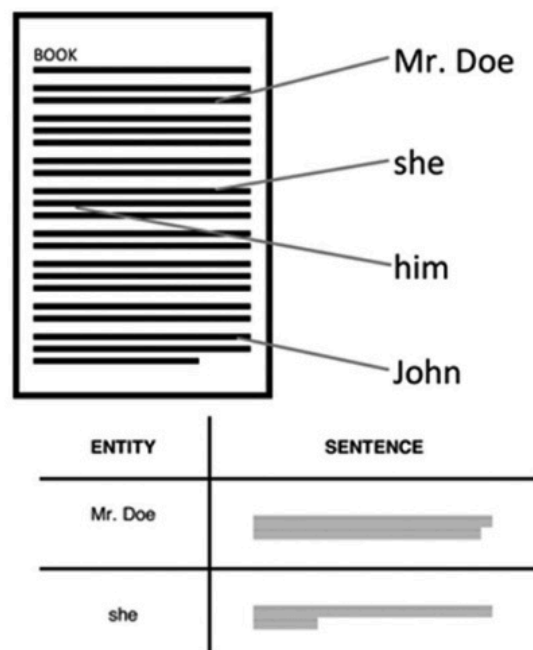
possessions of an entity

Pre-Trained Models

Knowledge Extraction

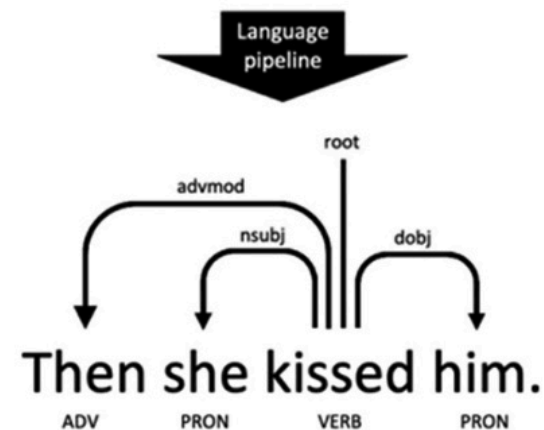
spaCy

1. Identify gendered entities

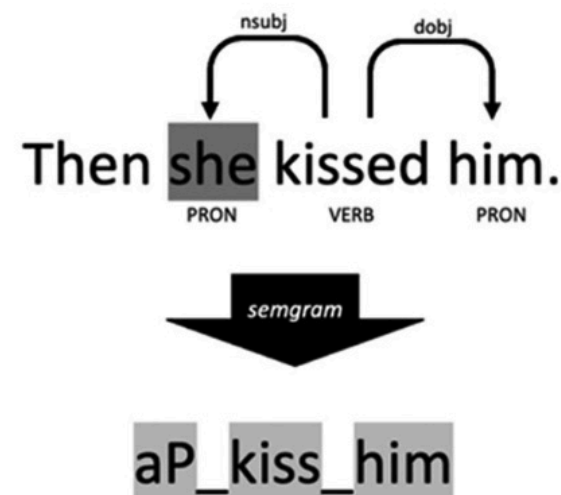


2. Annotate syntactic relations and POS tags

Then she kissed him.



3. Extract semantic motifs



4. Represent gendered interaction in a book

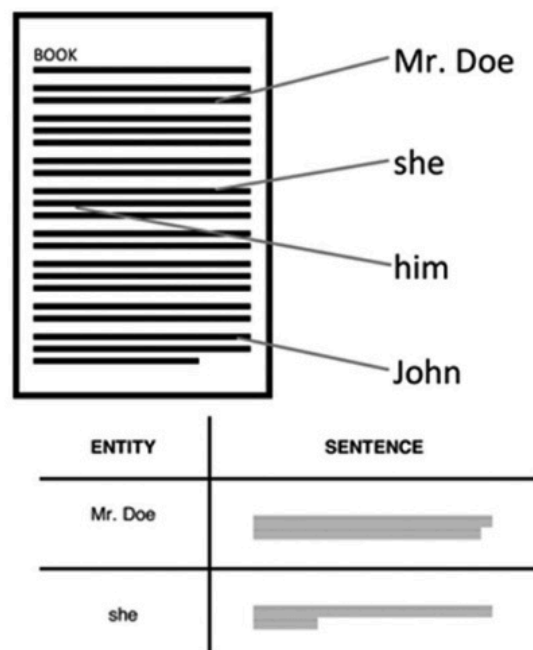
BOOK	kiss	motif_2	motif_3
FEMALE → MALE	1	0	...
FEMALE → FEMALE	0	2	...
MALE → FEMALE	0	0	...
MALE → MALE	0	1	...

Pre-Trained Models

spaCy

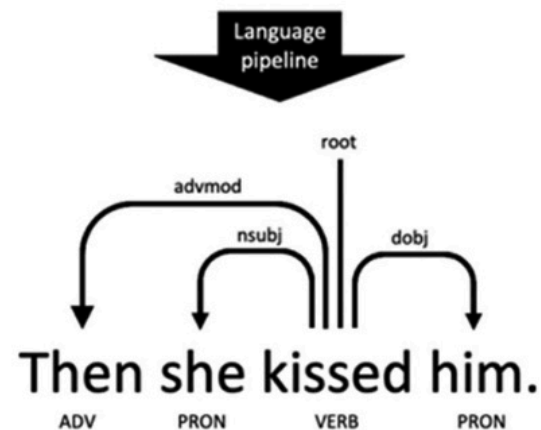
Knowledge Extraction

1. Identify gendered entities

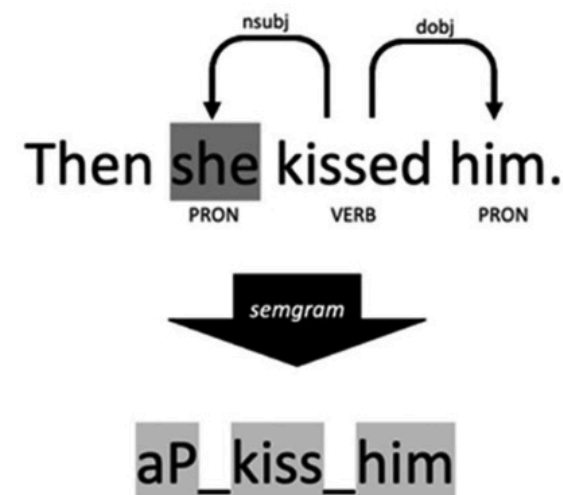


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BOOK			
	kiss	motif_2	motif_3
FEMALE → MALE	1	0	...
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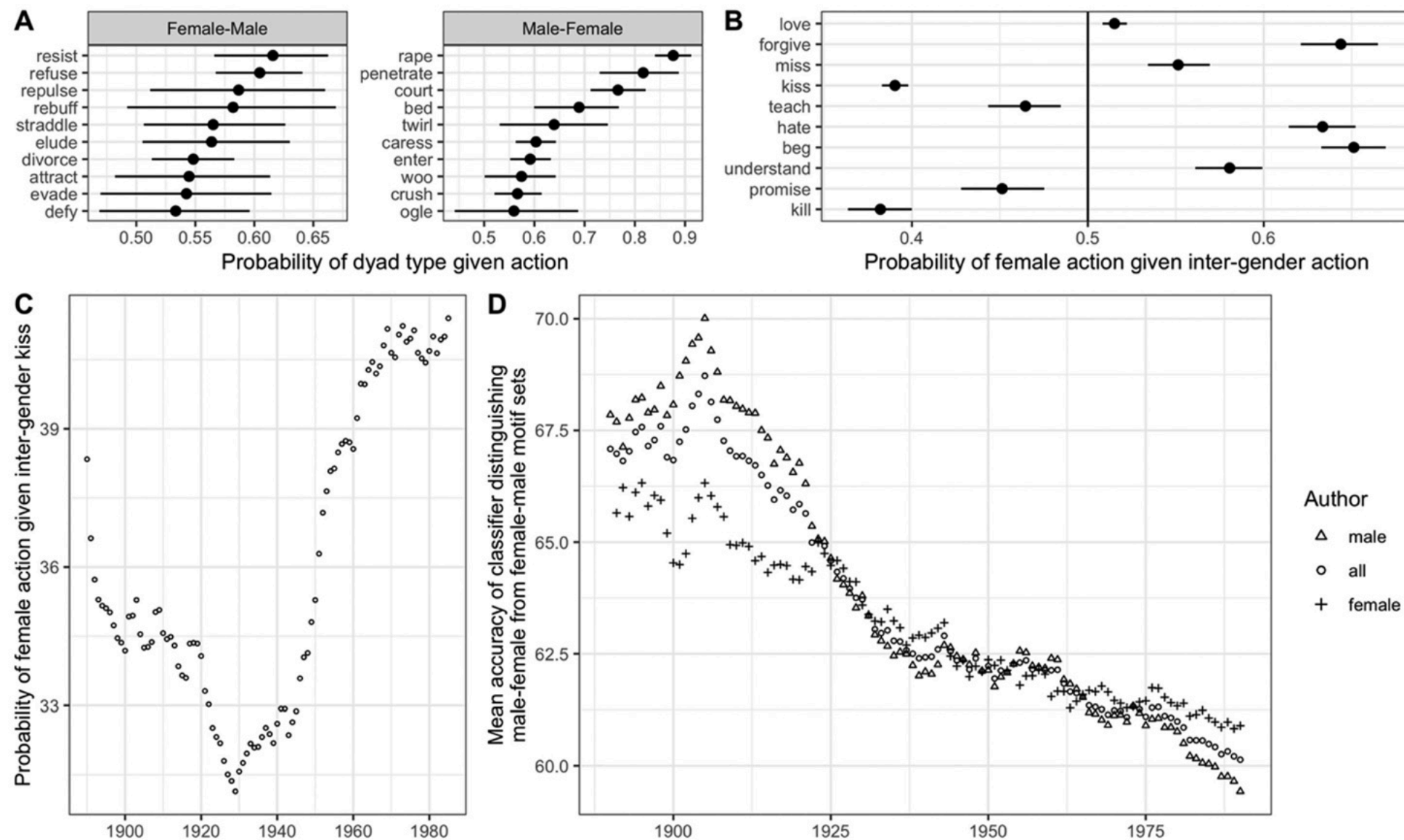
	Male patient	Female patient	Total
Male actions	17.3%	39.1%	56.4%
Female actions	32.2%	11.5%	43.7%
Total	49.5%	50.6%	

Stuhler, 2022

Pre-Trained Models

Knowledge Extraction

spaCy



Stuhler, 2022

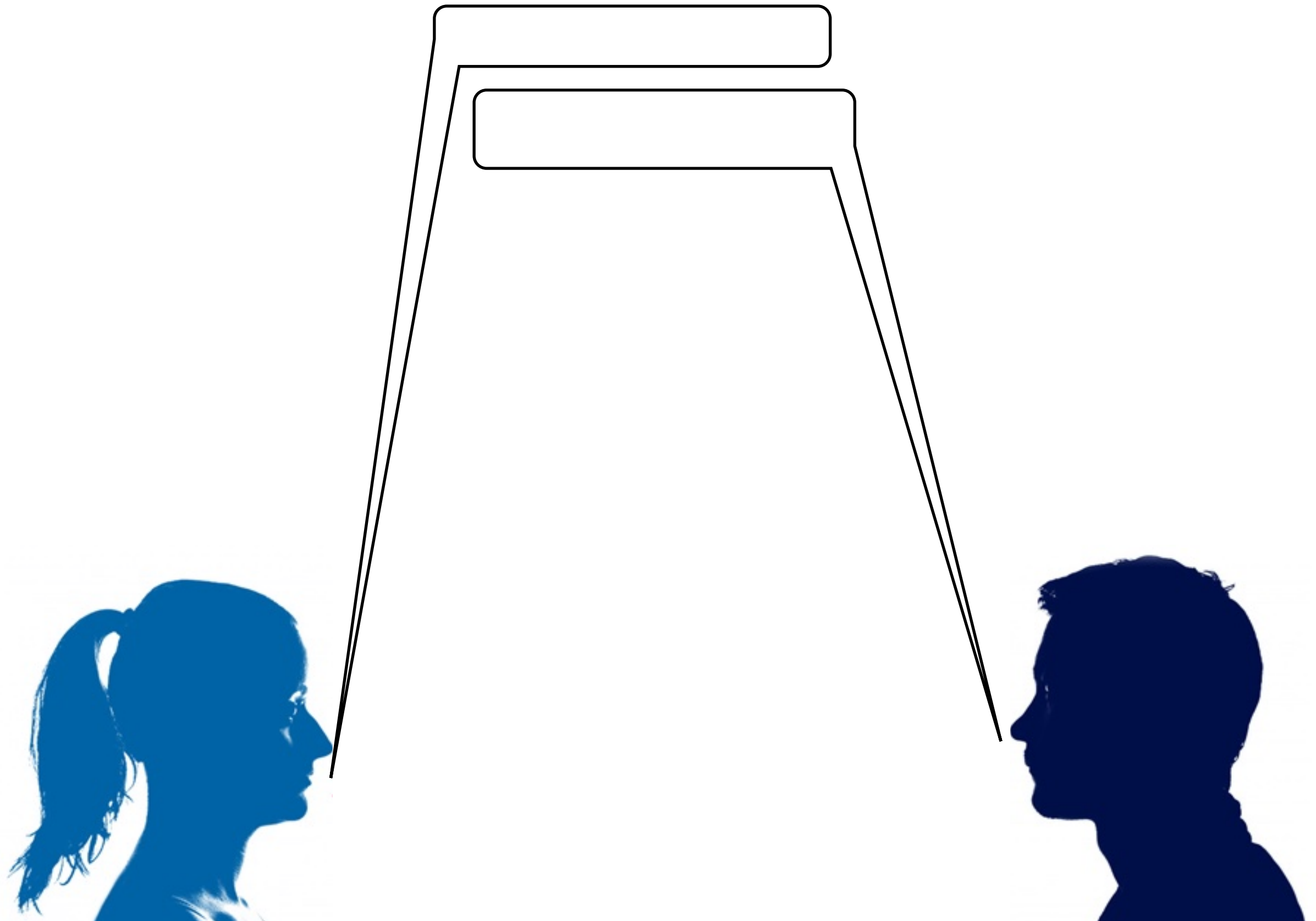
Conversations



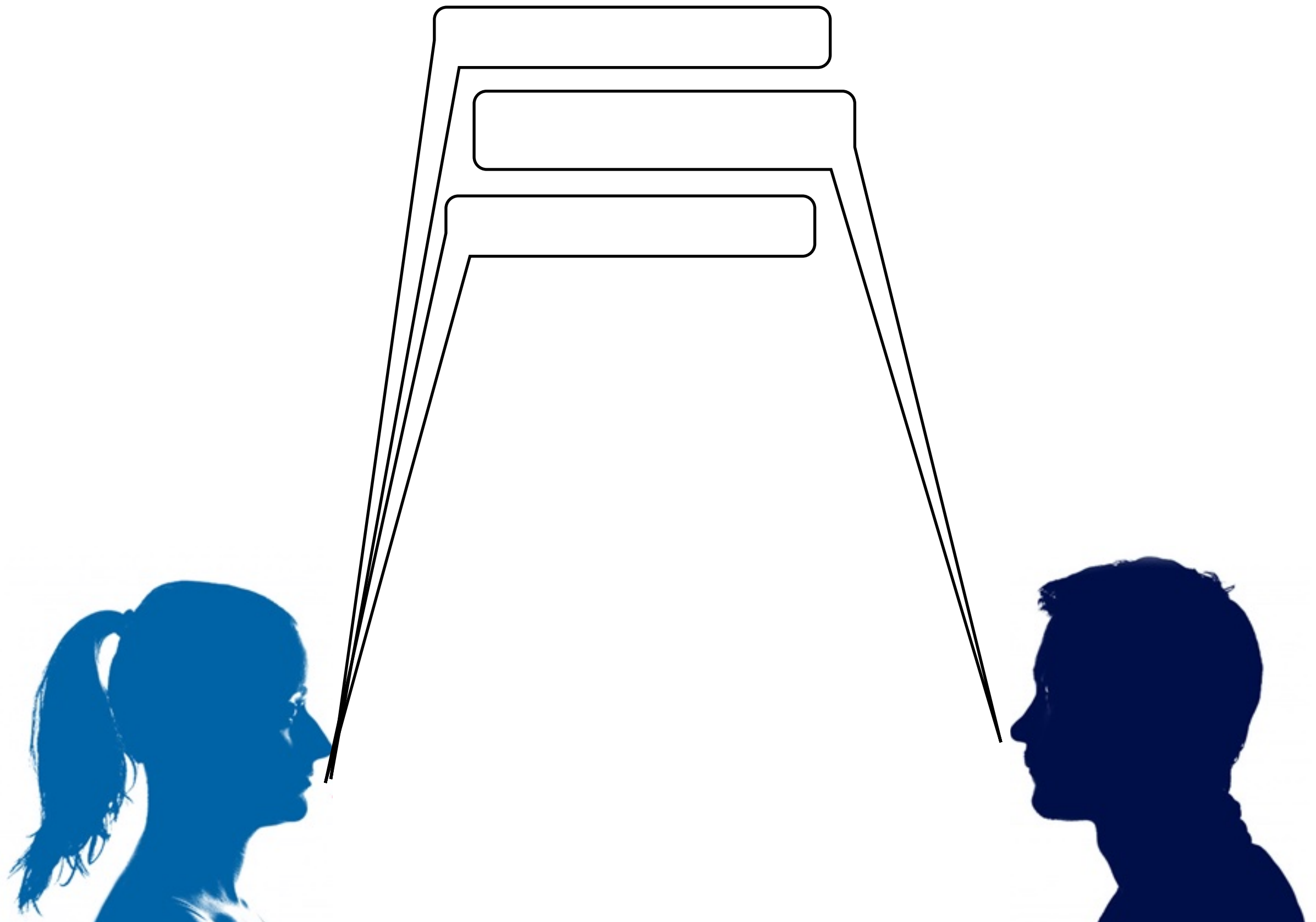
Conversations



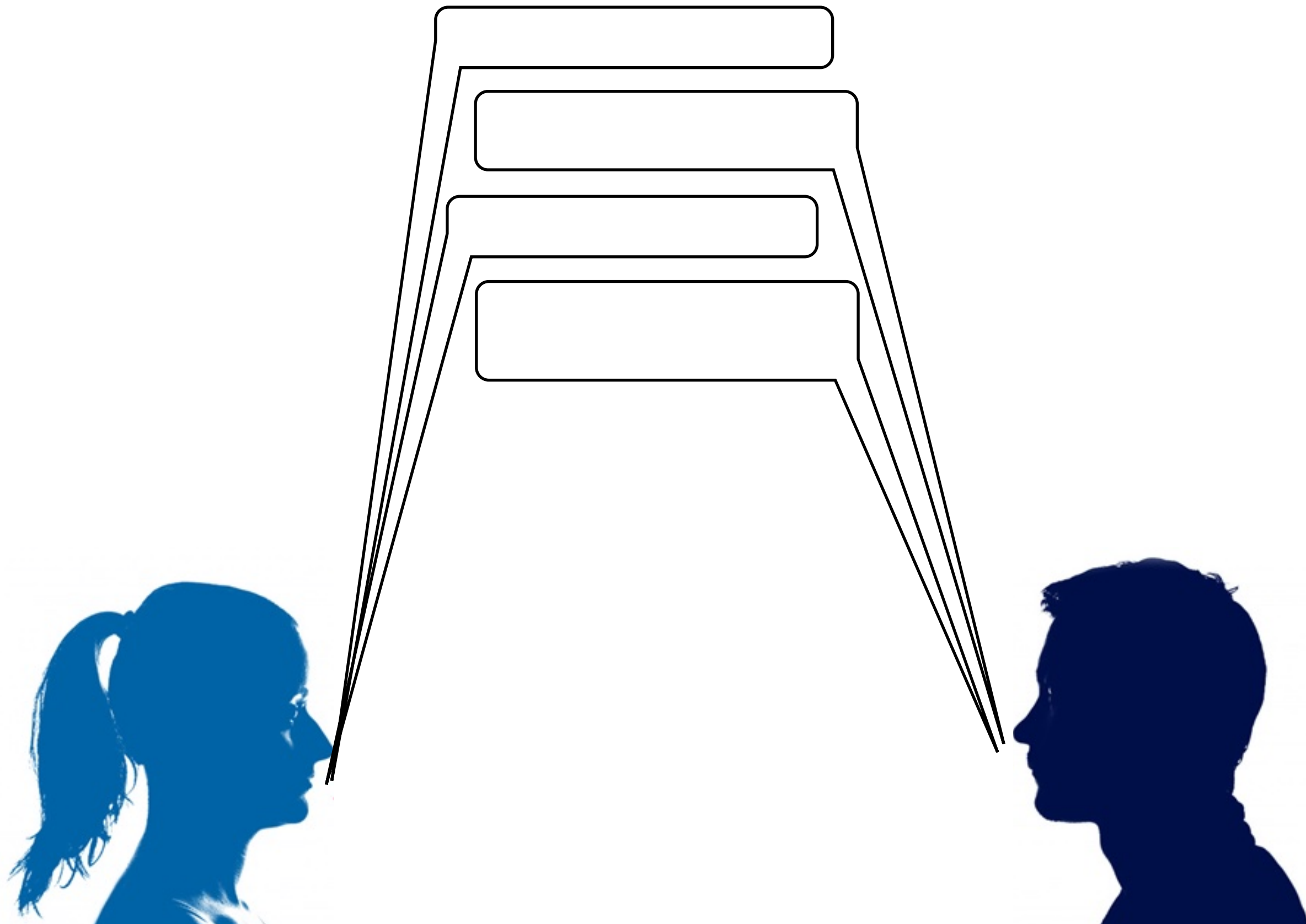
Conversations



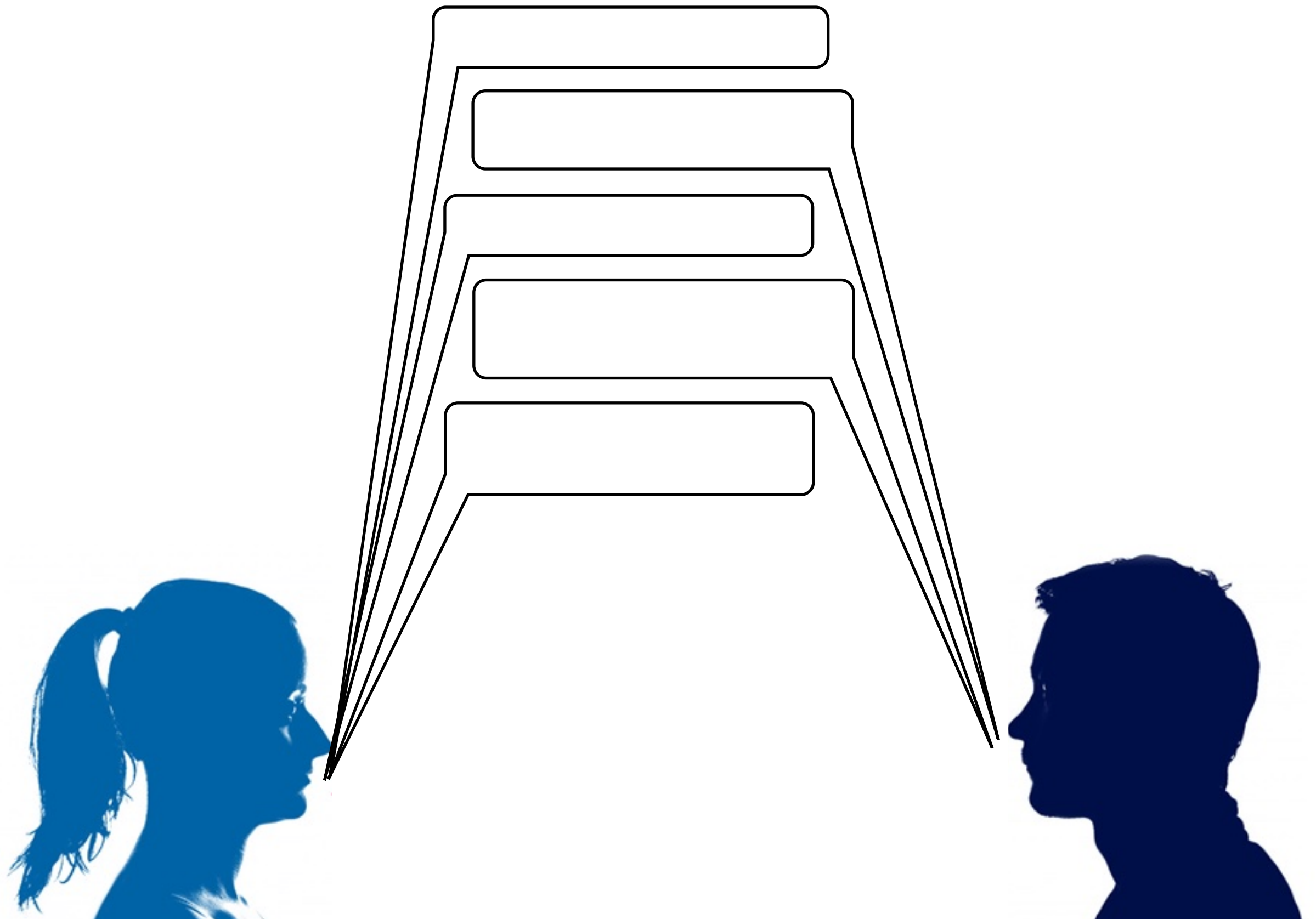
Conversations



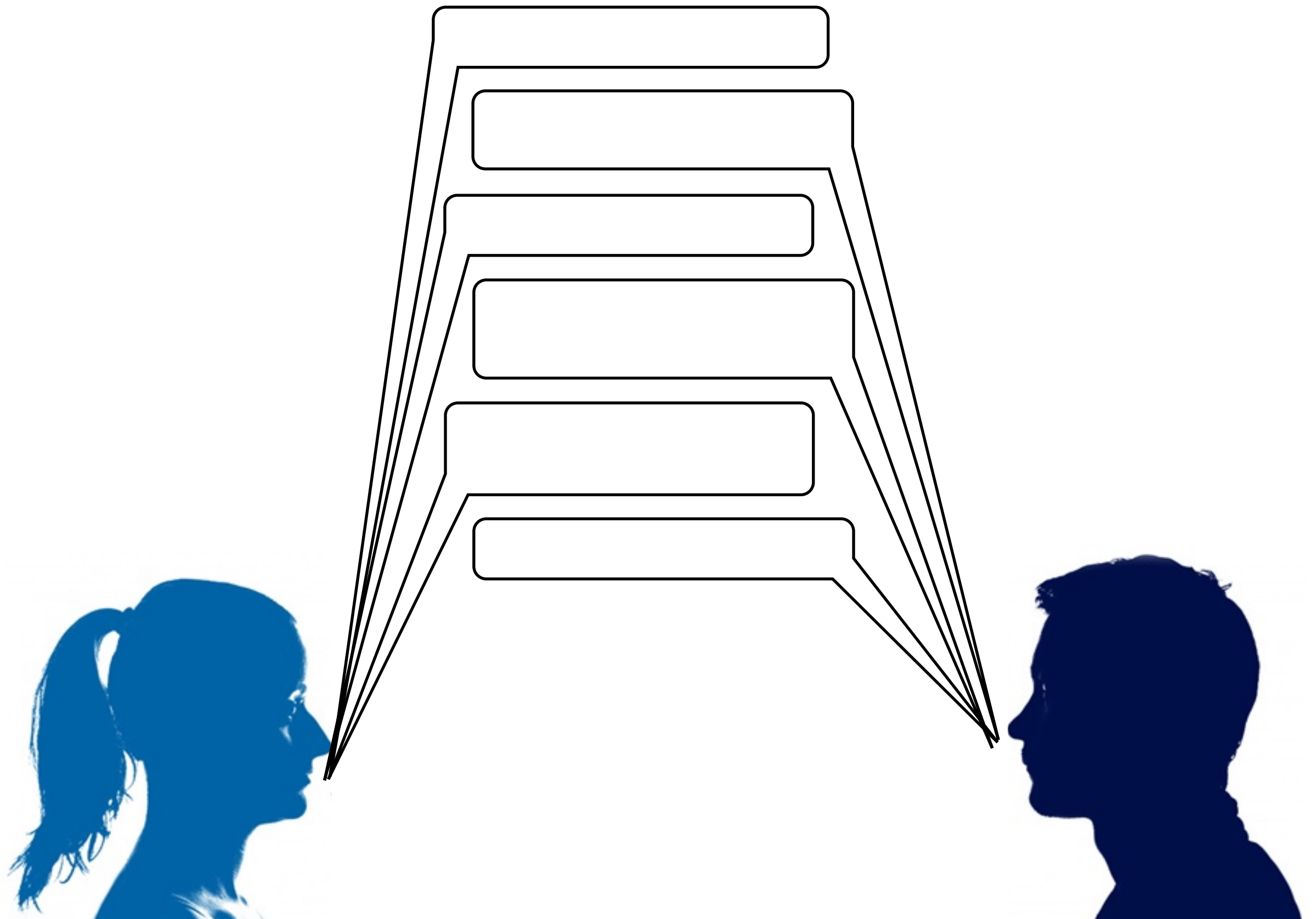
Conversations



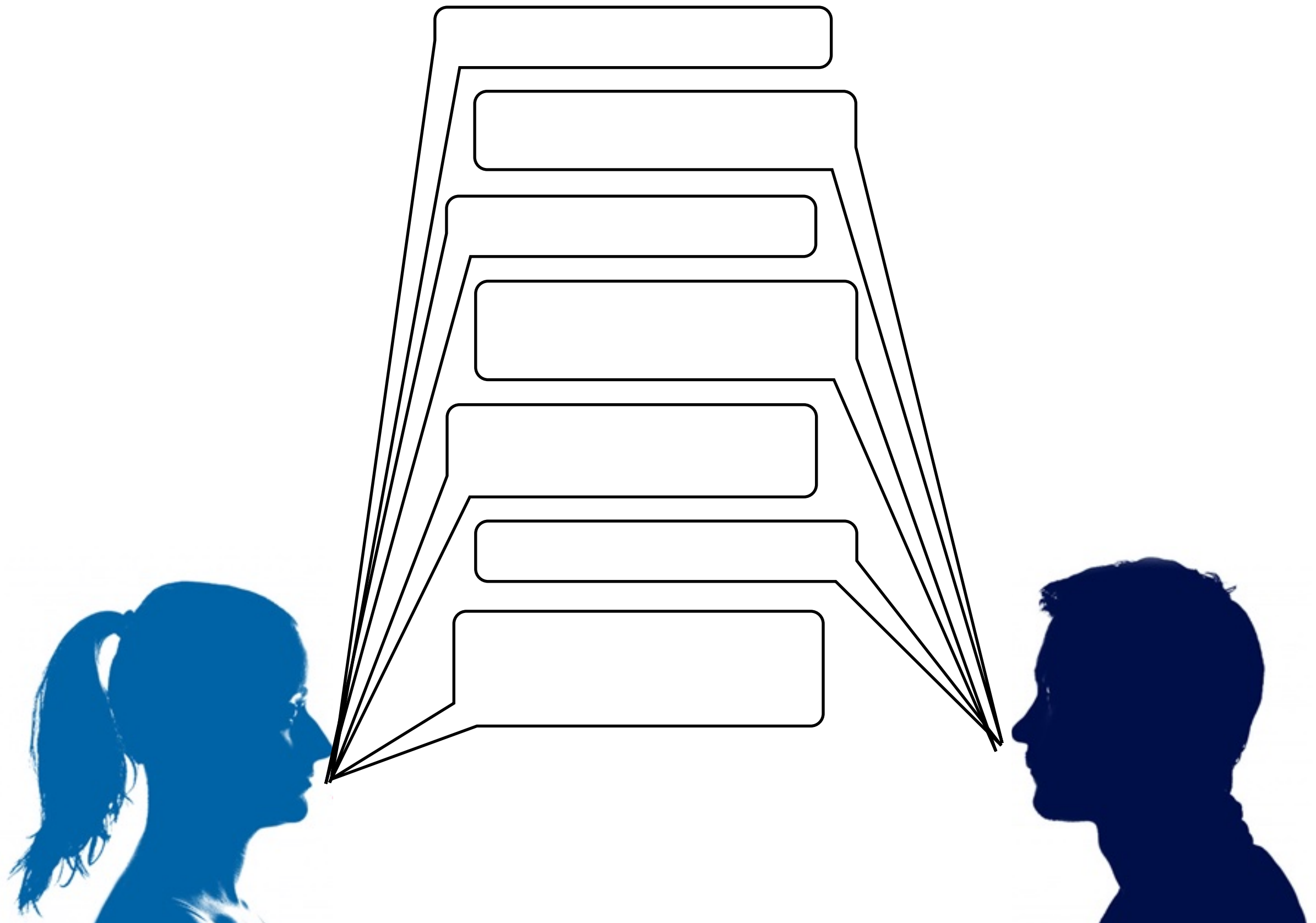
Conversations



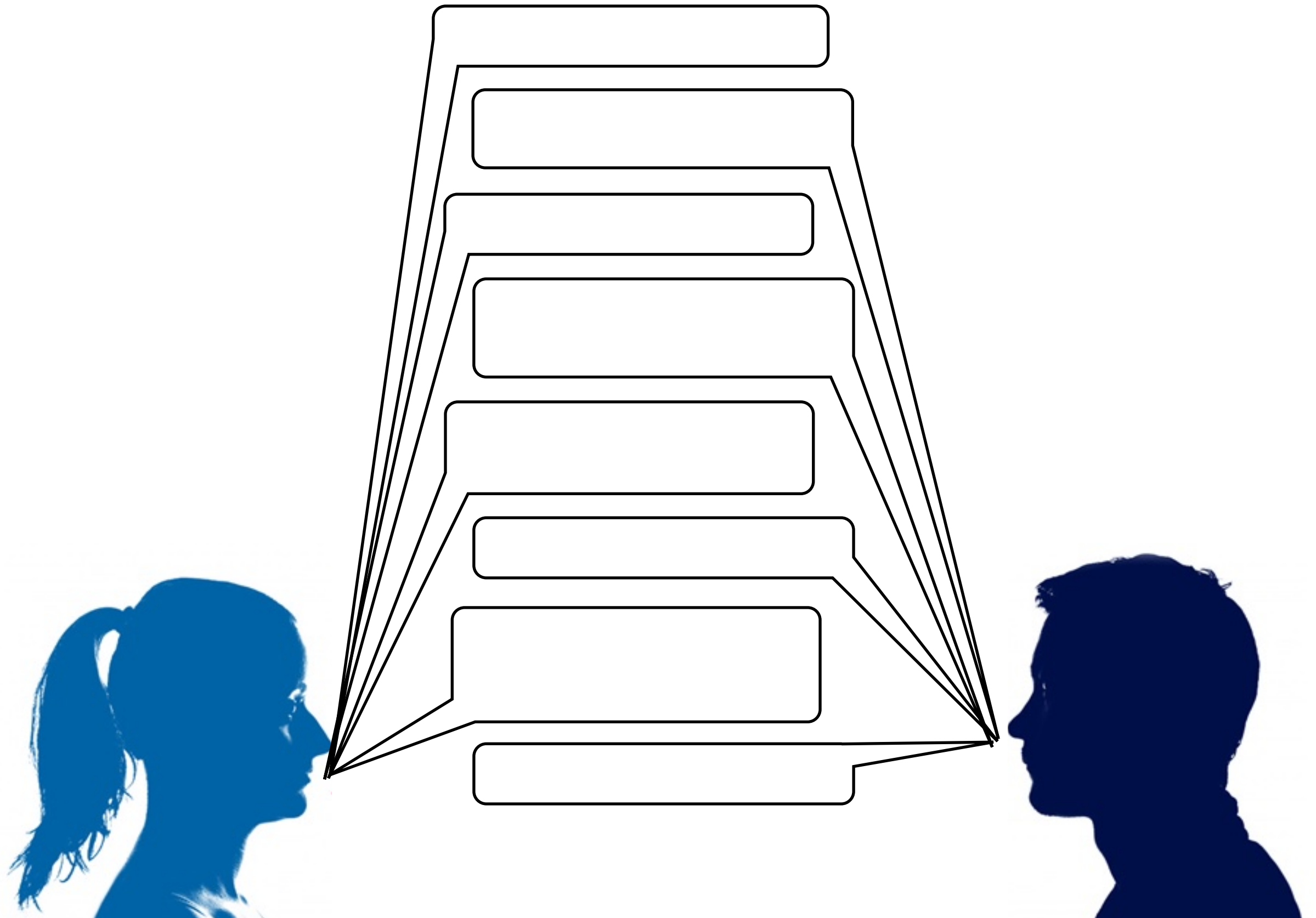
Conversations



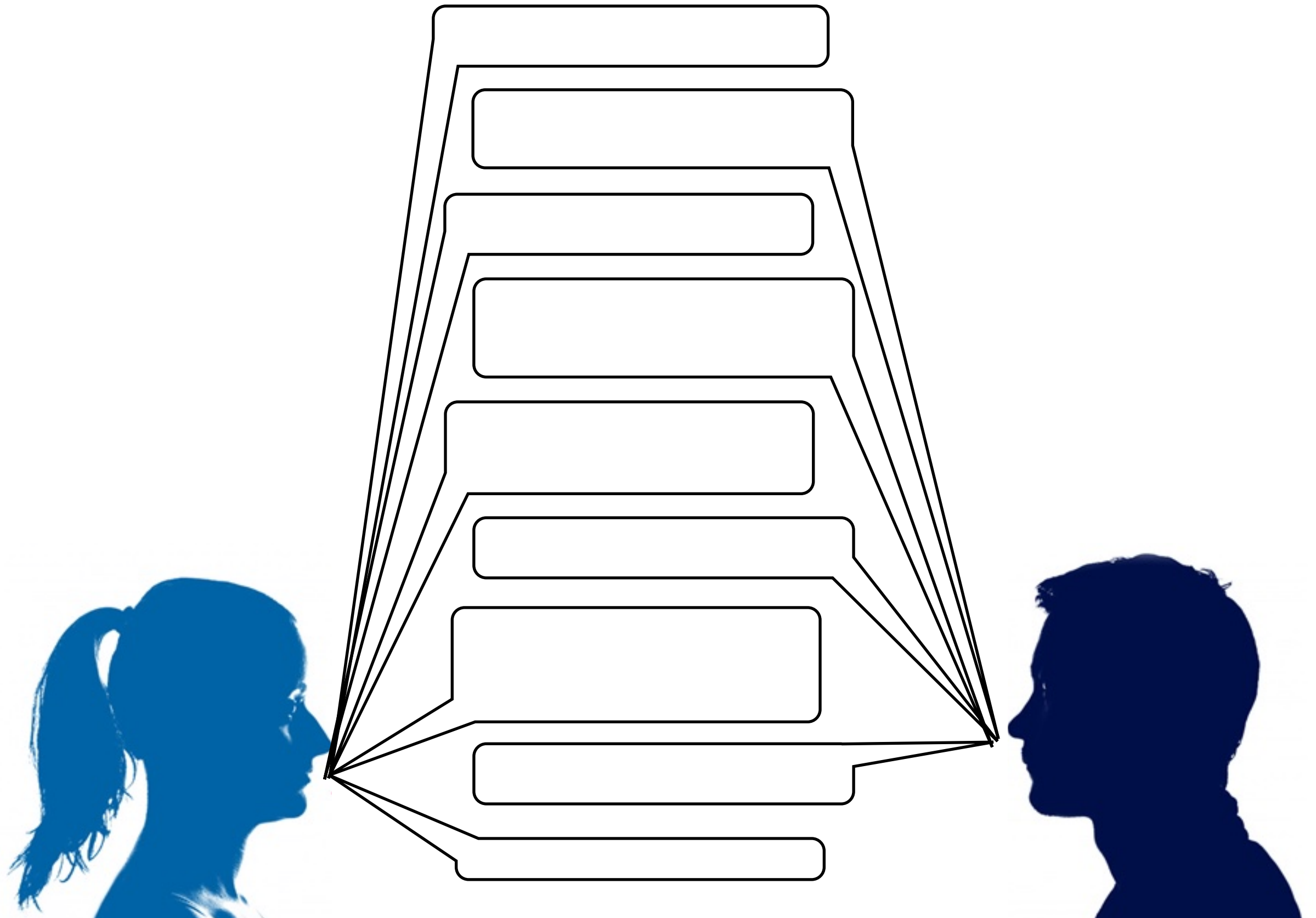
Conversations



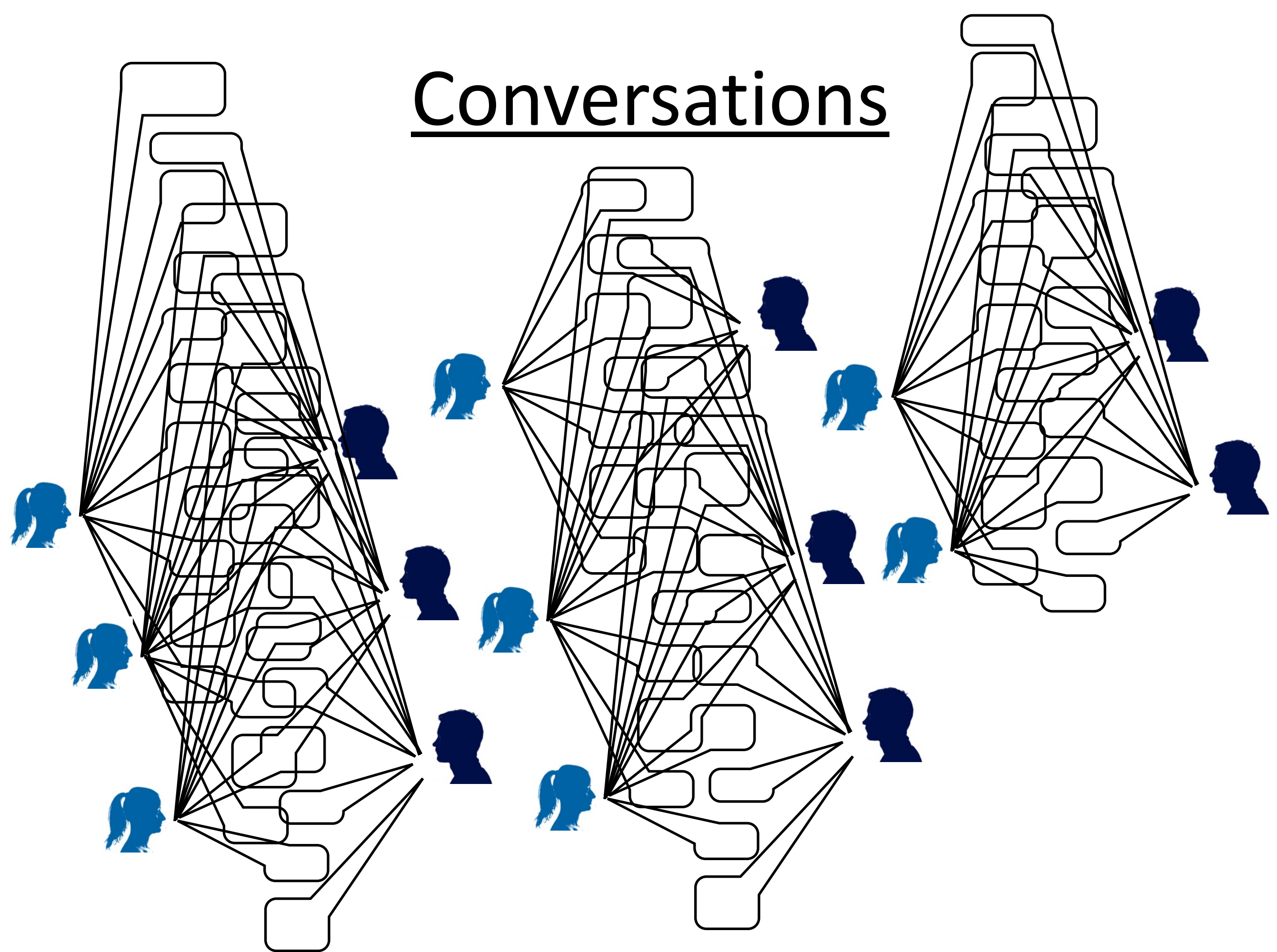
Conversations



Conversations



Conversations



Conversational Challenges

Compared to single author texts (news articles, product reviews, etc.)

Conversational Challenges

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1. Multiple speakers, each with different metadata

Conversational Challenges

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2. Generated responsively -> sequence matters!

Conversational Challenges

Compared to single author texts (news articles, product reviews, etc.)

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3. Less edited -> more pronouns, disfluencies, interruptions, etc.

Conversational Challenges

Compared to single author texts (news articles, product reviews, etc.)

1. Multiple speakers, each with different metadata
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4. Common ground maintenance: repairs, clarifications,
backchannels, affirmation, acknowledgment, etc

Conversational Challenges

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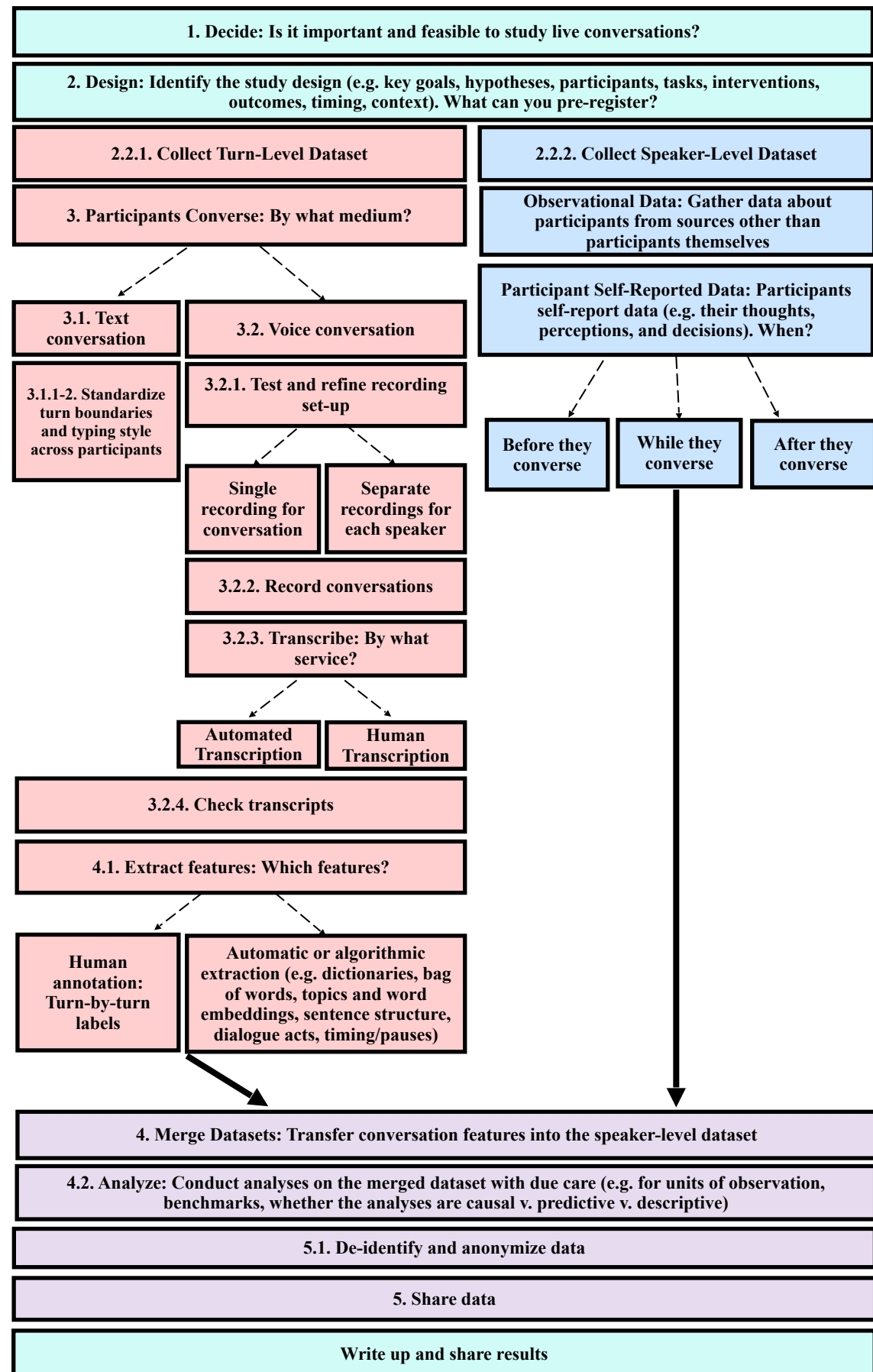
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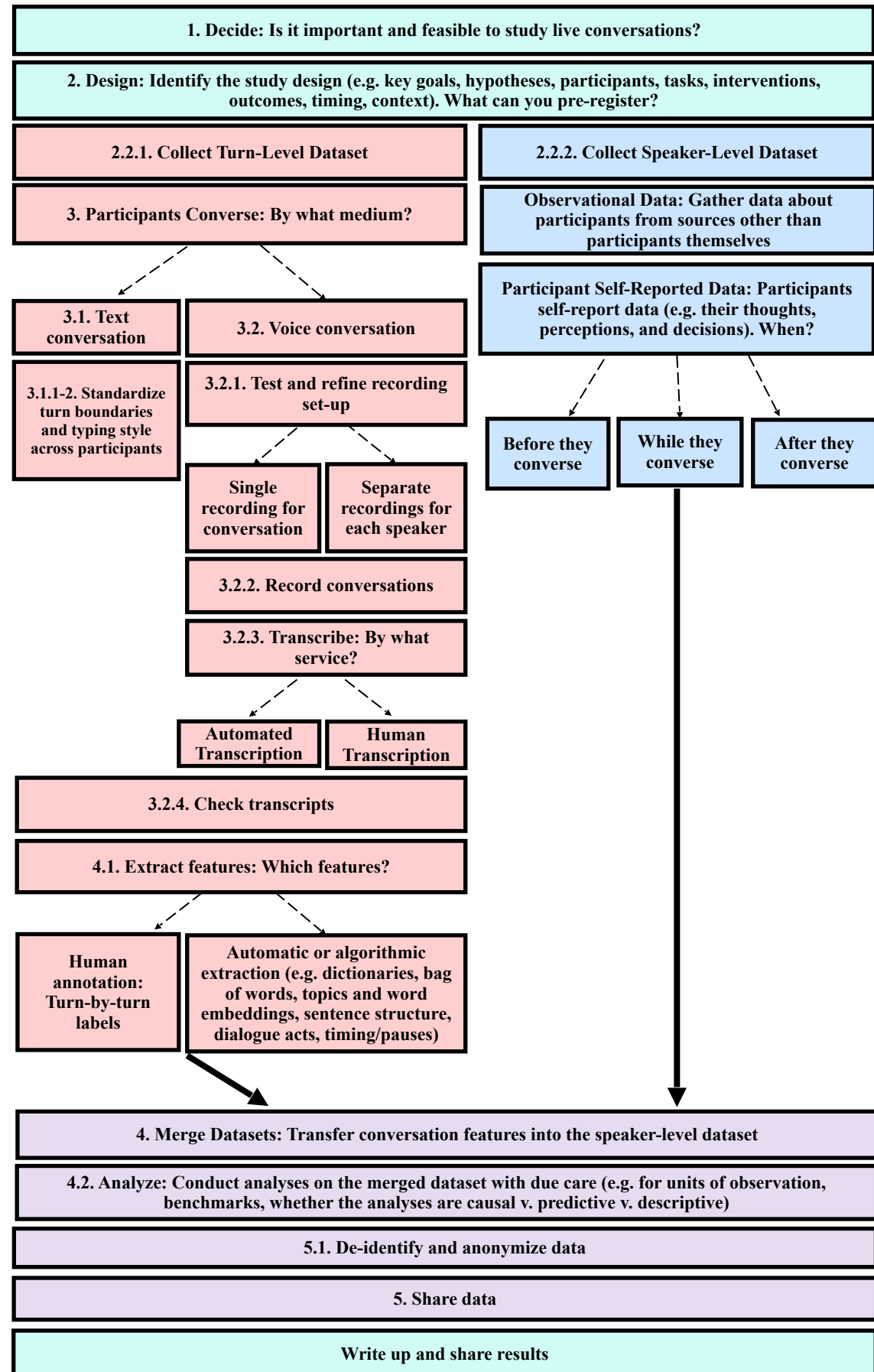
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4. Common ground maintenance: repairs, clarifications,
backchannels, affirmation, acknowledgment, etc
5. Many different topics in a single conversation
6. Many different goals between/within speakers
7. Many non-verbals - prosody, body posture, faces...

Research on Conversations



*A Practical Guide to
Conversation Research*
(Yeomans, Boland, Collins,
Abi-Esber & Brooks, 2023)

Research on Conversations



It's hard!

*A Practical Guide to
Conversation Research*
(Yeomans, Boland, Collins,
Abi-Esber & Brooks, 2023)

Let's keep the conversation going

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